Long-twist volume limits

for arborescent knots built by tangle composition

A specified construction class, its finite invariant,
and the hyperbolic volume after filling

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Research manuscript with an executable certificate companion

Abstract

We prove the volume conjecture for a specified recursive family of arborescent knot diagrams with sufficiently many twists in each designated region. The construction starts with two-strand tangles surrounded by auxiliary crossing circles. It uses vertical stacking, quarter-turns, and attachment of one new elementary tangle. Two boxes, each built from at least two elementary pieces by the stated rule, are joined by a cylindrical closure. We define these operations geometrically before introducing their notation. After choosing a parity pattern that gives one main component, filling the auxiliary circles produces the knots of the theorem.

For each fixed construction, every sufficiently large tuple of independently signed filling integers gives a hyperbolic knot whose reduced Kashaev invariant has exponential growth equal to its knot complement volume. The integers are fixed before the root-of-unity level tends to infinity, and the limit includes both parities and composite levels. Only auxiliary circles are filled; the knot cusp remains unfilled.

The proof starts from the finite matrices defining the invariant. Their polynomial algebra retains derivative data at repeated roots. Positive multiplication coefficients give an angle maximization problem, and explicit ideal hyperbolic cells realize its complete value on the actual augmented link. Rigorous rational inequalities control the signed complementary color terms throughout the recursive construction. A holomorphic asymptotic for the remaining finite sum is then inserted into the exact filling transform. The contour argument retains its boundary contributions and all Fourier modes, and its leading action is identified with the volume after filling. Detailed appendices and an executable certificate companion support the proof. The theorem concerns the stated construction class; it does not classify all arborescent knots.

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Chapter 1

The knots, the theorem, and the proof

1.1 What the theorem compares

The volume conjecture asks whether the growth of a sequence of finite knot invariants recovers hyperbolic volume. For each integer $N \geq 2$, the invariant is evaluated at a root of unity and produces a complex number. For a hyperbolic knot, the complement carries a complete finite-volume hyperbolic metric. The proposed comparison is

$$\frac{2\pi}{N} \log |\langle K \rangle_N| \longrightarrow \mathrm{Vol}(S^3 \setminus K).$$

The normalization of $\langle K \rangle_N$ matters: throughout this paper the unknot has value one, and Chapter 2 specifies the matrices and traces defining the invariant. Kashaev introduced the finite invariant in [1] and the volume proposal in [2]. Its relation with the root-of-unity colored Jones polynomial is described in [3].

We prove this limit for the family of knot diagrams constructed below, once every designated twist region has sufficiently many twists. The family permits arbitrary finite branching and nesting of tangles, subject to a precise recursive rule. We first explain the geometric operations and only then give symbols for the rule. A construction expression records an actual diagram: it includes the order of the pieces and the positions where they are joined.

There are two different limiting questions, and our theorem uses only one at a time. Choose a finite construction, choose the signs and sufficiently large integer numbers of twists, and hold all these choices fixed. Then let N tend to infinity. The theorem does not let the diagram or its twist numbers grow with N. Different twist regions may have different signs and arbitrarily different large magnitudes.

1.2 Building the diagrams from four-ended tangles

A *two-string tangle* is a pair of arcs in a ball whose four endpoints are fixed on its boundary. We draw its ball as a box, with two endpoints at the bottom and two at the top. Closed curves may also be present inside a box. In the augmented diagrams used in the proof, some of those curves are distinguished auxiliary circles. Composition can create further closed main

components, so the eventual requirement that the filled diagram be a knot must be checked rather than assumed.

Label the boundary points in their cyclic order by

$$0 = \text{bottom left}, \quad 1 = \text{bottom right}, \quad 2 = \text{top right}, \quad 3 = \text{top left}.$$

We use three operations, all performed on the drawn tangles with these boundary positions retained.

Vertical stacking. Place a box F below a box G. Join the two top endpoints of F to the two bottom endpoints of G by short, uncrossed arcs: F_2 joins G_1 and F_3 joins G_0. Denote the resulting box by FG. Its exposed endpoints, in the order just defined, are (F_0, F_1, G_2, G_3). Changing parentheses only changes the heights of boxes; it does not interchange their order.

Quarter-turn. Rotate the entire box clockwise through 90 degrees in the diagram plane, carrying every strand and auxiliary circle with it. Write $\mathsf{T}F$ for the result. In terms of the old endpoints, its new ordered boundary is (F_1, F_2, F_3, F_0). The algebraic evaluation implements this operation with the ordinary cup and cap matrices. It is not an arbitrary permutation of matrix entries.

Cylindrical closure. Join the exposed bottom-left endpoint to the top-left endpoint outside the box, and do the same on the right, without adding crossings. Write $C(F)$ for this closed diagram. Thus the closing pairs are 0–3 and 1–2. In $C(FG)$ the two connections between F and G and the two exterior return arcs are all part of the definition. This closure differs from joining the two top endpoints together and the two bottom endpoints together, which we call a *cap closure*.

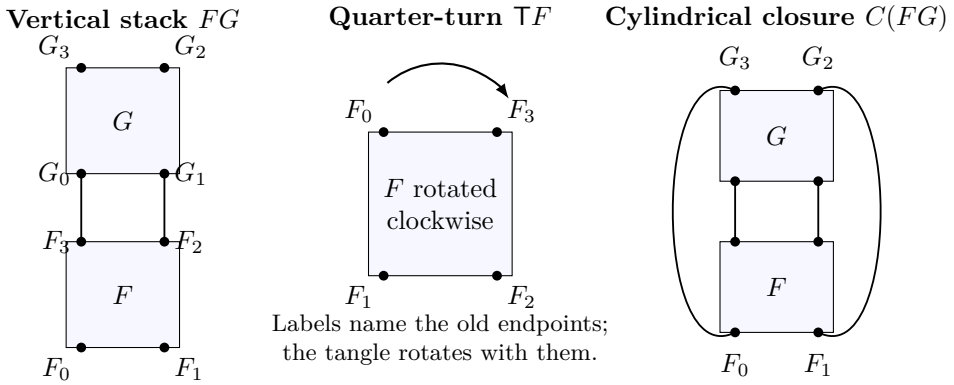


Figure 1.1: The actual connections defining the construction. Each box contains an arbitrary tangle; its interior is suppressed so that the boundary order is visible. The quarter-turn carries the tangle with the box. The exterior arcs in the last panel are the cylindrical closure.

The elementary augmented box has two through strands and a small auxiliary circle surrounding them. The circle bounds a disk pierced once by each strand; this disk fixes the framing used for surgery. Denote the box by L, and denote its quarter-turn by $h = \mathsf{T}L$. Every

occurrence of h in a construction is a separate copy with its own auxiliary circle. We call such an occurrence an *elementary piece*; later, when discussing its position in a construction tree, we also call it a *leaf*.

To obtain the final knot, replace each auxiliary circle by twists of the strands passing through its disk. Let R be one positive half twist of the two through strands; R^{-1} is the negative crossing. An integer power R^n is a stack of $|n|$ such crossings with the indicated sign. At elementary piece ℓ we retain $\varepsilon_\ell \in \{0, 1, -1\}$ half twists and perform the disk-framed integral filling with parameter p_ℓ. The filled piece is exactly

$$\mathsf{T}\big(R^{\varepsilon_\ell - 2p_\ell}\big). \tag{1.1}$$

A full twist contains two crossings, which accounts for the factor two. This formula fixes the sign convention for p_ℓ throughout the proof. After every auxiliary circle is filled, the remaining closed diagram has no auxiliary components. The main component is never filled; its complement is the manifold whose volume occurs in the theorem.

1.3 Which compositions are included

Start by stacking two elementary pieces. From an already constructed box, one may take a quarter-turn, stack it with another already constructed box, or attach one new elementary piece at its bottom or top. Repeat these operations any finite number of times. We call a box obtained in this way *completed*. This is a name for the construction rule, not an assertion that a hyperbolic metric has already been constructed. In particular, a single elementary piece is not completed.

Definition 1.1 (The recursive construction class)**.** The symbol E ranges over the completed boxes just described. Their concise recursive description is

$$E ::= hh \;\mid\; \mathsf{T}(E) \;\mid\; E_1E_2 \;\mid\; hE \;\mid\; Eh. \tag{1.2}$$

Here ::= means "can be built by one of the following operations," and the vertical bars separate the alternatives. Each written h denotes a distinct elementary piece. A closed diagram in the main theorem is $C(FG)$ with both F and G completed. Retain the original order of all endpoints, the crossing-disk framings, and one marked point on the main component for evaluating the reduced invariant.

For example, $C((hh)(hh))$ uses four elementary pieces, with two in each of the final boxes. A longer allowed construction is $C((h\mathsf{T}(hh))(hh))$: first build the two-piece box hh, turn it, attach one elementary piece, and close it against another two-piece box. Repeating $E_{r+1} = h\mathsf{T}(E_r)$ from $E_0 = hh$ gives arbitrarily long nested constructions. Branching is equally allowed, since any two completed boxes can be stacked. Starting instead with any stack h^s, $s \geq 2$, gives the same class.

The last condition in Definition 1.1 is important: both boxes joined by the final cylindrical closure must be completed. The proof uses it to place every stationary leaf parameter strictly above $1/4$, as shown in Lemma 3.5. This paper does not extend its theorem by replacing that condition with a more general feasibility test.

The filled elementary pieces in (1.1) are rational tangles. Ordered stacking and quarter-turns of rational tangles produce Conway algebraic, or arborescent, links. Thus the construction

gives a specified family within that class; it is not a classification of all arborescent knots. The choice of ε must make the final diagram one component and create no closed main component inside a box. Section 4.7 gives a finite endpoint-pairing test and proves that a suitable choice exists for every allowed construction. Changing p_ℓ adds full twists and therefore does not change this component test.

1.4 The long-twist theorem

Theorem 1.2 (Long-twist volume limit)**.** *Fix a finite expression $C(FG)$ as in Definition 1.1 and a half-twist pattern ε giving a knot. There is a finite number $P(F,G,\varepsilon)$ such that, for every independently chosen tuple of signed integers $\mathbf{p}=(p_\ell)$ with $|p_\ell|>P(F,G,\varepsilon)$, the resulting knot $K_{F,G,\varepsilon}(\mathbf{p})$ is hyperbolic and*

$$\lim_{N\to\infty}\frac{2\pi}{N}\log\left|\langle K_{F,G,\varepsilon}(\mathbf{p})\rangle_N\right|=\operatorname{Vol}\bigl(S^3\setminus K_{F,G,\varepsilon}(\mathbf{p})\bigr). \tag{1.3}$$

The level tends to infinity through all integers, including even and composite levels. The expression, signs and twists are fixed before that limit. The threshold depends on the fixed expression and sign pattern, but not on a comparison between the magnitudes or signs of the individual sufficiently long twists.

In concrete terms, one first chooses the entire finite arrangement of boxes and the parity of the crossings in each elementary piece. The theorem then supplies one threshold for that arrangement. Each filling integer may independently be positive or negative and exceed that threshold in absolute value. No ratio between different filling integers is bounded. Once these integers have been chosen, the conclusion concerns the single resulting knot and the sequence of its invariants as N grows.

The reduced invariant $\langle\cdot\rangle_N$ has unknot value one. Chapter 2 also compares it with the full modified one-cut normalization: that normalization adds the single factor i^{N+3}/N and a unit framing character, so the exponential limit is unchanged. This factor belongs only to the marked main component. The auxiliary circles are evaluated by ordinary traces.

The isolated closures $C(h^s)$, $s\geq 3$, form a separate inherited family; Section 6.4 works out $C(h^3)$ explicitly. The isolated $C(hh)$ is excluded: its even-level evaluation vanishes and its odd-level contribution lies on a flat boundary. General closures with a single elementary piece on one side, such as $C(hE)$, are also outside Theorem 1.2. For instance, $C(h\mathsf{T}(h^s))$ places the inner complete leaf parameters at $1/4$, outside the strict interval used in this paper's local asymptotic argument. These exclusions describe the scope of this proof, not an assertion that no other methods apply to those knots.

1.5 A guide to the proof and its terminology

The proof first studies the augmented diagram, in which the auxiliary circles are still present, and then performs their fillings by an exact finite identity. A *color* is the integer label specifying the representation assigned to an auxiliary circle. At level N these labels range from 1 to N. Before filling, the evaluation therefore depends on a tuple of colors. We call this finite, color-dependent evaluation the *source*. The *complete-color source* means that every auxiliary color equals N; the term is distinct from completeness of a hyperbolic metric.

The algebra used to calculate a source has repeated roots. At a double root, both a polynomial's value and its derivative matter. We refer to the derivative data as the *first normal data*, in the elementary algebraic sense of the coefficient of $z - c$ modulo $(z - c)^2$. Keeping these data is essential at both parities of N.

Expanding each elementary piece in a polynomial basis gives a finite sum. An admissible assignment of integer indices to that sum is a *state*. An internal *wire* is an index shared by two factors; it is not an additional physical strand. A *product gate* is the coefficient for one stacking operation with two input indices and one output index. After scaling indices by N, these coefficients have exponential growth described by a real function, the *action*. A state is *strict* when all its nonforced angle inequalities are strict. The maximizer of the complete-color action determines explicit ideal hyperbolic cells. The cells have quadrilateral faces along which they are glued. The order of the four vertices specifies the face map; Chapter 4 explains how these auxiliary gluing faces reconstruct the actual closed link diagram.

For colors other than N, the exact elementary evaluation splits into a principal part and a signed complementary part. We call them the *core* and the *alias*; the identities defining them appear in (2.17)–(2.18). Chapter 5 proves that every term containing an alias is exponentially smaller than the leading complete action, even after arbitrary allowed composition. This is where the finite rational certificates enter the proof.

Near the remaining strict maximum, the finite sum has a holomorphic asymptotic of the form $N^{3/2}a(\chi)e^{N\Phi(\chi)}$, with a nonzero. We call this local analytic contribution the *packet*. The word includes a specified finite sum and a controlled error; it does not mean a formal stationary-phase expression. The complex color parameters χ are also the parameters of a marked deformation of the augmented hyperbolic structure. Identifying their peripheral holonomies fixes the absolute volume, not just its differential.

Finally, the filling identity sums the source over all colors with oscillatory coefficients. We analyze the whole sum, including the boundary integrals created when contours move and every integer Fourier mode. These modes are called *charges*. The leading charges add without cancellation, and their critical action equals the volume after filling. This last step is necessary because that volume is smaller than the complete augmented volume; an estimate obtained by taking absolute values of the filling coefficients would not recover it.

1.6 Conventions and verification

All logarithms on a positive real interval are real. Complex logarithms and square roots continue those values on the explicitly chosen neighborhoods. We use $\operatorname{arccot} X \in (0, \pi)$ for real X and $\mathrm{i}^2 = -1$. Constants may depend on the fixed construction, its selected strict stationary point and a fixed analytic neighborhood, but not on N. A polynomial factor means a bound by N^C with such a fixed exponent.

The appendices give the finite identities and detailed analytic estimates. The certificate files contain rational candidate points and exact domain subdivisions; the verifier proves their supporting inequalities using outward rational bounds. Numerical optimization is not part of verification. Chapter 9 explains which claims are analytic proofs, which depend on these finite certificates, and which are merely checked by optional experiments.

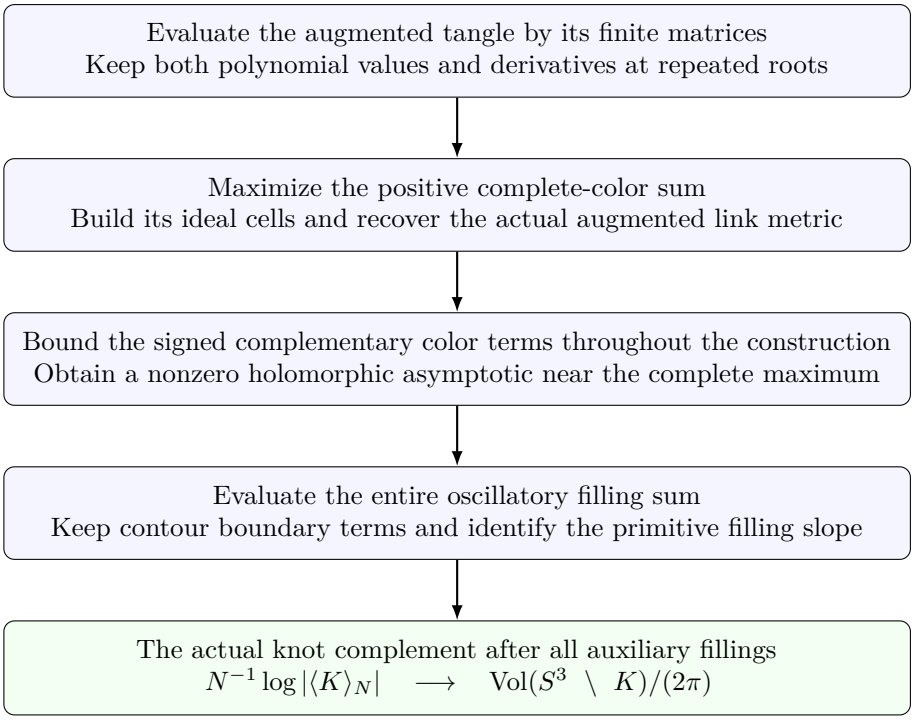


Figure 1.2: The four comparisons in the proof. The last comparison analyzes the full oscillatory filling sum and identifies its critical action with the volume of the resulting knot complement.

Chapter 2

Evaluating the invariant in a finite polynomial algebra

We now turn the tangle operations of Chapter 1 into finite matrix calculations. A strand carries a finite-dimensional vector space, a crossing acts on the two corresponding factors, and joining endpoints means contracting the appropriate indices. Closing an auxiliary circle takes a trace. Leaving one marked main strand open until the final contraction gives the reduced normalization.

The purpose of the polynomial algebra introduced below is to make these contractions compositional. Every two-strand operator that occurs here is a polynomial in one matrix, the pair Casimir Ω. Stacking becomes multiplication of polynomials, while the quarter-turn has a particularly simple action in the Newton basis. The last section then expresses an arbitrary filling power in the basis of colored auxiliary-circle operators. All of these statements are finite identities at each level; asymptotics begin only in Chapter 3.

2.1 Crossings and the one-cut normalization

Fix $N \geq 2$, and set

$$d = N - 1, \qquad q = e^{\pi \mathrm{i}/N}, \qquad Q = q^2, \qquad x = q^d, \qquad \kappa = q^{d^2/2}, \qquad \rho = q - q^{-1}. \tag{2.1}$$

There is no restriction to prime or odd N. The main strands carry the N-dimensional vector space V_N; this is what "complete color" means at this stage. The operators H, K, E, F encode its quantum-group action. They are concrete matrices, which we specify so that the crossing convention can be checked. One convenient choice of basis is $v_0, \dots, v_d$ and

$$Hv_j = (d - 2j)v_j, \quad K = q^H, \quad Ev_j = [j]_q^2 v_{j-1}, \quad Fv_j = v_{j+1}, \qquad [j]_q = \frac{q^j - q^{-j}}{q - q^{-1}}.$$

Missing endpoint vectors are zero. The finite universal crossing, including its Cartan factor, is

$$\mathscr{R} = q^{H \otimes H/2} \sum_{k=0}^{d} \frac{q^{k(k-1)/2} \rho^k}{[k]_q!} E^k \otimes F^k. \tag{2.2}$$

Every denominator is nonzero. The physical braided crossing is obtained with the fixed exchange and transpose convention; the inverse is the actual matrix inverse. The framing and

enhancement are part of the definition, not an unspecified factor appended to an asymptotic formula.

For calculations in the polynomial algebra it is useful to change basis on this same space. We call the new coordinates balanced; they do not define a different invariant. In this basis the operators are

$$\mathcal{E}|a\rangle = (Q^{-a} - 1)|a-1\rangle, \quad \mathcal{F}|a\rangle = (1 - Q^{a+1})|a+1\rangle, \quad K|a\rangle = xQ^{-a}|a\rangle. \tag{2.3}$$

The ordinary cup and cap, and the one marked cap, have entries

$$p_{ab} = Nx^{-b}\delta_{a+b,d}, \qquad \ell_{ab} = \frac{x^a}{N}\delta_{a+b,d}, \qquad m_{ab} = \frac{x^b}{N^2}. \tag{2.4}$$

The cup creates two endpoints and the cap joins them. Their two snake identities say that a bend followed by the opposite bend evaluates to a straight strand. The different marked cap implements the reduced invariant and satisfies $mp = 1$. The diagonal operator $g = K^{1-N} = (-1)^d K$ is the *pivot*, the correction inserted when an ordinary strand is closed by a trace. For a two-strand operator A commuting with the pair quantum-group action, the original cylindrical evaluation is

$$\phi(A) = \frac{1}{N}\,\mathrm{Tr}_{V_N \otimes V_N}\big[(I \otimes g)A\big]. \tag{2.5}$$

The factor $1/N$ in this expression is the single cut average. It is not repeated at the auxiliary circles.

Proposition 2.1 (Normalization)**.** *Let A_{raw} be the scalar obtained from the original crossings, turns, ordinary auxiliary traces and the single mark. In the enhanced reduced Kashaev convention,*

$$|\langle L\rangle_N| = |A_{\mathrm{raw}}(L)|.$$

In the full modified one-cut convention the modulus is $N^{-1}|A_{\mathrm{raw}}(L)|$. The changes of recorded component framing contribute only unit phases at the complete color.

Proof. The exact Fourier comparison is recorded in Appendix A.1. It is a simultaneous similarity of every adjacent braid generator and its inverse, together with an exact equality of enhancements. The factorial changes of basis cancel on every internal or traced wire. On the one remaining head the partial trace is scalar: it commutes with E, F, K, whose distinct weights and nonzero adjacent coefficients force scalarity. Hence the open-factor similarity cancels as well.

For clarity, the complete modified dimension is evaluated as a whole meridian germ. With the original meridian $M = -qq^{\alpha} = e^u$,

$$d_N(\alpha) = -q^{N(1-N)/2}\frac{q^{\alpha+1} - q^{-\alpha-1}}{q^{N\alpha} - q^{-N\alpha}} = \mathrm{i}^{N+3}\frac{\sinh u}{\sinh Nu}.$$

Its removable value at $u = 0$ is i^{N+3}/N. Exactly one component receives this factor. The twist at completeness is $q^{-d^2/2} = \kappa^{-1}$, of modulus one. These facts prove the asserted absolute normalization without estimating any large change-of-basis matrix. □

2.2 The full pair algebra

An operator on two strands must commute with the joint action of the quantum-group generators if it is to represent one of our invariant tangle boxes. The formulas below specify that joint action. The subsequent lemma identifies all such commuting operators; this is the meaning of the term *commutant*. The balanced pair generators and Casimir are

$$\mathcal{E}_{12} = \mathcal{E}_1 K_2 + \mathcal{E}_2, \quad \mathcal{F}_{12} = \mathcal{F}_1 + K_1^{-1}\mathcal{F}_2, \quad K_{12} = K_1 K_2, \quad \Omega = \mathcal{F}_{12}\mathcal{E}_{12} + qK_{12} + q^{-1}K_{12}^{-1}. \tag{2.6}$$

Let $\mathsf{C}_n(z) = 2T_n(z/2)$, where T_n is the usual Chebyshev polynomial. Thus $\mathsf{C}_0 = 2$, $\mathsf{C}_1 = z$ and $\mathsf{C}_{n+1} = z\mathsf{C}_n - \mathsf{C}_{n-1}$. Define

$$P_N(z) = \mathsf{C}_N(z) + 2 = \prod_{j=0}^{d}(z - c_j), \qquad c_j = q^{2j+1} + q^{-2j-1}, \qquad \mathcal{H}_N = \mathbb{C}[z]/(P_N). \tag{2.7}$$

The list of roots includes multiplicity. Since $c_{d-j} = c_j$, all distinct roots are double except the simple root -2 at odd N.

Lemma 2.2 (Full commutant)**.** *The endomorphisms of $V_N \otimes V_N$ commuting with the three pair generators are exactly $\mathbb{C}[\Omega] \simeq \mathcal{H}_N$.*

Proof. Let W_T be the span of $|a, T - a\rangle$. The K_{12} eigenvalue one occurs only on W_d. The matrices $\mathcal{E}_{12} : W_{T+1} \to W_T$ for $T < d$, and $\mathcal{F}_{12} : W_T \to W_{T+1}$ for $T \geq d$, have triangular maximal minors with nonzero diagonals. Consequently the entire pair is generated from W_d by these raising and lowering operators.

On W_d, the Casimir is a symmetric tridiagonal matrix with adjacent entries $-(1-Q^{a+1})^2 \neq 0$. Its first basis vector is cyclic: the kth power has last possible coordinate k, with coefficient equal to the nonzero product of the first k adjacent entries. An operator commuting with this block is therefore a unique polynomial of degree below N in that block. Subtracting the same polynomial in Ω from a pair intertwiner gives zero on W_d, hence zero everywhere by generation.

The elementary quantum-shift product gives $P_N(\Omega) = 0$. The cyclic block has minimal polynomial of degree N; its continuant, or the generic highest-weight decomposition followed by regular specialization, gives precisely (2.7). This proves the claim with all first normal directions intact. □

At a double root c, reduction modulo $(z - c)^2$ leaves

$$f(z) = f(c) + f'(c)(z - c) \pmod{(z - c)^2}.$$

Thus both the value and its first derivative are independent coordinates of the operator. Throughout the paper, retaining the "full algebra" or the "normal data" means retaining this derivative as well. A polynomial can vanish at every distinct root and still define a nonzero operator in $\mathcal{H}_N$. Evaluating only at roots would therefore lose part of the invariant.

2.3 The actual turn and the Newton basis

We choose a polynomial basis adapted to successive Casimir roots, including their multiplicities. Its normalization is designed so that turning a physical box reverses the basis index exactly.

The same basis makes the final closure a coefficient extraction. These are the two facts that will allow us to follow every internal summation index through a large construction. Put

$$s_r = \prod_{u=1}^{r} 2\sin\frac{\pi u}{N}, \quad s_0 = 1, \qquad B_r(z) = \prod_{j=0}^{r-1}(c_j - z), \qquad \widehat{B}_r(z) = \frac{B_r(z)}{s_r^2}, \quad 0 \le r \le d. \tag{2.8}$$

The quarter-turn is the literal contraction

$$\mathsf{T}(A) = (I^{\otimes 2} \otimes \ell)(I \otimes A \otimes I)(p \otimes I^{\otimes 2}).$$

Its entries in the recorded all-primal chart are

$$\mathsf{T}(A)_{(a,b);(c,e)} = x^{a-e} A_{(b,d-e);(d-a,c)}.$$

The cups and caps are those of (2.4). A different source-normalized rotation can carry an additional power of x; throughout this manuscript T is the unscaled genuine turn.

Proposition 2.3 (Newton turn and closure)**.** *In the full algebra,*

$$\mathsf{T}\widehat{B}_r = \widehat{B}_{d-r}, \qquad \phi\left(\sum_{r=0}^{d} \alpha_r \widehat{B}_r\right) = \alpha_d, \qquad \phi(\widehat{B}_i \widehat{B}_j) = \delta_{i+j,d}. \tag{2.9}$$

The finite verification is given in Appendix A.2. It uses the physical turn, its nonzero adjacent charge-block entries and the finite quantum-binomial recurrence. In particular, (2.9) is not a generic-q continuation of a formula checked only on distinct root values. The turn is linear and involutive in this basis, but is not multiplicative.

The same closure can be displayed directly in Hermite coordinates:

$$\phi(f) = (-1)^d \sum_{c \ mdouble} \big((4-c^2)f'(c) - cf(c)\big) + \mathbf{1}_{N \ modd} f(-2). \tag{2.10}$$

Equivalently,

$$\phi(f) = (-1)^d N^2 [z^d](f \bmod P_N). \tag{2.11}$$

For example, at a double root c, writing $Q_c = P_N/(z-c)^2$ gives $Q_c(c) = N^2/(4-c^2)$ and $Q_c'(c)/Q_c(c) = c/(4-c^2)$ from the Chebyshev differential equation. The residue of f/P_N is exactly the corresponding term of (2.10) after multiplication by $(-1)^d N^2$. Summing residues proves (2.11); the simple odd root has $P_N'(-2) = N^2$. The sole shortest tridiagonal path from the first to the last coordinate has product $(-1)^d N^2$, which also identifies this scalar with the original endpoint matrix element.

2.4 Positive multiplication with every normal retained

Suppose two stacked boxes contribute basis indices i and j. Their product is a linear combination of output indices r. The following coefficient formula gives both the allowed triples (i, j, r) and their weights. Calling this local operation a *gate* is only shorthand for this one multiplication; the geometric operation remains the vertical stack.

Theorem 2.4 (Multiplication law)**.** *If $i+j<N$, then*

$$\widehat{B}_i\widehat{B}_j=\sum_{r=\max(i,j)}^{i+j}\mu_{ijr}\widehat{B}_r,\qquad \mu_{ijr}=\frac{s_{i+j}s_r}{s_is_js_{r-i}s_{r-j}s_{i+j-r}}>0. \tag{2.12}$$

If $i+j\geq N$, the product is zero in the full algebra. In particular, a product gate has precisely the integer support

$$i+j\leq d,\qquad \max(i,j)\leq r\leq i+j. \tag{2.13}$$

Proof. At a generic parameter ζ, put $t=-(\zeta-\zeta^{-1})^2$. The unnormalized coefficients are

$$B_iB_j=\sum_r t^{i+j-r}\frac{[i]![j]![i+j]!}{[r]![r-i]![r-j]![i+j-r]!}\,B_r.$$

Multiplication by c_i-z and the identity $(c_i-z)B_r=B_{r+1}+(c_i-c_r)B_r$, with $c_i-c_r=t[r+i+1][r-i]$, reduce induction to

$$[i+1][i+j+1]=[r][r-j]+[r+i+1][i+j+1-r].$$

This follows by expanding the four powers of ζ. Endpoint coefficients follow from the same zero factors. At the principal root, all factorial indices in the case $i+j<N$ are below N. Since $[u]!=s_u/(2\sin(\pi/N))^u$, normalization by $s_i^2s_j^2/s_r^2$ gives (2.12).

For $i+j\geq N$, use the exact polynomial identity $B_N/B_j=B_{N-j}$, a consequence of $c_{d-u}=c_u$. Then B_iB_j is divisible by $B_N=(-1)^NP_N$ with multiplicities. This proves the zero assertion without dropping the first normal directions. □

2.5 The literal crossing circle and its colors

A color j on an auxiliary circle means that the circle carries a j-dimensional representation. The main strands still carry V_N. We must compute the resulting operator on those two main strands, including the trace around the auxiliary circle, before replacing that circle by twists. The answer is a Chebyshev polynomial in Ω, with the sign fixed by the pivot. For $1\leq j\leq N$, the auxiliary module is the ordinary j-dimensional highest-weight module. Its loop is the actual block-braiding word $[2,1,1,2]$, with the intermediate strand dimensions retained, followed by its ordinary pivot K_j^{1-N}. There is no auxiliary $1/j$ or $1/N$.

Proposition 2.5 (Circle identity)**.** *The resulting two-head operator is*

$$L_j=(-1)^{j-1}U_{j-1}(\Omega/2). \tag{2.14}$$

Here $U_0=1$, $U_1(z/2)=z$, and $U_{n+1}(z/2)=zU_n(z/2)-U_{n-1}(z/2)$.

Proof. At generic q, evaluate the closed monodromy on a target highest vector of weight n. In one direction the positive raising terms annihilate that vector; in the return direction a positive lowering term cannot return to its highest coefficient. The two Cartan factors and the auxiliary pivot give

$$\sum_{a=0}^{j-1}q^{(n+1)(j-1-2a)}=U_{j-1}((q^{n+1}+q^{-n-1})/2).$$

The generic pair highest vectors have distinct Casimirs and descendants of total dimension N^2, so this proves the full matrix identity. All finite crossing denominators have indices below N, and the identity specializes regularly. Finally $K_j^{1-N} = (-1)^{j-1}K_j$ at the root, proving (2.14), including its nilpotent coefficients. □

In particular $L_N = (-1)^d P_N'(\Omega)/N$. The trace identity

$$\phi(L_N f) = N \sum_{j=0}^{d} f(c_j) \tag{2.15}$$

counts the repeated roots. It follows first by Lagrange interpolation for a polynomial with distinct roots and then by polynomial continuation in its coefficients. At even levels $L_N^2 = 0$; at odd levels $L_N^2 = N^2 E_{-2}$, with the simple-node projector E_{-2}. Thus complete color alone cannot be inverted to reconstruct a filling.

Let $M = \lfloor d/2 \rfloor$ and $h_j = \mathsf{T} L_j$. The complete leaf has the exact positive expansion

$$h_N = \sum_{v=0}^{M} a_{N,v} \widehat{B}_v, \qquad a_{N,v} = \frac{N^2 s_{2v}}{s_v^4} > 0. \tag{2.16}$$

For $0 \le h \le M$, with $h_0 = 0$, the full color decomposition is

$$h_{N-h} = \sum_{v=h}^{M} \frac{N^2 s_{2v}}{s_v^2 s_{v-h} s_{v+h}} \widehat{B}_v + (-1)^N h_h. \tag{2.17}$$

For $1 \le h \le \lfloor N/2 \rfloor$,

$$h_h = \sum_{r=0}^{h-1} (-1)^{h-1+r} \frac{s_r^2 s_{h+r}}{s_{2r+1} s_{h-r-1}} \widehat{B}_{d-r}. \tag{2.18}$$

These identities separate two different ranges of Newton indices. The first term in (2.17), called the *core*, has positive coefficients before a half twist and occupies the lower indices. The second term, called the *alias*, is an exact signed contribution from the complementary color; formula (2.18) puts it in the higher Newton indices. An alias is therefore part of the finite invariant, not an error introduced by approximation. Appendix A.3 derives both formulas from the finite Chebyshev generating function. Chapter 5 will prove when the contribution of aliases is exponentially smaller after the complete tangle is assembled.

2.6 Half twists and exact filling

The preceding formulas describe a circle with no retained crossing. We next insert one positive or negative crossing in its original disk framing. This changes both the coefficients and their phases. We then expand a full integer twist power in the circle basis. Keeping these two steps separate ensures that the physical power is exactly $R^{\varepsilon-2p}$, with the retained half twist counted once. A retained half twist acts on a turned leaf by $\mathsf{T}_\varepsilon(f) = \mathsf{T}(R^\varepsilon \mathsf{T} f)$, for $\varepsilon = \pm 1$. Its exact transport is

$$\mathsf{T}_\varepsilon \widehat{B}_v = \kappa^\varepsilon (-1)^v \sum_{r=0}^{v} q^{\varepsilon v(r+1)} \frac{s_v}{s_r s_{v-r}} \widehat{B}_r. \tag{2.19}$$

The triangular formula retains the full Cartan phase and all normals. Its derivation uses the finite continuant expansion and root factorial reflection; the two signs are coefficient conjugates in the real Newton basis. A half-twisted alias may occupy every Newton coordinate, so the pure alias support cannot be reused after (2.19).

The N circle polynomials $L_1, \dots, L_N$ form a basis of $\mathcal{H}_N$. The full crossing has a Hermite polynomial $R = \kappa r_N(\Omega)$ with

$$r_N(c_k) = (-1)^k q^{k(k+1)}, \qquad \frac{r'_N(c_k)}{r_N(c_k)} = -\frac{d-2k}{2(q^{2k+1} - q^{-2k-1})}$$

at each double node, and the stated value at the odd simple node. Fourier–Hermite interpolation therefore gives, for every integer p,

$$R^{-2p} = \kappa^{-2p} \sum_{j=1}^{N} \beta_{j-1}^{-p} L_j, \qquad \sum_{j=1}^{N} |\beta_{j-1}^{-p}| \le 2N(1 + |p|). \tag{2.20}$$

The notation β_{j-1}^{-p} denotes the full interpolation coefficient, not a power of a scalar eigenvalue. Its value and derivative data are specified in Appendix A.4.

Insert (2.20) at every original auxiliary disk, tensoring with the spectators before taking the unchanged closure. If $\mathcal{A}_N(\mathbf{h})$ is the actual mixed source with colors $N - h_\ell$, then

$$\langle K(\mathbf{p}) \rangle_N = \chi_N \kappa^{-2\sum p_\ell} \sum_{j_1, \dots, j_H = 1}^{N} \left(\prod_{\ell} \beta_{j_\ell - 1}^{-p_\ell} \right) \mathcal{A}_N(N - j_1, \dots, N - j_H). \tag{2.21}$$

Here $|\chi_N| = 1$ in the reduced convention. This is an all-input finite identity. The remainder of the proof analyzes this complete sum, rather than replacing it by the complete-color term.

Chapter 3

From the finite sum to an angle maximization problem

At complete auxiliary color, every elementary coefficient and every multiplication coefficient is positive. We can therefore determine its exponential growth by maximizing the growth rate of an individual summand. This chapter makes that statement precise in three steps: determine the exact allowed integer indices, pass to their scaled compact polytope, and prove that the resulting action has a unique maximum away from every nonforced boundary. The same functions will then be recognized as hyperbolic volumes.

The distinction between an exact integer constraint and its limiting inequality matters here. We use $d = N - 1$ in every finite relation. The replacement by a continuous variable is made only after those relations have been solved, so no parity condition or half-lattice factor is lost.

3.1 Following the allowed indices through the construction

Expand an expression into its individual leaves, turns and binary products. A leaf has the positive support $[0, M] \cap \mathbb{Z}$. A turn replaces an index r by $d - r$. A product is governed by (2.13). Thus the source is a finite sum over a fixed tree of integer labels, with a positive coefficient at every admissible state in the complete pure case. The closure selects output d.

The finite support can be computed without evaluating any transcendental number. If a positive expression has interval support $[l, u] \cap \mathbb{Z}$, its turn has support $[d - u, d - l] \cap \mathbb{Z}$. If two children have supports $[l_1, u_1]$ and $[l_2, u_2]$, their product is zero when $l_1 + l_2 > d$. Otherwise its support is

$$[\max(l_1, l_2),\ \min(d, u_1 + u_2)] \cap \mathbb{Z}. \tag{3.1}$$

To verify that no integer in this interval is missing, fix such an output r. Choose $i \le r$ and $j \le r$ in the two intervals with $r \le i + j \le d$. Starting at the smallest permitted inputs and increasing them one at a time produces every intermediate sum up to $\min(d, u_1 + u_2)$. The lower bound $r \ge \max(l_1, l_2)$ ensures that both individual caps $i, j \le r$ leave enough room. The coefficient is then positive by Theorem 2.4.

At a cylindrical closing product, $i + j = d$ is forced and $\mu_{ij,d} = 1$. Delete this gate only after retaining that exact relation. It is the unique forced flat gate in the completed-root construction.

3.2 Strict output types

The word *strict* means that a state lies in the interior of every support inequality that has not been forced by closure. It is useful to classify a box by the possible values of its scaled output index. The letters in the following table name intervals, not new kinds of physical tangles. The table is a concise way of checking that the recursive construction leaves room for positive angles at every step. The limiting support inequalities for a retained product are

$$x > 0, \quad y > 0, \quad \max(x, y) < z < x + y < 1. \tag{3.2}$$

A strict leaf has $0 < x < 1/2$. A turn has output $1 - x$. For arbitrary syntax made from h, turn and product, the possible strict output intervals are

$$L = (0, 1/2), \qquad H = (1/2, 1), \qquad A = (0, 1), \qquad \varnothing.$$

Turn exchanges L and H and fixes A. Product has the following table; an empty child makes the product empty.

	L	H	A
L	A	H	A
H	H	$\varnothing$	H
A	A	H	A

For example, a high child forces the output above $1/2$, and two high inputs violate $x + y < 1$. For two low inputs and any prescribed $z \in (0, 1)$, choose both in $(z/2, \min(z, 1/2))$. The other entries follow by the same interval choices. At closure a strict reduced state exists exactly when the two intervals meet after reflection $y \mapsto 1 - y$.

Lemma 3.1 (Strict feasibility)**.** *Every completed expression has a strict state at every output $b \in (0, 1)$. Every completed/completed cylindrical closure has a strict reduced state.*

Proof. For hh, choose equal leaves $a \in (b/2, \min(b, 1/2))$. For a completed product choose its two outputs in the same interval and use induction in each child. For hE or Eh, use that number for the bare leaf and for the completed child's output. A turn uses the child's strict state at $1 - b$. At the closing product prescribe $1/2, 1/2$ and remove its forced gate. All remaining inequalities are strict. □

The distinction between strict support and every boundary coordinate is useful. At even N, hh has top support $2M = d - 1$, not d. This missing boundary coordinate does not affect the strict rounding argument below. Conversely, an empty strict interval does not imply a zero source at every level: $\mathsf{T}h\,\mathsf{T}h$ is zero at even levels but has a boundary-only odd contribution.

3.3 The primitive integer lattice

A lattice is called primitive here when the remaining free indices range over the ordinary integer lattice, with no hidden congruence restriction. We prove this by solving the closing relation with coefficient one. This gives both the correct number of summation variables and an admissible integer state near any fixed strict real state at every large level. Suppose there are H leaves. The uncontracted tree has $H - 1$ product gates. Remove the final forced gate

and contract every turn by $r \mapsto d - r$. There are then H leaf outputs and $H - 2$ retained product outputs, with one closing relation. Solving that relation for one output in one of the two disjoint root subtrees leaves

$$K = 2H - 3 \tag{3.3}$$

free integer coordinates. Every other index is an integer affine function of these coordinates with bounded, tree-dependent coefficients. No relation $2r = d$ is introduced.

Choose a strict reduced state and round its d-scaled free coordinates to integers. Reconstruct the eliminated indices exactly. The errors are bounded, whereas every strict inequality has a fixed positive slack times d. Therefore all constraints hold for every sufficiently large N, with either parity. For an open prescribed output, first fix r_N/d in its actual strict interval and use a continuous strict state at that output; then round the remaining coordinates. This gives a genuine all-level lower sequence. It does not rely on a subsequence whose denominator happens to divide d.

3.4 The action and its uniform finite approximation

The factorials in the finite coefficients are products of sines. Their logarithms are Riemann sums for a single integrable function. We use its integral J to express the exponential rates of an elementary piece and a stacking operation. The word *action* below means this logarithmic growth function, with the normalization fixed by the finite coefficients. Define

$$J(t) = \int_0^t \log(2 \sin \pi u)\, \mathrm{d}u, \qquad 0 \le t \le 1, \tag{3.4}$$

with its continuous endpoint values. Reflection gives $J(1 - t) = -J(t)$ and $J(1/2) = 0$. The leaf and gate actions are

$$g(x) = -2J(x) - 2J(1/2 - x) = J(2x) - 4J(x), \tag{3.5}$$

$$\begin{aligned} m(x, y, z) = {} & J(x + y) + J(z) - J(x) - J(y) \\ & - J(z - x) - J(z - y) - J(x + y - z). \end{aligned} \tag{3.6}$$

Let $\mathcal{A}_E$ be the sum of these terms over the leaves and retained gates, on the compact polytope defined by the non-strict support inequalities, the turn relations and the closing relation.

Lemma 3.2 (Uniform factorial estimate)**.** *Uniformly for* $0 \le r < N$*,*

$$\log s_r = NJ(r/N) + O(\log N).$$

For a fixed tree, every nonzero complete state of scaled labels $\mathbf{r}/d$ *has weight*

$$\log w_N(\mathbf{r}) = N\mathcal{A}_E(\mathbf{r}/d) + O_E(\log N). \tag{3.7}$$

Proof. Away from the endpoints, compare the sum of $\log(2 \sin \pi u)$ with its integral on consecutive mesh cells. Near zero and one, subtract $\log u$ or $\log(1 - u)$; the remaining function has bounded derivative, and the logarithmic part is controlled by factorials. The total discrepancy is $O(\log N)$, uniformly in the stopping index. Apply this estimate to (2.12) and (2.16). The shifts between r/N and r/d, and between M/N and $1/2$, change NJ by at most $O(\log N)$ because J has modulus of continuity $O(\delta |\log \delta|)$. There are a fixed number of factors. □

There are at most N^K states. The upper bound from (3.7), together with strict-state rounding at the maximizing state once strictness is proved, will give the complete-source limit.

3.5 Tetrahedral concavity and the boundary barrier

The geometric content of the action appears in its second derivatives and in its boundary behavior. Each local term is a sum of ideal-tetrahedron volumes. Their concavity will give uniqueness; their infinite one-sided derivative at a nontrivial flat angle will prevent a maximum from escaping to the boundary. We check both points because concavity alone would allow a boundary maximum. For $a+b+c=1$, $a,b,c\geq 0$, put

$$V(a,b,c) = -\pi\big(J(a)+J(b)+J(c)\big).$$

For positive angles this is the volume of the ideal tetrahedron with dihedral angles $\pi a, \pi b, \pi c$. It extends continuously by zero to the boundary. Its Hessian on the positive angle simplex is negative definite. This can be checked directly from $J''(t) = \pi\cot\pi t$: after eliminating c, the determinant is positive by the cotangent identity for three angles summing to π, while both diagonal entries of the volume Hessian are negative.

For a product write

$$p = z-x, \quad q = z-y, \quad r = x+y-z, \quad w = 1-x-y, \qquad p+q+2r+w = 1.$$

Substitution and reflection of J give

$$\pi g(x) = 2V(x, 1/2-x, 1/2), \tag{3.8}$$

$$\pi m(x,y,z) = V(y,x,w) + V(p,q,2r+w) + V(p+q,r,r+w). \tag{3.9}$$

Every triple in these expressions sums to one. Thus the entire action is concave on its original shared-label polytope.

Proposition 3.3 (Unique strict maximum)**.** *After deleting the forced closing gate, the complete action has a unique strict maximizer. Its Hessian on the primitive coordinates is negative definite. The fixed-output profile $f_E(b) = \max \mathcal{A}_E$ is real analytic on $0<b<1$ for every completed expression.*

Proof. Move from any boundary point toward the strict state of Lemma 3.1. A tetrahedron of type $(0, a, 1-a)$, $0<a<1$, has derivative $+\infty$ in this direction, since $-J(t) = -t\log t + O(t)$ near zero. A vertex-flat tetrahedron $(0,0,1)$ has a finite derivative: its two logarithmic singularities cancel. A positive tetrahedron also has a finite derivative. Hence no atom contributes a negative infinite term.

An endpoint wire descends through turns and products to a nonvertex-flat atom. At output zero both product inputs are zero; at output one they sum to one, giving $(y,x,0)$ unless an endpoint child permits further descent. A finite descent terminates at a leaf with atom $(0,1/2,1/2)$. Once all proper wires are interior, each remaining product boundary $p=0$, $q=0$, $r=0$ or $w=0$ exposes a nonvertex-flat atom in (3.9). A leaf at $1/2$ does the same. Thus every nonforced boundary point has a direction of positive infinite derivative and cannot maximize.

At a strict point the sum of the tetrahedral Hessians is strictly negative on every nonzero reduced tangent. Indeed, annihilating all leaf angle maps and all three product angle maps annihilates every leaf and product wire; turns then annihilate their wires. The only removed gate was already determined by its two children and the closing relation. Strict concavity gives uniqueness, and the implicit-function theorem on each fixed-output slice gives analyticity. □

Corollary 3.4 (Complete source limit)**.** *For the positive complete scalar $Z_{E,N}$,*

$$\lim_{N\to\infty}\frac{1}{N}\log Z_{E,N}=F_0,\qquad F_0=\max\mathcal{A}_E.$$

Proof. The number of states is polynomial, so Lemma 3.2 gives the upper bound. Round the strict maximizer in the primitive lattice. Its positive state weight gives the matching lower bound at every sufficiently large level. □

3.6 Boundary parameters and the leaf interval

For an open completed box, fix its output parameter b and maximize over all its other indices. The resulting function is the profile $f_E(b)$. Its derivative measures how the optimum changes when the output is varied. We package that derivative and b into three positive-branch coordinates (B,X,Y). Chapter 4 identifies their ratio with the shape of an ordered quadrilateral attachment face. For now they provide an efficient exact recursion for locating the stationary leaf parameters. The ordered output metric will be described in Chapter 4. Its useful algebraic coordinates are

$$B=\cot\pi b,\quad \varkappa=e^{-f_E'(b)},\quad X=\sqrt{\frac{1+B^2}{1+\varkappa}},\quad Y=\sqrt{\varkappa}\,X. \tag{3.10}$$

They satisfy $X^2+Y^2-B^2=1$. A leaf has $(B,X,Y)=(X,X,1)$. At a stationary product, the two children share X, and direct differentiation of (3.6) gives

$$B_o=\frac{B_1B_2+X^2-1}{B_1+B_2},\qquad Y_o=\frac{Y_1Y_2}{B_1+B_2},\qquad B_1+B_2>0. \tag{3.11}$$

A turn sends (B,X,Y) to $(-B,Y,X)$.

Lemma 3.5 (Bounds on the boundary parameters)**.** *The boundary parameters of every completed box satisfy $-Y<B<X$. In a completed/completed closure every completed descendant has $0<X,Y<1$, and every leaf satisfies*

$$\frac{1}{2}<X_\ell=\cot\pi a_\ell<1,\qquad \frac{1}{4}<a_\ell<\frac{\arctan 2}{\pi}. \tag{3.12}$$

Proof. The base hh has $B=X-1/(2X)$ and $Y=1/(2X)$. Turns preserve the claimed inequality. For a single attachment, (3.11) becomes

$$B_o=X-\frac{1}{X+B},\qquad Y_o=\frac{Y}{X+B}.$$

For $X\ge 1$, the child inequality forces $B>\sqrt{X^2-1}$ and the result follows. For $X<1$, let $h_0=\sqrt{1-X^2}$ and $\alpha=\arccos X$. Express a child as $(B,Y)=(h_0\cot\beta,h_0\csc\beta)$. A bare leaf has $\beta=\alpha$, a completed child has $\beta>\alpha$, and product adds the angles, with their sum below π. This proves the strict upper bound and the lower bound $B_o+Y_o=h_0\cot((\beta_1+\beta_2)/2)>0$. It also proves the completed/completed step when $X<1$. When $X\ge 1$, both completed children satisfy $\sqrt{X^2-1}<B_i<X$. Hence $0<X-B_i<X-\sqrt{X^2-1}\le 1$, so $(X-B_1)(X-B_2)<1$. Formula (3.11) then gives $B_o<X$, while $B_o>0$ proves the lower bound.

At closure the boundary triples of the two boxes are (B, X, Y) and $(-B, X, Y)$. Applying both inequalities gives $|B| < \min(X, Y)$; the hyperboloid relation implies $X, Y < 1$. In the angle chart, parent $Y < 1$ puts the sum of child angles below $\pi - \alpha$, while each child angle is at least α. Every completed child therefore has $Y < 1$, and turns exchange the bounds. Finally, at a single attachment $Y_o < 1$ gives $1 < 2X(X + B) < 4X^2$. At an hh base it gives $1/(2X) < 1$. Thus every leaf has $X > 1/2$, proving (3.12). □

For a fixed tree the inequalities give a compact neighborhood of every selected leaf inside $(1/4, 1/2)$. This is the precise input needed for the complex source analysis. It is not a uniform bound over an unbounded family of trees.

Chapter 4

Constructing the complete augmented hyperbolic metric

An ideal hyperbolic polyhedron has its vertices on the sphere at infinity. Truncating such a vertex by a horosphere gives a Euclidean polygon; after the face pairings, these polygons form the cross sections of the cusp ends. We shall build one polyhedron from the stationary angles and use two copies of it to recover the complement of the actual augmented diagram.

There are three separate verifications. The cells must fit isometrically along their ordered faces. Their labeled face identifications must recover the original link exterior. Finally, the Euclidean cusp polygons must glue without dilation, so that the metric is complete. A numerical agreement of volumes, or an agreement of face counts alone, would not establish these facts.

4.1 Why the local volume identity is not enough

The angle identity (3.9) proves concavity, but an abstract sum of tetrahedral volumes is not yet the volume of a link complement. We construct the cells, identify the ordered input maps, and then compare the closed colored boundary with the literal projection cut. This supplies both the topology and the degree of the geometric realization. The construction does not require a polyhedron existence theorem; Rivin's variational approach [4] provides useful context, but is not a premise of this explicit argument.

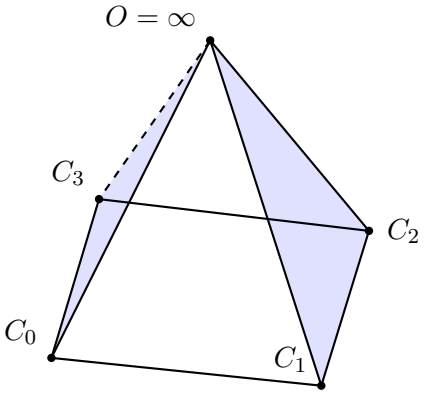
4.2 The leaf pyramid

We use the upper-half-space model of hyperbolic three-space, with boundary $\mathbb{C} \cup \{\infty\}$. An elementary piece will contribute an ideal pyramid. Its two shaded faces correspond to the two sides of a half crossing disk; the white faces come from regions of the projection sphere. These colors record different kinds of face pairings. The remaining quadrilateral is a gluing face used while the construction is still open. Its four ideal vertices are ordered: that order determines which vertex must be sent to which when two faces are identified. In upper half-space, place the apex at $O = \infty$ and the ordered vertices of the gluing face at

$$(C_0, C_1, C_2, C_3) = (1, e^{2\pi i a}, -1, -e^{2\pi i a}), \qquad 0 < a < 1/2. \tag{4.1}$$

Their convex hull is an ideal quadrangular pyramid. Its four apex edges are right angled. The base edge angles alternate πa and $\pi(1/2-a)$. A diagonal of the base gives the two tetrahedra of (3.8), so its volume is $\pi g(a)$. Color the side triangles (O, C_1, C_2) and (O, C_3, C_0) shaded; the other two side triangles are white. The base is the output gluing face.

Call edges $01, 23$ of an output gluing face *free*, and edges $12, 30$ *cap*. A leaf has free angles πa and cap angles $\pi(1/2-a)$. A completed cell will have free angles πb and cap angles $\pi(1-b)$.



quadrilateral gluing face

Schematic hyperbolic pyramid.

All five vertices are ideal. The shaded triangular faces are (O, C_1, C_2) and (O, C_3, C_0).

In the boundary plane: $C_0 = 1$, $C_1 = e^{2\pi i a}$, $C_2 = -1$, $C_3 = -e^{2\pi i a}$.

Figure 4.1: One leaf contributes two shaded crossing-disk triangles and one ordered quadrilateral gluing face. The one-leaf cap angle is $\pi(1/2-a)$; replacing it by the completed cap angle would give the wrong seam.

4.3 The multiplication prism

For a strict product define p, q, r, w as in Chapter 3. In the boundary plane set

$$v_0 = 1, \quad v_1 = e^{2\pi i p}, \quad v_2 = e^{2\pi i (p+r)}, \quad v_3 = e^{2\pi i (p+r+q)}. \tag{4.2}$$

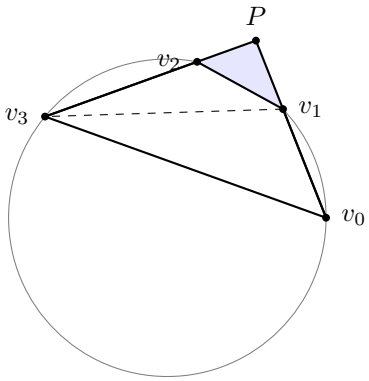
The consecutive half-arcs are $p, r, q, r+w$. Let P be the intersection of the lines (v_0, v_1) and (v_2, v_3). The Euclidean triangle (P, v_1, v_2) has positive angles $\pi y, \pi x, \pi w$, and the sine rule gives

$$P = v_1 + \frac{\sin \pi x \sin \pi r}{\sin \pi w \sin \pi p}(v_1 - v_0) = v_2 + \frac{\sin \pi y \sin \pi r}{\sin \pi w \sin \pi q}(v_2 - v_3). \tag{4.3}$$

In particular $|P| > 1$. Attach the tetrahedron (∞, P, v_1, v_2) to the quadrangular pyramid $(\infty, v_0, v_1, v_2, v_3)$, on opposite sides of their common face. The two coplanar face pairs merge. The result is a convex ideal triangular prism with white triangular faces (P, v_1, v_2) and (∞, v_0, v_3) and three quadrilateral gluing faces, with the indicated vertex orders

$$F_x = (\infty, v_0, v_1, P), \quad F_y = (\infty, v_3, v_2, P), \quad F_z = (v_0, v_1, v_2, v_3). \tag{4.4}$$

Its edge angles, divided by π, are



Boundary-plane construction.
$P = (v_0v_1) \cap (v_2v_3)$ lies outside the unit circle.
The sixth ideal vertex is ∞.
The three quadrilateral gluing faces have ordered vertices (∞, v_0, v_1, P), (∞, v_3, v_2, P) and (v_0, v_1, v_2, v_3).

$p = .12, \quad r = .10, \quad q = .17, \quad w = .51$

Figure 4.2: The exact boundary-plane construction of a strict prism at one rational angle tuple. The lines and circle are the defining support data; the drawing is not a perspective view of the three-dimensional hyperbolic cell.

Edges	Angle
$Pv_1,\ \infty v_0$	x
$Pv_2,\ \infty v_3$	y
$v_1v_2,\ v_0v_3$	$1 - z$
$P\infty$	w
v_0v_1	p
v_2v_3	q

The support half-spaces are the three vertical planes over the lines through P and over (v_3, v_0), and the hemispheres through the two remaining face circles. Formula (4.3) places the two attached pieces on the required sides. Equivalently, their seam is locally convex because its boundary angle sums are at most π. The volume is $\pi m(x, y, z)$: triangulate the pyramid across v_0v_2 and cancel the complementary intermediate Lobachevsky terms; the attached tetrahedron supplies $V(y, x, w)$.

4.4 Stationarity matches whole ordered faces

For cyclically ordered points on a circle or line, define

$$\varkappa(a, b, c, d) = \frac{(a - b)(c - d)}{(a - d)(b - c)} > 0.$$

Three ordered vertices can be sent to $(\infty, 0, 1)$, so this number determines the ordered ideal quadrilateral. A cyclic shift or the relevant reversal replaces it by its reciprocal. Direct

substitution gives

$$\varkappa(F_x) = \frac{\sin \pi x \sin \pi r}{\sin \pi w \sin \pi p}, \qquad \varkappa(F_y) = \frac{\sin \pi y \sin \pi r}{\sin \pi w \sin \pi q},$$
$$\varkappa(F_z) = \frac{\sin \pi p \sin \pi q}{\sin \pi z \sin \pi r}. \tag{4.5}$$

These are e^{-m_x}, e^{-m_y} and e^{-m_z}, respectively. The leaf metric is $\tan^2 \pi a = e^{-g'(a)}$.

The lower and upper stack maps are exactly

$$(C_0, C_1, C_2, C_3) \mapsto (v_0, v_1, P, \infty), \qquad (C_0, C_1, C_2, C_3) \mapsto (\infty, P, v_2, v_3). \tag{4.6}$$

Both invert the shape parameter of the gluing face. At each internal wire, stationarity says that the child output derivative and parent input derivative sum to zero. Hence $\varkappa_{\text{child}}\varkappa_{\text{parent}} = 1$, precisely the isometry condition for (4.6). This verifies the full face map, rather than only an edge-product equation.

4.5 Convex assembly and the closed cut

Attach child cells to the two prism inputs on opposite supported sides. A leaf cap edge meets a prism edge with total angle $\pi/2$; a completed cap edge gives π and merges two white faces. The common input edge has angle $\pi(w + x + y) = \pi$. The two free output seams have angles $\pi(p + x) = \pi z$ and $\pi(q + y) = \pi z$, while the cap output angles are $\pi(1 - z)$.

These inequalities prove local convexity at each finite seam. To justify global convexity directly, a length-minimizing sequence between two finite points of the union stays in a compact hyperbolic ball. A shortest path exists. Local convexity makes it an ambient geodesic at every finite point, so it is the unique ambient segment and lies in the union. The argument concerns finite points. To control the ideal boundary as well, observe that a strict quadrilateral gluing face is a maximal supported face. Cells attached on opposite sides can have no extra ideal intersection away from its supporting circle. An extra vertex on that circle would enlarge the supported face across one of its strictly nonflat edges. This is excluded by the strict output angles. Appendix D gives the corresponding incidence argument in detail.

There is also a combinatorial issue: repeated identifications must not collapse two corners of one leaf. At every open stage the four exposed corners are still unpaired original endpoints. A product consumes two disjoint pairs and never exposes those endpoints again; a turn only permutes the four remaining endpoints. Thus each consumed pair is permanent, and later gluing cannot extend it into a chain of identifications. The root pairs endpoints from the two disjoint final boxes. Each leaf therefore retains four distinct corners and its original apex. Together these geometric and combinatorial checks show that every completed box gives the claimed convex ideal cell, with right-angled edges outside the gluing face and two distinct shaded triangles per elementary piece.

At the final closure, the outputs are b and $1 - b$. Stationarity gives equal shape parameters for their ordered gluing faces. The gluing is

$$F : C_j \longmapsto G : C_{3-j} \qquad (j \bmod 4). \tag{4.7}$$

It implements both the stack interface and the two outside closing wires. Free and cap angles each sum to π for two completed children. After merging flat white seams, the result is a

convex right-angled ideal polyhedron P_E. It has $2H$ shaded triangles, $H+2$ white faces, $3H$ ideal vertices and $6H$ edges. These counts are checks, not the proof of topology.

For topology, cut the literal augmented diagram along its projection sphere and the recorded half crossing disks. In each projection ball the auxiliary semicircles are boundary-parallel along those half disks. Drilling them and then cutting returns a ball. A disk contributes precisely the two shaded triangles (O, C_1, C_2) and (O, C_3, C_0). The remaining regions are white. The original stack identifies lower corners $2, 3$ with upper corners $1, 0$ and preserves the exposed order (F_0, F_1, G_2, G_3). Turns rotate that order, and the final closure pairs 0 with 3 and 1 with 2. These rules determine every oriented white facial walk. They are exactly the rules induced by (4.6) and (4.7). The labeled sphere cellulation therefore agrees with the projection cut; its identification extends over each ball. Reversing the cuts gives the stated link exterior with degree one.

4.6 Complete cusps and half-twist partners

The polyhedron is only one of the two balls obtained by cutting along the projection sphere. We now describe how the two balls are reassembled. A complete cusp requires more than a total angle of 2π: the face maps must return pointwise around every edge, and their restrictions to the horospheres must have no dilation. The following pairing and length checks establish those conditions on the original labeled diagram. Take two oppositely oriented copies of P_E. Pair white faces between copies by their projection labels. In a pure leaf, pair its two shaded triangles within each copy, fixing O and sending $C_1 \mapsto C_0$, $C_2 \mapsto C_3$. The edge words alternate white and shaded gluings, contain four right angles, and return pointwise: the copied white maps and the inverse triangle maps cancel. Thus there is no translation along an edge hidden by an angle count.

At each ideal vertex, a horosphere cuts a Euclidean rectangle. Scale it so that its two opposite shaded sides have length one. Every shaded triangle then has the same canonical corner horocycles. All shaded and white pairings preserve the decoration. Uniform shrinking makes the horospheres disjoint. The links glue without dilation to the actual compact cusp tori, while their complement is compact. The resulting metric is complete and has volume

$$\operatorname{Vol}(Y_E^0) = 2\operatorname{Vol}(P_E) = 2\pi F_0. \tag{4.8}$$

A half twist rotates one side of the twice-punctured disk by π, exchanging both punctures and the upper/lower half disks. Hence the shaded maps become

$$S_0^+ \to S_1^- : \ (O, C_1, C_2) \mapsto (O, C_3, C_0), \qquad S_1^+ \to S_0^- : \ (O, C_3, C_0) \mapsto (O, C_1, C_2).$$

They go across the two copies, not within one copy. White maps are unchanged. The explicit edge matrices in Chapter 7 show pointwise cancellation of all three odd edge words. The length-one triangle decorations are also preserved, so every fixed half-twist pattern has the same complete volume $2\pi F_0$. The two half-twist signs have the same unmarked pairing but different collar paths; those paths are kept for filling.

4.7 A constructive knot choice

The hyperbolic construction initially keeps all main components. To obtain a knot after the auxiliary fillings, we must choose the retained half twists so that the final main diagram is

connected. This can be checked without any geometry. A box connects its four boundary points in one of three pairings; the table below follows the two actual identity arcs of a stack. It also records the only way stacking creates a closed internal main component. For the four tangle endpoints $(0, 1, 2, 3)$ in cyclic order, write

$$I = (0, 3)(1, 2), \qquad U = (0, 1)(2, 3), \qquad X = (0, 2)(1, 3).$$

A pure leaf gives U and an odd leaf gives X. Turn exchanges I, U and fixes X. Tracing the two stack wires gives

lower/upper	I	U	X
I	I	U	X
U	U	U plus an internal circle	U
X	X	U	I

Only $U \cdot U$ creates an internal component. The base hh offers $\{I, U\}$ without such a component. Every product of two feasible two-element subsets of $\{I, U, X\}$, after deleting the forbidden choice, still offers at least two elements; the same holds when one child is the leaf set $\{U, X\}$. Turn permutes the choices. Induction gives at least two loop-free choices for every completed expression. At the root choose U or X, whose cylindrical closure is one component, and backtrack the finite choices. Either sign realizes an odd leaf. Full twists have even endpoint permutation, so all subsequent auxiliary fillings preserve the one main component.

Chapter 5

Controlling the signed complementary color terms

The positive complete-color calculation does not by itself estimate the invariant after filling, because the exact filling formula uses every auxiliary color. For those colors an elementary piece contains the core and alias terms defined in Chapter 2. After a retained crossing, their coefficients can oscillate. We need a bound on the entire contribution of every selection containing at least one alias.

The bound is obtained by comparing logarithmic coefficient rates with tangent planes at the complete stationary state. Internal tangent terms cancel when boxes are stacked. A positive loss left at one elementary piece or one small block then survives all later compositions. The reason for treating blocks is that an individual alias need not lie below its leaf tangent throughout the permitted interval.

5.1 Coefficient envelopes on the true support

Let C_j^{ε} and A_j^{ε} denote, respectively, the core polynomial and the alias polynomial of an elementary piece of color j with retained half-twist sign ε. For a polynomial P, the notation $[\widehat{B}_r]P$ means its coefficient in the Newton basis, not an evaluation at a root. Write each colored elementary piece as core plus alias, using (2.17) and then applying (2.19) when required. A low physical color has no core. A high physical color has the displayed positive core before its half twist. The resulting signed arrays are always propagated through the full Newton multiplication law.

The endpoint-uniform factorial estimates and the selected complete half-twist recurrence give the following coefficient bounds, uniformly in every physical color and Newton index:

$$\begin{aligned} &|[\widehat{B}_r]C_j^{\varepsilon}| \le \exp\{N[\widetilde{g}(r/N) + e_N]\}, && 0 \le r \le M, \\ &|[\widehat{B}_r]A_j^{\varepsilon}| \le N^C e^{Nk(r/N)}, && 0 \le r \le d, \\ &\quad \widetilde{g}(u) = g(\max(u, 1/4)), \qquad k(u) = -2J(u/2). \end{aligned} \tag{5.1}$$

Here $e_N \to 0$ uniformly over colors and indices. No polynomial prefactor is asserted for the core through spectral collisions. Zero coefficients satisfy the same bounds. The first profile is constant below $1/4$ and agrees with g above it; its derivative joins continuously at zero.

The second follows directly from the triangular transport: the logarithmic rate of a term with upper index $v/N = w$ is

$$J(w) - J(u) - J(w-u), \qquad w \geq \max(u, 1/2).$$

Its maximum is at $w = (1+u)/2$. Reflection and duplication of J reduce that value to $-2J(u/2)$. The untransformed alias coefficients in (2.18) are bounded by $N/2$, so they have no additional exponential cost. The detailed finite derivation and the core recurrence estimate are in Appendix B.1.

The gate domain is the entire tetrahedron

$$D = \{(u, v, w) : u, v \geq 0,\ \max(u, v) \leq w \leq u + v \leq 1\}. \tag{5.2}$$

In particular, an aliased input can exceed $1/2$. All three functions $\widetilde{g}$, k and m are concave on their stated domains. For k, this is $k''(u) = -(\pi/2)\cot(\pi u/2) < 0$; for m, it follows from (3.9). This global concavity is what makes a finite supporting-plane certificate possible.

5.2 Why a block estimate is needed

At a pure leaf a, the tangent deficit against an alias is

$$I(a) = \min_{0 \leq u \leq 1} \{g(a) + g'(a)(u - a) - k(u)\}.$$

It can be negative in part of the interval allowed by (3.12). Thus an induction that assigns a positive loss to every individual leaf would fail. The multiplication gate provides a compensating contribution. We first treat an entire hh seed, then a new leaf together with the gate attaching it to a completed child.

Lemma 5.1 (Two-leaf block)**.** *At the pure stationary hh block, write*

$$X = \cot \pi a, \quad \cot \pi b = X - \frac{1}{2X}, \quad F_2(b) = 2g(a) + m(a, a, b), \quad \lambda = 2\log(2X^2).$$

For $1/2 \leq X \leq 1$, each of the two alias dual envelopes satisfies

$$\max_{D,\ v \leq 1/2} \{k(u) + \widetilde{g}(v) + m(u, v, w) - \lambda w\} \leq F_2(b) - \lambda b - \frac{1}{100}, \tag{5.3}$$

$$\max_{D} \{k(u) + k(v) + m(u, v, w) - \lambda w\} \leq F_2(b) - \lambda b - \frac{1}{100}. \tag{5.4}$$

Proof. Let $P(\lambda)$ be the pure dual maximum with both inputs at most $1/2$. At (a, a, b), $m_u = m_v = -2\log X$, $m_w = 2\log(2X^2)$ and $g'(a) = 2\log X$. Concavity therefore gives $P(\lambda) = F_2(b) - \lambda b$.

Choose a rational feasible pure point p and a rational strict point v for an alias action H. The pure value bounds P below; concavity gives the global upper tangent $H(w) \leq H(v) + \nabla H(v) \cdot (w - v)$. The vertices of D are

$$(0, 0, 0),\ (1, 0, 1),\ (0, 1, 1),\ (1/2, 1/2, 1/2).$$

Cutting by the core condition $v \le 1/2$ gives instead

$$(0,0,0),\ (1,0,1),\ (1/2,1/2,1/2),\ (0,1/2,1/2),\ (1/2,1/2,1).$$

Partition $\lambda/\log 2 \in [-2,2]$ into sixteen intervals of length $1/4$. The certificate supplies one pure point and two alias tangent points on each interval. For every vertex and both interval endpoints it verifies

$$A(p) - \lambda p_w - H(v) - \nabla H(v)\cdot(w-v) + \lambda w_w > \frac{1}{100}.$$

The expression is affine in the slope and in w, so these checks cover the whole interval and polytope. There are 288 rationally enclosed inequalities. Their least certified lower bound is

$$\frac{21000210290230963809378576600597934432 9}{50000000000000000000000000000000000000000000} > \frac{1}{25}.$$

Appendix C proves the enclosure rules and specifies the finite data and replay. No optimizer, sampled covering, or assertion about the alias maximizer is used in verification. □

5.3 A leaf attached to an arbitrary completed child

Suppose the new elementary piece is stacked with a box that has already been constructed. We retain all the internal structure of that box through its profile f_E and its output tangent. The comparison below therefore has one leaf variable, one output variable for the existing box, and the output of their stack. It is not an estimate for two bare leaves. Let (B,X,Y) be the selected boundary triple of the child and (B_p,X,Y_p) the boundary triple of the parent. Put

$$\begin{aligned} a = \operatorname{arccot} X/\pi, \quad b = \operatorname{arccot} B/\pi, \quad c = \operatorname{arccot} B_p/\pi, \qquad B_p = X - \frac{1}{X+B}, \\ \mu = f_E'(b) = 2\log(X/Y), \qquad \lambda = f_{hE}'(c) = \mu + 2\log(X+B). \end{aligned} \tag{5.5}$$

The bounds of Lemma 3.5 give $1/2 < X < 1$ and $1/(2X) - X < B < X$. Define

$$P(\mu,\lambda) = \max_{D,\ u\le 1/2} \{g(u) + m(u,v,w) + \mu v - \lambda w\}, \tag{5.6}$$

$$Q(\mu,\lambda) = \max_{D} \{k(u) + m(u,v,w) + \mu v - \lambda w\}. \tag{5.7}$$

Only the new leaf is restricted below half in P. The child variable v retains its whole interval; it is represented by its output tangent, not by a second bare leaf.

Lemma 5.2 (Attachment gap)**.** *At every proper attachment in the completed-root grammar,*

$$P(\mu,\lambda) - Q(\mu,\lambda) \ge \frac{1}{2500}. \tag{5.8}$$

The pure maximum in (5.6) *is attained at* (a,b,c) *from* (5.5)*.*

Proof. The three derivatives there are $m_u = -2\log X$, $m_v = -\mu$, $m_w = \lambda$. Thus the pure objective is stationary, and concavity gives its global maximum. The same is true with $\widetilde{g}$ replacing g, because $a > 1/4$ and the two functions agree near a.

For $1/2 < X \le 2/3$, both slopes belong to $[-2\log 2, 0]$. Indeed, either child or parent has $Y < 1$, so $2\log(X/Y) > 2\log X > -2\log 2$, while $Y^2 \ge 1 - X^2 > X^2$ gives the upper bound zero. On this entire square an exact certificate proves $P - Q > 1/1000$. It uses 55 dyadic rectangles, a rational pure point and a global alias tangent on each rectangle. For the four slope corners and four vertices of D, it checks

$$g(p_1) + m(p) - \{H(v) + \nabla H(v) \cdot (w - v)\} \\ + \mu(p_2 - w_2) - \lambda(p_3 - w_3) > \frac{1}{1000}, \qquad H = k + m. \quad (5.9)$$

The 880 inequalities and exact recursive square coverage imply the global assertion. The least lower bound is

$$\frac{5543719176001987489251797268402076337 3}{10^{40}} > \frac{1}{1000}.$$

This uses the same outward rational engine as the block certificate.

For $X \ge 2/3$, the elementary one-leaf estimate is

$$k(u) \le g(a) + g'(a)(u - a) - \frac{1}{2500}, \qquad 0 \le u \le 1. \quad (5.10)$$

Here is a direct proof of its endpoint margin. For $a \ge \arctan\sqrt{2}/\pi$, the minimum deficit occurs at $u = 1$, and it equals

$$g(a) + (1 - a)g'(a) = \frac{2}{\pi}\,\mathrm{Ti}_2(X) + \log X, \quad \mathrm{Ti}_2(X) = \int_0^X \frac{\arctan t}{t}\,\mathrm{d}t.$$

It decreases with a. At $X = 2/3$, four alternating terms give $\mathrm{Ti}_2(2/3) > 1708762/2679075$; the logarithm series gives $\log(3/2) \le 177391/437500$. With $\pi < 22/7$, the difference is greater than $57269/136687500 > 1/2500$. For $1/4 \le a \le \arctan\sqrt{2}/\pi$, the interior minimizing alias index is $u = (2/\pi)\arcsin(\tan^2(\pi a)/2) > a$, so the deficit decreases with a and its endpoint is larger than the same bound. The exact rational checks are included in the reproduction suite.

Finally add the tangent of the concave gate and the child linear term. The three variable coefficients cancel by stationarity, proving (5.8). □

5.4 The whole-tree induction

For a fixed choice of which elementary pieces contribute aliases, evaluate that contribution through the original construction tree. Such a choice is called an *alias subset*, or sector. There are only finitely many subsets because the construction is fixed. We first bound each subset separately, retaining an affine tangent in the open output index; only at the final closure are those output terms canceled. For every completed subtree at its selected output b_E, write $\lambda_E = f_E'(b_E)$. We prove, uniformly over all colors and all output indices,

$$|[\widehat{B}_r]E_S| \le \exp(N\,[f_E(b_E) + \lambda_E(r/N - b_E) - \delta_S + e_N])\,, \quad e_N \to 0, \quad (5.11)$$

where S is a specified alias subset, $\delta_\varnothing = 0$, and $\delta_S > 0$ when $S \ne \varnothing$. Lemma 5.1 supplies the base. A turn reverses the index, negates the slope, and adds only the fixed affine correction

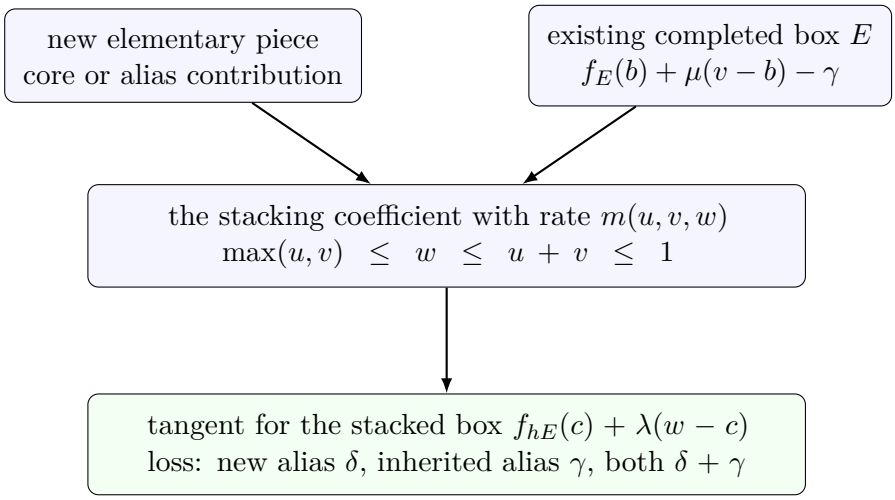


Figure 5.1: The child retains its output tangent and its original index. The new estimate is a leaf–gate comparison; it does not substitute a leaf for the child.

from $d/N = 1 - 1/N$. A completed/completed product adds both child tangents and the gate tangent. The internal derivatives cancel, while any inherited loss remains.

For hE, the exact split is

$$C_{hE} = C_h \star C_E, \qquad A_{hE} = A_h \star C_E + C_h \star A_E + A_h \star A_E,$$

where $\star$ is (2.12) with $i + j \le d$. A new alias loses at least $1/2500$ by Lemma 5.2; a child alias carries its old loss; when both occur the losses add. A core leaf uses the tangent of $\widetilde{g}$ with zero new loss. Thus the parent may take the minimum of its child loss and $1/2500$. The order Eh has the same algebraic bound without moving a coupon through a turn.

At the root, the two output tangents cancel on $i + j = d$. The finite affine shifts contribute $O(1)$ to the exponent. There are only polynomially many wire states and finitely many alias subsets for a fixed tree. We obtain constants $\gamma > 0$ and C such that the total aliased source satisfies

$$|\mathcal{A}_N - \mathcal{A}_N^{\mathrm{core}}| \le N^C e^{N(F_0 - \gamma)} \tag{5.12}$$

uniformly over all physical colors.

The all-core majorant also has a unique maximum at the complete strict state. On its high-index neighborhood it agrees with the strictly concave complete action. If another global maximum existed, concavity would make the segment joining it to the strict maximum constant, contradicting the negative Hessian near that maximum. Compactness therefore gives a fixed loss outside a fixed state window. Inside the window, the exact local core bound has a strict loss when any complementary color h_ℓ is at least a fixed positive multiple of N. Combining the two alternatives yields

$$\max_\ell h_\ell \ge \delta N \quad \Longrightarrow \quad |\mathcal{A}_N(\mathbf{h})| \le N^C e^{N(F_0 - \gamma_\delta)}. \tag{5.13}$$

The uniform global envelope and the strict color estimate are proved in Appendix B.2. Equations (5.12) and (5.13) are the global estimates required for filling. They concern the entire physical color range, not only a formal small-color series.

Chapter 6

A holomorphic asymptotic near the complete maximum

The preceding chapter removes the alias terms and the states far from the complete maximum with an exponential error. What remains is a finite sum over a fixed neighborhood of that maximum. We now continue its color parameters to complex values and compute its leading term. This finite holomorphic function is the *packet* used later in the filling integral.

The order of the construction matters. We first define the finite function by continuing the exact elementary coefficients, with their original phases. We then compare it with analytic index interpolants to exponential accuracy and apply a lattice saddle argument. An error of order N^{-1} on the real grid would not suffice: after an oscillatory filling transform, such an error could be larger than the desired leading answer.

6.1 A finite holomorphic object

Let $\mathbf{k} \in \mathbb{Z}^K$ be the primitive wire coordinates from (3.3). Every other wire has the exact form

$$r_w = N\ell_w(\mathbf{x}) + \delta_w, \qquad \mathbf{x} = \mathbf{k}/N, \tag{6.1}$$

with an integral affine function ℓ_w and a fixed integer δ_w. In particular, each use of $d = N - 1$ contributes to the offsets. Choose a closed rectangular box about the strict complete maximizer, lying wholly inside the retained support. By Lemma 3.5, every leaf lies in a fixed compact subinterval of $(1/4, 1/2)$.

For pure leaves, the exact color coefficient on this box is

$$\frac{N^2 s_{2r}}{s_r^2 s_{r-h} s_{r+h}}, \qquad t = h/N.$$

Use the canonical factorial continuation for the paired arguments $r - h, r + h$. For a piece with a retained crossing, use its finite continuant sum, meaning the solution of the displayed second-order recurrence fixed by the finite upper endpoint. Retain the continuous quadratic phase q^{h^2}, often called a chirp, and the recurrence branch selected by that exact sum. The precise construction, including the endpoint source anchor and exponential interpolation error, is proved in Appendix B.1. In particular, no interpolation is chosen merely because it agrees to order N^{-1} on the real grid.

Define $C_N(\mathbf{t})$ by the *finite* rectangular wire sum of the exact continued-color leaf factors and the original integer gate coefficients. At each fixed integer wire index, only the color is continued: the signed leaf remains its finite source sum with the canonical chirp. Index interpolants are introduced afterward as a proof device and replace this exact window only with exponentially small error. This is the definition in (B.42). It is holomorphic on a fixed complex polydisk and separately even in every t_ℓ. It agrees with the original source in the small physical color corridor up to the exponentially small errors established in Chapter 5 and the local interpolation proof.

6.2 The selected local action

The pure action depends on a scaled Newton index x and a scaled color deficit t. A retained half twist changes the relation between this pure parameter and the complex parameter χ that will become a peripheral coordinate. The following implicit change of variables describes that relation. Its branch is fixed by the finite recurrence, so both its derivative and its additive constant are determined. For a pure leaf the action is

$$g(x,t) = J(2x) - 2J(x) - J(x-t) - J(x+t). \tag{6.2}$$

For a signed half twist, the selected action f_ε is an analytic shear of this action. Put

$$\chi = t - \frac{\varepsilon g_t(x,t)}{2\pi\mathrm{i}}, \qquad f_\varepsilon(x,\chi) = g(x,t) - \frac{\varepsilon g_t(x,t)^2}{4\pi\mathrm{i}}, \qquad \varepsilon \in \{0,\pm 1\}. \tag{6.3}$$

For $\varepsilon = 0$ this is the identity. For $\varepsilon \neq 0$, the implicit map is nonsingular near the selected complete leaf. The finite recurrence selects this branch and fixes its additive constant; the relation is not used to choose an arbitrary solution after taking a semiclassical limit.

Let $F_\varepsilon(\mathbf{x},\boldsymbol{\chi})$ be the sum of the f_{ε_ℓ} and the unchanged gate actions. The implicit-function theorem gives a unique nearby stationary point $\mathbf{x}_*(\boldsymbol{\chi})$, and define

$$\Phi_\varepsilon(\boldsymbol{\chi}) = F_\varepsilon(\mathbf{x}_*(\boldsymbol{\chi}),\boldsymbol{\chi}).$$

The same relation holds after eliminating all internal wires:

$$\chi_\ell = t_\ell - \frac{\varepsilon_\ell \partial_{t_\ell}\Phi_0}{2\pi\mathrm{i}}, \qquad \Phi_\varepsilon(\boldsymbol{\chi}) = \Phi_0(\mathbf{t}) - \sum_\ell \frac{\varepsilon_\ell(\partial_{t_\ell}\Phi_0)^2}{4\pi\mathrm{i}}.$$

At fixed χ, differentiation of (6.3) gives $(f_\varepsilon)_x = g_x$ and $(f_\varepsilon)_\chi = g_t$. Thus the wire stationary equations are the pure equations evaluated at the implicitly changed colors. This proves the global shear by the envelope theorem, with the complete value F_0 as its anchor.

6.3 Offsets, powers of the level, and the Gaussian

The exponential action alone is not enough to prove noncancellation. We also need a nonzero leading coefficient and its exact power of N. Integer shifts such as $d = N - 1$ affect that coefficient even though they disappear from the limiting action. We keep these shifts before

applying the Gaussian integral in the free lattice coordinates. On a compact complex strict window, the factorial expansion is

$$G_N(N\alpha + c) = \sqrt{2N \sin \pi\alpha}\, e^{NJ(\alpha)} (2 \sin \pi\alpha)^c (1 + O(N^{-1})). \tag{6.4}$$

The square roots and logarithms continue from positive real arguments. The integer shift c is essential. For a signed leaf the corresponding offset factor is $e^{\delta \partial_x f_\varepsilon}$, evaluated at the changed critical point, not at the complete point or with g in place of f_ε.

Each leaf has prefactor $N^{1/2}$; each of the $H-2$ retained gates has prefactor $N^{-3/2}$. Hence the summand has N^{3-H}. A $K = 2H-3$ dimensional lattice Gaussian contributes $N^{K/2}$, leaving $N^{3/2}$. More explicitly, if

$$I_N(\mathbf{x}, \boldsymbol{\chi}) = N^{3-H} b_\varepsilon(\mathbf{x}, \boldsymbol{\chi}) e^{N F_\varepsilon(\mathbf{x}, \boldsymbol{\chi})} (1 + O(N^{-1})),$$

then the leading packet amplitude is

$$a_\varepsilon(\boldsymbol{\chi}) = \frac{(2\pi)^{K/2} b_\varepsilon(\mathbf{x}_*(\boldsymbol{\chi}), \boldsymbol{\chi})}{\sqrt{\det[-\partial_{\mathbf{x}}^2 F_\varepsilon(\mathbf{x}_*, \boldsymbol{\chi})]}}. \tag{6.5}$$

Every factor in b_ε is specified by (6.4), the finite signed recurrence, and the offsets (6.1).

Theorem 6.1 (Exact corridor packet)**.** *For the fixed completed-root expression there are a fixed complex polydisk, a nonzero holomorphic a_ε, and a separately even holomorphic Φ_ε such that*

$$C_N(\boldsymbol{\chi}) = N^{3/2} a_\varepsilon(\boldsymbol{\chi}) e^{N\Phi_\varepsilon(\boldsymbol{\chi})} (1 + O(N^{-1})) \tag{6.6}$$

uniformly there. The physical source has the comparison

$$|\mathcal{A}_N(\mathbf{h}) - \eta_N C_N(\mathbf{h}/N)| \le N^C e^{N(F_0 - \gamma)}, \qquad 0 \le h_\ell \le \delta N, \tag{6.7}$$

where $\eta_N = \exp(\mathrm{i}\pi d \sum_\ell \varepsilon_\ell/4)$ is the retained unit phase. At zero,

$$\Phi_\varepsilon(0) = F_0, \qquad \partial_{\chi_\ell} \partial_{\chi_j} \Phi_\varepsilon(0) = -\frac{2\pi X_\ell}{1 - \varepsilon_\ell \mathrm{i} X_\ell} \delta_{\ell j}. \tag{6.8}$$

The statement holds at both parities and all sufficiently large levels.

Proof. The local finite interpolation and its exponentially small error are proved in Appendix B.1. Apply the finite complex lattice Laplace lemma of Appendix B.1.8 to the full rectangular window. That lemma uses exact cotangent residues, bounds every vertical side and nonconstant cotangent mode, and translates the ordinary integral to the unique complex critical point. It gives (6.6) and (6.5).

At completeness, the signed multiplier relative to the positive amplitude is

$$\prod_{\varepsilon_\ell \neq 0} \sqrt{\sin \pi a_\ell} \exp(-\mathrm{i}\varepsilon_\ell \pi a_\ell/2),$$

which is nonzero. The positive amplitude is positive by the strict Gaussian. For pure colors, separate evenness makes the mixed wire–color derivative vanish at zero, and direct differentiation of (6.2) gives $-2\pi X_\ell$. The shear transforms it to (6.8). Its real part is strictly negative. The source estimates of Chapter 5, together with the finite-window local error, prove (6.7). All lattice offsets are fixed integers and no parity-dependent lattice density occurs. □

After shrinking the polydisk, this implies a source-specific Gaussian upper bound

$$|C_N(\mathbf{x}+\mathrm{i}\mathbf{y})| \le N^C \exp N\{F_0 - a_0\|\mathbf{x}\|^2 + B_0\|\mathbf{y}\|^2\}, \qquad a_0 > 0. \tag{6.9}$$

This bound is needed for the finite Fourier contours. A pointwise asymptotic on the physical grid would not imply it.

6.4 An exact three-leaf calibration

For $C(h^3)$ choose the three leaf indices i, j, k as free coordinates. The surviving gate output is $d-k$, so its limiting output is $1-x_3$ with integer offset -1. The complete phase is

$$F(\mathbf{x}) = \sum_{\ell=1}^{3} g(x_\ell) + m(x_1, x_2, 1-x_3).$$

Its stationary point is $x_\ell = a = \arctan 2/\pi$. The negative Hessian is

$$A = \frac{25\pi}{4}(I_3 + \mathbf{1}\mathbf{1}^T), \qquad \sqrt{\det A} = \frac{125\pi^{3/2}}{4}.$$

The three unshifted leaf amplitudes are $\sqrt{10}/8$, the unshifted gate amplitude is $5\sqrt{10}/16$, and the gate offset is $e^{-m_z} = 4$ at completeness. Thus $b = 125/512$ and $a_0 = \sqrt{2}/64$. The complete rate is $F_0 = 2v_8/\pi$, so the augmented volume is $4v_8$. At a changed color the offset is e^{-m_z} evaluated at the changed critical point; it is not the constant four.

The alias proof for this isolated case uses the exact symmetric kernel

$$\phi(\widehat{B}_i\widehat{B}_j\widehat{B}_k) = \frac{s_{i+j}s_{i+k}s_{j+k}}{N s_i s_j s_k s_{i+j+k-d}}$$

on $i+j, i+k, j+k \le d$ and $i+j+k \ge d$, and zero elsewhere. It follows from (2.12) and $s_r s_{d-r} = N$. For one or two aliases, algebraic reassociation groups the aliased leaves into a two-leaf block and uses Lemma 5.1 at $X = 1/2$; the remaining core tangent cancels the output slope, with only $\log 4/N$ from the closing shift. For three aliases, $k \le v_3/\pi$ and $m \le 3v_3/\pi$, while $6v_3 < 2v_8$. The exact rational margin and its elementary bounds are included in the reproduction suite. This is a source identity, not a tangle move. The one-leaf cap seam in the geometric closure is $\pi/2$, as in Chapter 4. These observations supply the same packet and filling argument for the isolated three-leaf case.

Chapter 7

Peripheral curves and the volume after filling

We now identify the complex action from Chapter 6 with a deformation of the actual augmented link complement. A cusp cross section is a torus. A small loop going around the corresponding link component is its meridian, and the original crossing disk specifies the longitude used in the surgery convention. These curves must be primitive: together they form an integral basis, rather than nontrivial multiples of a basis.

A *marked deformation* keeps those curves, their orientations, and the continuously selected holonomy logarithms. The complex-valued function G below is a volume primitive: its derivative records the peripheral holonomies, and an explicit boundary correction recovers the volume of the representation. The complete metric fixes its additive constant. This information is what connects the critical point of the filling integral to the knot complement with the intended surgery slopes.

7.1 A metric modulus is not yet a peripheral coordinate

The real boundary shape parameter $e^{-f'_E(b)}$ describes an ordered ideal quadrilateral. To identify the filling action with volume, one must also construct the nearby representation on the same link exterior, specify the two auxiliary peripheral curves, and fix the additive constant of its regulator. Matching only a Hessian or an unmarked cusp shape would not suffice.

Appendix D gives the local rational face maps and their compositions. The construction starts at the complete metric of Chapter 4. It continues the leaf apex flags and the prism coordinates in the selected source variables. The input and output maps obey the nine local prism compatibility identities, so the white-face map is independent of the sequence of glued cells. The pure and signed shaded pairings give pointwise edge relations. The main cusps stay parabolic. The resulting framed representation is holomorphic on the same sufficiently small parameter neighborhood used by the packet.

Proposition 7.1 (Marked source primitive)**.** *For the completed-root source, with the original disk framings and half-twist signs, the selected deformation has centered primitive auxiliary logarithms*

$$U_\ell = -2\pi\mathrm{i}\chi_\ell, \qquad V_\ell = -2\partial_{\chi_\ell}\Phi_\varepsilon. \tag{7.1}$$

Its absolute complex volume primitive can be chosen as

$$G = 2\pi\mathrm{i}\Phi_\varepsilon, \qquad dG = \frac{1}{2}\sum_\ell V_\ell\, dU_\ell, \qquad \Im G(0) = \mathrm{Vol}(Y_E^\varepsilon) = 2\pi F_0. \tag{7.2}$$

The half-twist contribution is the specified quadratic correction $-\varepsilon_\ell V_\ell^2/8$ in G. The physical filling power is $\varepsilon_\ell - 2p_\ell$.

Proof. The local rational identities and original peripheral paths are proved in Appendix D. Differentiating the shear (6.3) gives the same boundary covectors as the pure source after the primitive meridian shear. Across each paired quadrilateral gluing face the two internal covectors cancel, leaving only the auxiliary boundary terms. Thus the differential is (7.2). The complete polyhedron fixes its imaginary constant, so no arbitrary additive volume normalization remains. The sign of the meridian shear is determined by the original collar path, not merely by the unmarked half-twist pairing. □

For perspective, in the leaf coordinate $E = e^{2\pi\mathrm{i}a}$ the complete disk longitude has translation $-2(E+1)$. The two signed meridian collar paths have translations -2 and $2E$, differing by that longitude. Each pair spans the area of the two cusp rectangles and is primitive. Their relation to the quantum sign is checked with the recorded crossing convention. This explains why the two signs can have the same complete volume but different filling equations.

7.2 Geometric completion for long primitive slopes

Solving a matrix equation that kills a peripheral element does not alone construct a smooth filled metric. We need to show that the geometric end closes around a solid-torus axis with angular period exactly 2π. The marked curves and small logarithms make that check possible. We first solve the equations near the complete structure and then use the explicit end construction in Appendix D. The local deformation is a representation of the actual augmented-link group near its oriented complete holonomy, with a bounded finite collection of framed vertices and face maps. The strict complete shapes remain nondegenerate in a small neighborhood. The primitive equations

$$U_\ell + p_\ell V_\ell = -2\pi\mathrm{i} \tag{7.3}$$

have a unique nearby solution for all independently sufficiently large signed p_ℓ. One can see the local existence by writing $1/p_\ell$ as independent small parameters and applying the implicit-function theorem to the cusp equations. The sign and the integer on the right side are fixed by the primitive collar, not chosen after solving a formal equation.

The geometric completion argument in Appendix D constructs a smooth solid-torus completion at each filled cusp on this branch. The remaining main cusp stays complete. It applies on one fixed neighborhood, hence requires only a lower bound on each $|p_\ell|$, with no comparison between them. The resulting complete metric is on the topology obtained by the specified auxiliary disk fillings. After the knot pairing choice and all such fillings, that topology is $S^3 \setminus K_{F,G,\varepsilon}(\mathbf{p})$.

Set

$$S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}) = \Phi_\varepsilon(\boldsymbol{\chi}) + \frac{\pi\mathrm{i}}{2}\sum_\ell \frac{(\chi_\ell - 1)^2}{p_\ell}. \tag{7.4}$$

Its critical equations are exactly (7.3), by (7.1). The regulator with the solid-torus correction gives

$$2\pi\Re S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}_*) = \mathrm{Vol}(S^3 \setminus K_{F,G,\varepsilon}(\mathbf{p})). \tag{7.5}$$

This is an equality on the selected, correctly marked branch. Chapter 8 proves that the full finite invariant has a nonzero leading contribution with precisely this action.

Chapter 8

Evaluating the complete oscillatory filling sum

The geometric saddle has now been identified, but the invariant is still the full finite sum over auxiliary colors in (2.21). This chapter proves that the saddle actually contributes to that sum with a nonzero coefficient, and that all other contributions are smaller. In particular, we do not infer the knot volume from the complete-color growth rate.

We convert the exact Hermite interpolation coefficients into a Fourier expansion. Its integer indices are called charges. The zero charge vanishes by an exact boundary identity; the charges with absolute value one give the leading term. To justify that conclusion, we control both the infinite charge tails and the boundary integrals introduced by moving contours. The latter are the *endpoint currents* mentioned below.

8.1 Replacing the color sum by its controlled local part

For the fixed diagram, Chapters 5 and 6 give a global physical-color gap, an exponentially accurate separately even corridor packet, and a uniform holomorphic asymptotic. In this chapter write $b = H$ for the number of filled circles, $\varphi = \Phi_\varepsilon$, and

$$\varphi''(0) = 2\pi\mathrm{i}\,\mathrm{diag}(\tau_1, \dots, \tau_b), \qquad \tau_\ell = \frac{\mathrm{i}X_\ell}{1 - \varepsilon_\ell \mathrm{i}X_\ell}, \qquad \Im\tau_\ell > 0.$$

The different cusps need not have equal τ_ℓ. The packet exponent is $\mu = 3/2$.

The coefficient mass bound in (2.20) shows that the global source error and the corridor comparison error cost only a polynomial in N for fixed twists. They remain exponentially below e^{NF_0}. This removal takes place before folding colors: an alias need not be even and is not artificially made so. It remains to analyze the exact even corridor.

Let $H_N = \lfloor \delta N \rfloor$, and define its finite Fourier polynomial

$$\mathcal{F}_N(\mathbf{z}) = \sum_{|h_\ell| \le H_N} C_N(\mathbf{h}/N) e^{2\pi\mathrm{i}\mathbf{h}\cdot\mathbf{z}}. \tag{8.1}$$

This is an entire, one-periodic function in each variable and is separately even. It is finite at every N; no infinite-lattice extension has been made.

8.2 An exact Fourier bound away from zero frequency

We record the finite contour estimate that controls the physical endpoints. In one coordinate suppose

$$|C_N(x+\mathrm{i}y)| \le N^C e^{N(F-ax^2+By^2)}$$

on a fixed tube. For a fixed $\phi_0 > 0$ and fixed nonnegative integer r,

$$\left| \sum_{h=-H_N}^{H_N} h^r C_N(h/N) e^{\mathrm{i}\phi h} \right| \le N^{C_r} e^{N(F-g)}, \qquad \operatorname{dist}(\phi, 2\pi\mathbb{Z}) \ge \phi_0, \tag{8.2}$$

for some $g > 0$.

To prove it, reduce ϕ to $[\phi_0, 2\pi - \phi_0]$ and use the rectangle in the unscaled variable z with real sides $\pm(H_N + 1/2)$ and imaginary sides $\pm\sigma N$. The exact residue formula is

$$\frac{1}{2\pi\mathrm{i}} \oint z^r C_N(z/N) e^{\mathrm{i}\phi z} (\pi \cot \pi z - \mathrm{i}\pi)\, \mathrm{d}z.$$

The kernel equals $2\pi\mathrm{i}/(e^{2\pi\mathrm{i}z} - 1)$. On the upper and lower horizontal sides the products with $e^{\mathrm{i}\phi z}$ are bounded by $4\pi e^{-\phi\sigma N}$ and $4\pi e^{-(2\pi-\phi)\sigma N}$. On a vertical half-integer side the bound is $2\pi e^{-\phi_0|\Im z|}$. Choose σ first so that $B\sigma^2$ is smaller than both half the horizontal loss and half the real boundary loss. All four sides then have a strict exponential gap. This proves (8.2); fixed derivatives are treated by the factor h^r. The argument is uniform in the other real colors and in a smaller complex neighborhood of any remaining parameters.

8.3 The total derivative and the zero-charge cancellation

The derivative terms in Hermite interpolation are especially important at this step. Together with the values at the roots, they combine into one total derivative on a finite midpoint grid. This exact identity determines the endpoint behavior and removes the zero Fourier mode. It is proved before any stationary-phase approximation is made. The exact Fourier–Hermite interpolation of Appendix A.4 has a useful combined form. In one color, for the normalized power r_N^{2P}, put

$$U_N(z) = e^{2\pi\mathrm{i}PNz^2} \sinh(2\pi\mathrm{i}z) \mathcal{F}_N(z), \qquad z_k = (k + 1/2)/N - 1/2.$$

Then the folded color pairing is exactly

$$\sum_{h=0}^{H_N} \beta_{N-h-1}^P C_N(h/N) = \frac{(-1)^d \kappa^{-P}}{2\pi\mathrm{i}N^2} \sum_{k=0}^{N-1} U_N'(z_k). \tag{8.3}$$

The derivative contains both the h term and the linear normal term from the Hermite data. Removing either would change this identity.

The endpoint values satisfy

$$U_N(1/2) = U_N(-1/2) = 0, \qquad U_N'(1/2) = U_N'(-1/2).$$

The periodic extension of U'_N is continuous and piecewise smooth. Two integrations by parts give an absolutely summable Fourier series at each fixed N. Midpoint Fourier selection followed by one integration by parts gives

$$(-1)^d\kappa^{-P}\sum_{m\neq 0}(-1)^{md}m\int_{-1/2}^{1/2}e^{2\pi\mathrm{i}N(Pz^2-mz)}\sinh(2\pi\mathrm{i}z)\mathcal{F}_N(z)\,\mathrm{d}z. \tag{8.4}$$

There is no zero charge: it vanishes from the two exact endpoint zeros. For an even source the terms at m and $-m$ are equal, because both m and the hyperbolic sine change sign. This equality precedes any saddle approximation.

Taking $P_\ell=-p_\ell$ in every color and restoring the Cartan factors gives the full corridor expression

$$\chi_N\eta_N(-1)^{bd}\kappa^{-\sum p_\ell}\sum_{\mathbf{m}\in(\mathbb{Z}\setminus\{0\})^b}(-1)^{d\sum m_\ell}\prod_\ell m_\ell \times\int_{[-1/2,1/2]^b}e^{-2\pi\mathrm{i}N\sum_\ell(p_\ell z_\ell^2+m_\ell z_\ell)}\prod_\ell\sinh(2\pi\mathrm{i}z_\ell)\mathcal{F}_N(\mathbf{z})\,\mathrm{d}\mathbf{z}. \tag{8.5}$$

This is a sum of commuting color-register functionals. No physical box is commuted through another box.

8.4 Uniform charge tails and their endpoint currents

A Fourier series can converge for each N without having a tail estimate uniform enough for an exponential limit. We therefore bound the tails directly. Moving the interval downward or upward adds two vertical boundary segments. Their difference vanishes to second order at height zero; this cancellation produces a summable power of the charge. We retain the pair together rather than discard either boundary segment. Absolute convergence at each N is not a uniform exponential tail estimate. We prove the latter before using a Gaussian approximation. By (8.2), $\mathcal{F}_N$ and its first few derivatives are exponentially small on fixed complex strips based at $z=\pm1/2$. Fix a small strip height σ using only this source estimate.

For $m>0$, shift the one-dimensional Fourier integral down by σ. On the horizontal side the source's physical support gives a bound with exponent

$$F_0-2\pi\sigma(m-|P|-\delta).$$

The two vertical sides must be paired. Their common Fourier endpoint phase is $(-1)^{mN}$, and their difference is

$$V_N(v)=U_N(1/2-\mathrm{i}v)-U_N(-1/2-\mathrm{i}v),\qquad 0\le v\le\sigma.$$

Periodicity of the finite Fourier polynomial and the exact endpoint relations give $V_N(0)=V'_N(0)=0$. Keeping the chirp's height-dependent cost, rather than bounding it at the top of the strip, gives

$$|V_N(v)|\le\mathrm{poly}(N,P)v^2e^{N(F_0-g_1)+2\pi|P|Nv}.$$

Therefore

$$\left|m\int_0^\sigma e^{-2\pi mNv}V_N(v)\,\mathrm{d}v\right|\le\frac{\mathrm{poly}(N,P)m}{N^3(m-|P|)^3}e^{N(F_0-g_1)},\qquad m>|P|. \tag{8.6}$$

This is summable for $m > 2|P| + 2\delta + 2$. The upper shift treats negative m. Neither vertical current has been set to zero.

In several variables, remove tails one at a time while all other chirps remain on real grids or finite real integrals. If D_i is the exact sampled derivative functional and D_i^{lo} its finite-charge restriction, use the exact identity

$$\prod_i D_i - \prod_i D_i^{\mathrm{lo}} = \sum_i \left(\prod_{j<i} D_j^{\mathrm{lo}} \right) (D_i - D_i^{\mathrm{lo}}) \left(\prod_{j>i} D_j \right).$$

The low-charge functionals have finite supremum norm; the sampled ones and their fixed derivatives cost powers of N and the fixed slopes. All spectator chirps have modulus one. Thus each term has the exponential gap of (8.6), without a factor $e^{N|p_j|}$ from a different cusp. This is the point that permits independently unequal long slopes.

8.5 The central Fourier packet

The implicit-function theorem defines $\boldsymbol{\xi}(\mathbf{z})$ by

$$\nabla\varphi(\boldsymbol{\xi}) + 2\pi\mathrm{i}\mathbf{z} = 0.$$

Set

$$\Psi(\mathbf{z}) = \varphi(\boldsymbol{\xi}(\mathbf{z})) + 2\pi\mathrm{i}\boldsymbol{\xi}(\mathbf{z}) \cdot \mathbf{z}, \qquad B(\mathbf{z}) = \frac{(2\pi)^{b/2} a_\varepsilon(\boldsymbol{\xi}(\mathbf{z}))}{\sqrt{\det[-\varphi''(\boldsymbol{\xi}(\mathbf{z}))]}}. \tag{8.7}$$

Both functions are separately even, $B(0) \neq 0$, and

$$\Psi(\mathbf{z}) = F_0 - \pi\mathrm{i} \sum_\ell \frac{z_\ell^2}{\tau_\ell} + O(\|\mathbf{z}\|^4). \tag{8.8}$$

Exact half-grid cotangent rectangles applied to (8.1) turn it into N^b times its finite real integral, with exponentially small boundary and nonconstant-mode errors. Translate that integral to the parallel real contour through $\boldsymbol{\xi}(\mathbf{z})$. The strict complex Gaussian estimate controls the connectors. Gaussian integration gives, uniformly on a fixed complex polydisk,

$$\mathcal{F}_N(\mathbf{z}) = N^{\mu+b/2} B(\mathbf{z}) e^{N\Psi(\mathbf{z})} (1 + O(N^{-1})). \tag{8.9}$$

This argument applies to the exact C_N, not to a pointwise real-grid approximation whose error might have uncontrolled Fourier behavior.

8.6 Local charges for independently signed slopes

After the tail estimate, only finitely many charges remain for the fixed filling tuple. Their saddles must be compared uniformly when the filling integers have different signs and magnitudes. We use the reciprocal slopes $1/p_\ell$ and the ratios m_ℓ/p_ℓ as independent parameters. This gives one neighborhood in which the saddle equations and the charge comparison are valid, instead of assuming the slopes are comparable. For a nonzero integer vector $\mathbf{m}$, define

$$T_{\mathbf{m}}(\mathbf{z}) = \Psi(\mathbf{z}) - 2\pi\mathrm{i} \sum_\ell (p_\ell z_\ell^2 + m_\ell z_\ell).$$

Write $e_\ell = 1/p_\ell$ and $t_\ell = m_\ell/p_\ell$. The saddle equations are

$$z_\ell = -\frac{t_\ell}{2} + \frac{e_\ell \Psi_\ell(\mathbf{z})}{4\pi \mathrm{i}}. \tag{8.10}$$

At $\mathbf{e} = 0$ their derivative in $\mathbf{z}$ is the identity, so one fixed polydisk in $(\mathbf{t}, \mathbf{e})$ has a unique holomorphic solution. Completing the square yields

$$T_{\mathbf{m}}(\mathbf{z}(\mathbf{t}, \mathbf{e})) = \frac{\pi \mathrm{i}}{2} \sum_\ell \frac{t_\ell^2}{e_\ell} + \widehat{B}(t_1^2, \dots, t_b^2, \mathbf{e}), \tag{8.11}$$

where the remainder is holomorphic even at $e_\ell = 0$. For real parameters the first term is purely imaginary. Put $b_\ell = \pi \Im \tau_\ell / |\tau_\ell|^2 > 0$. Equation (8.8) gives

$$\partial_{t_\ell^2} \Re \widehat{B}(0, 0) = -b_\ell/4.$$

On one smaller fixed real box these derivatives are at most $-b_\ell/8$. Integrating along the squared coordinates proves

$$\Re T_{\mathbf{m}}(\mathbf{z}_{\mathbf{m}}) \le \Re T_{\mathbf{1}}(\mathbf{z}_*) - \sum_\ell \frac{b_\ell}{8} \frac{m_\ell^2 - 1}{p_\ell^2} \tag{8.12}$$

for all local nonzero charges. The strict loss holds unless every $|m_\ell| = 1$. This is an exact analytic inequality on a fixed parameter box, not a conclusion from a truncated expansion with comparable slopes.

Separate evenness writes $\Psi_\ell = z_\ell Q_\ell(z_1^2, \dots, z_b^2)$. Thus

$$z_\ell = -\frac{t_\ell}{2(1 - e_\ell Q_\ell/(4\pi \mathrm{i}))}.$$

For a nonzero charge, $z_\ell \neq 0$; on the small neighborhood $\sinh(2\pi \mathrm{i} z_\ell) \neq 0$. The real part of the translated-contour Hessian stays negative definite. Hence the local Gaussian coefficient is nonzero for negative as well as positive slopes.

There are finitely many charges left after the tails were removed. A charge outside the local ratio box has some $|m_i/p_i| \ge t_0$. Split its real interval at $|z_i| = t_0/4$. Outside, the source Fourier bound has a fixed loss. Inside, $|m_i + 2p_i z_i| \ge |p_i| t_0/2$; a small imaginary displacement of the appropriate sign decreases the filling phase. The displacement can be chosen within the fixed source tube, and its joining segments are estimated with the same nonstationary inequality and the endpoint Fourier loss. Spectator coordinates are still real. This supplies a fixed exponential loss for every such nonlocal charge. For local saddles, the connector height is $O(|p_i|^{-1} |m_i/p_i|)$ by the displayed factorization, which keeps their boundary losses below the selected saddle. These estimates are detailed in Appendix B.3.

8.7 Noncancellation and the final action

The 2^b charges with $|m_\ell| = 1$ have identical complex critical action. Under $(m_\ell, z_\ell) \mapsto (-m_\ell, -z_\ell)$, the product $m_\ell \sinh(2\pi \mathrm{i} z_\ell)$, the parity character and the integral are unchanged. They add exactly. For these charges the factor $(-1)^{bd+d\sum m_\ell}$ in (8.5) is one.

Combining (8.9) with the b-dimensional Gaussian gives

$$\langle K(\mathbf{p})\rangle_N = \chi_N \eta_N \kappa^{-\sum p_\ell} N^{3/2} D(\mathbf{p}) e^{NT_{\mathbf{1}}(\mathbf{z}_*)}(1 + O_{\mathbf{p}}(N^{-1})), \tag{8.13}$$

where

$$D(\mathbf{p}) = \frac{2^b(2\pi)^{b/2} B(\mathbf{z}_*) \prod_\ell \sinh(2\pi \mathrm{i} z_{*,\ell})}{\sqrt{\det[-T_{\mathbf{1}}''(\mathbf{z}_*)]}} \neq 0. \tag{8.14}$$

The Gaussian branches are fixed by the translated original real contours. The selected real action tends to F_0 as all $1/p_\ell \to 0$. Enlarge the fixed slope threshold so that it lies above every source, endpoint, nonlocal-charge and interpolation error level. Equation (8.13) then holds for the whole source, not only for the corridor.

Finally $\Psi_\ell = 2\pi \mathrm{i} \xi_\ell$ and the charge-one saddle equation give $\xi_\ell = 1 + 2p_\ell z_{*,\ell}$. Substitution into the Legendre expression shows

$$T_{\mathbf{1}}(\mathbf{z}_*) = \varphi(\boldsymbol{\xi}) + \frac{\pi \mathrm{i}}{2} \sum_\ell \frac{(\xi_\ell - 1)^2}{p_\ell} = S_{\varepsilon,\mathbf{p}}(\boldsymbol{\xi}).$$

Equations (7.5), (8.13) and Proposition 2.1 prove Theorem 1.2. The construction fills precisely the auxiliary cusps. The surviving cusp is the knot cusp, so the right side is the knot complement volume.

Chapter 9

Checking the certificates and understanding the scope

The theorem has two kinds of proof obligations. Most are analytic or geometric arguments written in this manuscript. Two local inequalities are reduced to finite rational certificates supplied with the paper. This chapter explains how to inspect and replay those certificates, what a successful replay establishes, and what remains outside the theorem's construction class.

9.1 Three different kinds of evidence

The manuscript contains analytic proofs, exact finite certificates, and optional finite audits. They have different logical roles. The matrix identities, support and strictness arguments, geometric incidence, complex estimates and filling analysis are proved in the text. The rational certificates are finite parts of the proof: the continuum inequalities reduce to their exact feasibility, coverage and supporting-plane checks. Finite matrix or numerical asymptotic audits provide independent checks of formulas and normalizations, but do not establish a theorem by extrapolation.

The proof is given in the chapters and appendices of this manuscript. The repository also supplies the exact certificate data and verification programs. Those programs check the finite inequalities described below; they do not formally verify the analytic or geometric arguments.

9.2 Replaying the exact certificates

From the repository root, run

```
python3 repro/verify.py --output repro/results
```

The serial runner uses the standard Python library. It replays the two-leaf block certificate, the attachment square certificate and the optional endpoint certificate, compares their exact results with the recorded data, and checks the elementary rational leaf and isolated-three-leaf margins. Its report records the source and data hashes. Candidate generation and numerical optimization are not run by verification.

The principal proof files are

Certificate	Mathematical coverage
`verify_conway_two_leaf_block_alias_certificate.py`	Sixteen slope intervals; both alias types; 288 corner/vertex inequalities.
`verify_conway_single_leaf_attachment_certificate.py`	Exact 55-rectangle cover; 880 corner/vertex inequalities on the full child/parent slope square.
Endpoint certificate in the runner's manifest	Optional stronger margin at the monotone-response endpoint; not needed for the main theorem.

All paths are resolved relative to the repository or script directory, rather than an author's temporary working directory. The runner's proof-input guide lists the written dependencies separately from executable checks.

9.3 Building the two reading views

The TeX source is in `manuscript/`; its native illustrations are in `manuscript/figures/`. A TeX Live or MacTeX installation with `latexmk` builds the paper:

```
python3 tools/build_manuscript.py
```

The output is `manuscript/build/main.pdf`. The build manifest records the hashes of all TeX and bibliography inputs and the generated PDF. It is a typesetting record, not a mathematical certificate.

The Sphinx reading site and the existing Markdown book are built separately:

```
python3 tools/build_sphinx.py
python3 tools/build_book.py
python3 tools/check_book.py
```

Sphinx is written to `docs/sphinx/`; the Markdown book remains in `docs/`. Documentation builds do not execute research scripts. The pinned Sphinx environment is described in `repro/requirements-docs.txt`. Both reading views link the same authored PDF and preserve the distinction between the main theorem and broader open questions.

9.4 What has and has not been proved

The theorem is uniform in the level for each fixed finite completed expression and each fixed sufficiently long signed filling tuple. It includes both parity sequences and composite levels. The filling threshold is allowed to depend on the expression, its complete geometry and the half-twist pattern. It does not depend on a ratio between the sizes of the long slopes.

The theorem does not identify all arborescent knots with the completed grammar. It does not turn an equality in $\mathcal{H}_N$ into a tangle isotopy. It does not prove a quantum theorem for arbitrary short or rational fillings, and it does not infer volume from an unsigned bound on a larger source envelope. The main knot cusp stays unfilled in the conclusion.

The construction rule is a sufficient hypothesis used by this paper, not a characterization of arborescent knots. If one allows a single elementary piece on one side of the final closure, some complete stationary parameters can equal $1/4$. The recurrence analysis proved here assumes a

compact interval strictly above that value. Likewise, a different rule for composing tangles requires new global bounds for its signed color terms, even if its complete ideal cells can be constructed. Such extensions need their own stated hypotheses and proofs. The theorem in this manuscript remains exactly Definition 1.1 and Theorem 1.2; no broader result is incorporated by an informal change of terminology.

Appendix A

Finite algebra and normalization identities

This appendix proves the finite identities on which the asymptotic argument rests. The first calculation compares the chosen crossing matrices and reduced trace with the conventional invariant. The next two calculations establish the Newton basis, its genuine geometric turn, and the exact color decomposition. The last calculation expands an arbitrary twist power while keeping derivatives at repeated roots. No limiting argument is used in these identities.

A.1 The crossing comparison and enhancement

This section supplies the finite normalization calculation used in Proposition 2.1. Put $\omega = Q$ and

$$p_j = \prod_{h=1}^{j}(1-\omega^h), \qquad \bar{p}_j = \prod_{h=1}^{j}(1-\omega^{-h}), \qquad p_0 = 1.$$

Kashaev's unbraided four-index matrix may be written by introducing nonnegative residues

$$u = [j-i-1], \quad v = [i-l], \quad t = [l-k], \quad h = [k-j].$$

Its support conditions are equivalent to $u+v+t+h = N-1$, and on that support

$$R^{o}_{(i,j);(k,l)} = \frac{N\omega^{1-(t+h)(u+h)}}{p_u \bar{p}_v p_t \bar{p}_h}. \tag{A.1}$$

One free j and such a composition specify one entry. Let $F_{ij} = \omega^{ij}/\sqrt{N}$. Its full Fourier transform has only the ordinary-integer-conserving entries

$$((F^{-1}\otimes F^{-1})R^{o}(F\otimes F))_{(a-k,b+k);(a,b)} = \omega^{1+(a-k)(b+1)}\frac{p_a p_{b+k}}{p_{a-k}p_b p_k}, \tag{A.2}$$

for $0 \le k \le \min(a, N-1-b)$.

Here is the finite summation. Define

$$S_m(X) = \sum_{j=0}^{m}\frac{X^j}{p_j \bar{p}_{m-j}} = \frac{\prod_{j=1}^{m}(X-\omega^j)}{p_m}.$$

The equality follows by induction from $\bar{p}_m = (-1)^m \omega^{-m(m+1)/2} p_m$ and $(1-\omega^m)S_m = (X - \omega^m)S_{m-1}$. The free Fourier index imposes conservation modulo N. The two remaining factors are $S_{N-1-z}(\omega^{c-z})S_z(\omega^{z-a})$; their zeros force $c \le z \le a$. With $m = a - c$, the residual sum is

$$\omega^{1+c(b+1)} \frac{p_a}{p_c} \sum_{j=0}^{m} \frac{(-1)^j \omega^{j(j-1)/2} \omega^{(b+1)j}}{p_j p_{m-j}} = \omega^{1+c(b+1)} \frac{p_a}{p_c} \frac{(\omega^{b+1};\omega)_m}{p_m}.$$

If $b + m \ge N$, the product contains $1 - \omega^N$ and vanishes. Otherwise it is p_{b+m}/p_b, proving (A.2) with integer, rather than merely residue, conservation.

Set $G_j = p_j/(1-\omega)^j = q^{j(j-1)/2}[j]_q!$, $D_j = (-q)^j$, and $\gamma = \omega q^{-d^2/2}$. Substitution into (2.2) gives exactly

$$(F^{-1} \otimes F^{-1}) R^o (F \otimes F) = \gamma (DG \otimes G) \mathscr{R} (G^{-1} \otimes G^{-1} D^{-1}).$$

For the enhanced braid conventions take

$$B_K = P(R^o)^T, \quad \mu_K = -qS, \quad \alpha_K = -q, \qquad B_q = P\mathscr{R}^T, \quad \mu_q = K^{-1}, \quad \alpha_q = q^{(N^2-1)/2},$$

where $Sv_j = v_{j+1 \bmod N}$. On b strands, the single similarity

$$U_b = F^{-\otimes b} \bigotimes_{j=1}^{b} (G^{-1} D^{j-1})$$

works for every adjacent generator because the quantum crossing conserves integer weight and commutes with $D \otimes D$. It also sends $\mu_q^{\otimes b}$ to $\mu_K^{\otimes b}$, and $\gamma\alpha_q = \alpha_K$. The writhe characters cancel exactly. The partial traces commute with the remaining module generators; distinct K weights and nonzero F arrows imply scalarity. Thus every internal similarity cancels, including the one open factor of a reduced cut. This proves the finite enhanced comparison used in Chapter 2. Large factorial gauges are never estimated by their norms.

The primal/dual change of basis is $Bv_j = (-1)^j q^{j(j-N)} v_{d-j}$. Direct generator substitution conjugates K^{-1} to the primal pivot and the crossing to the recorded primal convention. At completeness $g_{\mathrm{ADO}} = (-1)^d K$ and the twist has the corresponding sign. A b-strand closure of writhe w and c components has $w \equiv b - c \pmod 2$, so the remaining enhancement ratio is $(-1)^{d(c-1)}$. Its modulus is one, and for a knot it is one in this convention.

A.2 The block polynomial and the physical Newton turn

For completeness we justify the block minimal polynomial used in Lemma 2.2. Before specialization, the ordinary highest-weight module of highest weight d has weights $d - 2a$. In the pair, a highest vector of total degree j is obtained by solving the bidiagonal raising equation. If its coefficient at $|a, j-a\rangle$ is u_a, the equation at $|a, j-a-1\rangle$ determines u_{a+1}/u_a as the negative ratio of the two nonzero raising entries. Thus for generic q there is one such vector in each degree $j = 0, \dots, d$. Its highest weight is $2d - 2j$, its Casimir is $q^{2d-2j+1} + q^{-2d+2j-1}$, and its successive lowering vectors are nonzero through the usual length. The dimensions $2d - 2j + 1$ sum to N^2. Distinct generic Casimirs make these submodules independent.

In degree $T \le d$, precisely the highest vectors $j \le T$ contribute one descendant each. Their eigenvalues give the block characteristic polynomial. At $q = e^{\pi \mathrm{i}/N}$ it becomes

$$\prod_{j=0}^{T}(z - c_j), \qquad c_j = q^{2j+1} + q^{-2j-1}.$$

All physical matrix entries specialize regularly. The nonzero adjacent entries of the tridiagonal block make a boundary vector cyclic, so this characteristic polynomial is also minimal even when roots coincide. At $T = d$ it is P_N. The elementary identity

$$\prod_{j=0}^{N-1}(z - q^{2j+1} - q^{-2j-1}) = 2T_N(z/2) + 2$$

can also be checked by putting $z = w + w^{-1}$: the roots correspond to $w^N = -1$, and the monic degree and leading coefficient agree. The generating property of W_d propagates its vanishing to the full pair.

Now $B_r(\Omega)$ annihilates every degree block W_T with $T < r$, with multiplicities, and has first-color bandwidth at most r. In the turned matrix its first-color displacement is $d - (a + b)$, so $\mathsf{T}B_r$ annihilates every W_T with $T < d - r$. Its representing polynomial is therefore divisible by B_{d-r}.

For the reverse degree bound, its first middle-block column is

$$(\mathsf{T}B_r)_{a0} = x^{a-d}(B_r(\Omega))_{(d-a,0);(d-a,0)},$$

which vanishes for $a > d - r$. A polynomial of degree $k < N$ in the cyclic tridiagonal middle block has a nonzero kth coordinate on the first basis vector, from its unique outermost path. Thus $\deg \mathsf{T}B_r \le d - r$, proving $\mathsf{T}B_r = C_r B_{d-r}$.

Use the physical entry from $(d - r, 0)$ to $(0, d - r)$. On the turned side it is the unique r-step path in the high-degree block, whose factors are

$$Q^{j-r}(1 - Q^{r-j})(1 - Q^{j+1}), \qquad 0 \le j < r.$$

Including the leading coefficient $(-1)^r$ gives s_r^2. The corresponding path for B_{d-r} in the low block gives s_{d-r}^2. All turn monomials in this entry equal one. Hence

$$\mathsf{T}B_r = \frac{s_r^2}{s_{d-r}^2} B_{d-r}, \qquad \mathsf{T}\widehat{B}_r = \widehat{B}_{d-r}.$$

Finally $s_r s_{d-r} = N$ and (2.11) give $\phi(\widehat{B}_i \widehat{B}_j) = \delta_{i+j,d}$: use degree below d, leading coefficient at d, and full divisibility by P_N above it. This proves Proposition 2.3 by physical support and exact paths.

A.3 Circle coefficients and the signed alias

We derive the coefficient formulas rather than infer them from values at the distinct Casimir roots. That distinction is necessary because the quotient algebra also contains first derivatives at double roots. The generating-function calculation first gives a polynomial identity at a

generic parameter; regular specialization then retains all those derivatives automatically. The generic Chebyshev–Newton identity is

$$U_{j-1}(z/2) = \sum_{r=0}^{j-1} (-1)^r \begin{bmatrix} j+r \\ 2r+1 \end{bmatrix}_q B_r(z). \tag{A.3}$$

To prove it, successively divide $1 - zt + t^2 = (1 - c_r t + t^2) + t(c_r - z)$. The formal identity is

$$\frac{1}{1 - zt + t^2} = \sum_{r \geq 0} \frac{(-t)^r B_r(z)}{\prod_{i=0}^{r} (1 - c_i t + t^2)}.$$

After r steps the remainder is divisible by t^{r+1}. The denominator factors as a consecutive q-string, and the Gaussian Pascal recurrence gives its reciprocal coefficient. Taking the coefficient of t^{j-1} gives (A.3). For $j \leq \lfloor N/2 \rfloor$, every factorial index is below N; specializing and turning proves (2.18).

For higher colors use the polynomial identity

$$U_{N-h-1} = \tfrac{1}{2}(\mathsf{C}_h U_{N-1} - \mathsf{C}_N U_{h-1}).$$

Since $\mathsf{C}_N = -2$ in the full algebra,

$$L_{N-h} = (-1)^h \tfrac{1}{2} \mathsf{C}_h L_N + (-1)^N L_h.$$

The vth turned coefficient of the first term is $(-1)^h N \sum_{k=0}^{d} \cos(h(2k+1)\pi/N) \widehat{B}_v(c_k)$. This finite sum can be evaluated without a zero denominator. Put $z = w + w^{-1}$. The factorization of B_v is the consecutive string

$$B_v(w + w^{-1}) = (-1)^v w^{-v} \prod_{j=0}^{2v-1} \left(w - q^{-(2v-1)+2j} \right).$$

For $0 \leq v, h \leq M$, all its Fourier exponents after multiplication by $w^{\pm h}$ have absolute value below N. Summing at $w = q^{2k+1}$ extracts only the constant coefficient. The finite Gaussian product gives that coefficient as zero for $v < h$, and as $(-1)^h s_{2v}/(s_{v-h} s_{v+h})$ for $v \geq h$. Dividing by s_v^2 and multiplying the two factors of N proves (2.17), including $h = 0$ and both parities.

For the half twist, multiplication by z in Newton coordinates is the lower bidiagonal matrix Z with $Z_{rr} = c_r$ and $Z_{r+1,r} = -4 \sin^2((r+1)\pi/N)$. The one-crossing matrix is

$$(R(Z))_{ij} = \kappa (-1)^i q^{i(j+1)} \frac{s_i}{s_j s_{i-j}}, \qquad i \geq j.$$

Its first column is the finite Newton expansion of the crossing. Substituting the two neighboring ratios shows that it commutes with Z; since the first coordinate is cyclic, this determines the full matrix. Reversal on both sides gives (2.19). Coefficient conjugation gives the inverse crossing because its full Hermite values and logarithmic derivatives are conjugate. This argument retains all normal directions.

A.4 Fourier–Hermite interpolation of an arbitrary twist

Ordinary interpolation prescribes values. Hermite interpolation also prescribes derivatives at a repeated node. Here the crossing power supplies both data, and the change of variable $z = w + w^{-1}$ turns the interpolation problem into a finite Fourier calculation. The two Fourier equations below solve simultaneously for the coefficients that ordinary value interpolation would fail to distinguish. For an integer ℓ, let $f_{N,\ell}$ be the degree-below-N remainder of r_N^ℓ modulo P_N. Negative powers exist because all its nodal values are nonzero. At $w_k = q^{2k+1}$ define

$$h_k = \big((-1)^k q^{k(k+1)}\big)^\ell, \qquad d_k = -\frac{\ell}{2}(d-2k)h_k.$$

At a double node, $d_k = (w_k - w_k^{-1}) f'_{N,\ell}(c_k)$; at the odd simple node both sides of the logarithmic derivative condition vanish without prescribing an extra derivative. Put

$$H_n = \frac{1}{N}\sum_k h_k w_k^{-n}, \qquad D_n = \frac{1}{N}\sum_k d_k w_k^{-n}, \qquad 0 \le n < N.$$

Then

$$f_{N,\ell}(z) = a_0 + \sum_{n=1}^{d} a_n \mathsf{C}_n(z), \qquad a_0 = H_0, \quad a_n = \frac{D_n + (N-n)H_n}{N}. \tag{A.4}$$

Indeed the Laurent polynomial $f(w + w^{-1})$ has symmetric coefficients a_n at $w^{\pm n}$. At $w_k^N = -1$ its values and logarithmic derivatives have aliases

$$H_n = a_n - a_{N-n}, \qquad D_n = n a_n + (N-n) a_{N-n}.$$

Solving gives (A.4). At the even middle index $H_{N/2} = 0$ and the derivative equation still determines the coefficient. Paired nodes give the required compatibility, and the odd simple root imposes exactly one condition. Thus no root or normal is omitted.

Set $a_N = a_{N+1} = 0$ and

$$\beta_j^{\ell/2} = (-1)^j (a_j - a_{j+2}), \qquad 0 \le j < N.$$

The identity $\mathsf{C}_n = U_n - U_{n-2}$ gives the circle expansion. Since $|h_k| = 1$ and $|d_k| \le |\ell| N/2$, one has $|a_0| \le 1$, $|a_n| \le 1 + |\ell|/2$, and

$$\sum_j |\beta_j^{\ell/2}| \le 2N(1 + |\ell|/2).$$

For $\ell = 2P$, these are the coefficients used in (8.3). To obtain that formula directly, substitute the two finite Fourier sums into $a_j - a_{j+2}$, set $j = N - h - 1$, and pair h with $-h$ in an even corridor. The difference of the two exponentials gives $\sinh(2\pi \mathrm{i} z_k)$; the differentiated data give the linear factor $2PNz_k + h$, and differentiating the hyperbolic sine gives its cosine. Their sum is precisely $U'_N(z_k)/(2\pi\mathrm{i})$. The constant phase is $(-1)^d \kappa^{-P}$ and the two averaging factors give N^{-2}. This verifies (8.3) with both normal terms and both parity conventions.

Appendix B

Detailed estimates for the color sums and filling contours

The local and global estimates have different purposes. Section B.1 constructs an exponentially accurate holomorphic approximation on the strict neighborhood used for the saddle. Section B.2 bounds every remaining core coefficient, including parameters where the recurrence eigenvalues meet. The last section treats the joining contours for the filling integral. Keeping these estimates separate prevents a local asymptotic expansion from being applied outside its proven domain.

B.1 The exact local recurrence and its holomorphic asymptotic

This section proves the local source estimate used in the completed-root theorem. Its hypothesis is a fixed compact interval

$$I = [\tau_0, \tau_1] \subset (1/4, 1/2) \tag{B.1}$$

for every retained complete leaf coordinate. All constants may depend on this interval and the fixed source tree. The color neighborhood and every complex index neighborhood below are fixed independently of the level. We work with the actual finite low core and its specified factorial continuation. No homogeneous solution of a recurrence is substituted for that source. In particular, an inverse-power approximation on the real index grid will not be used as if it were an exponentially accurate complex interpolation.

B.1.1 Canonical factorials and the literal mixed coefficient

To move a color parameter off the integer grid, we must specify a function that agrees with the finite coefficient and preserves its phase. The Gamma product below supplies that function for the sine factorials. Its exact shift identity will then allow the finite recurrence to remain valid at complex colors; agreement only in a leading asymptotic term would not be sufficient.

Retain

$$d = N - 1, \quad M = \lfloor d/2 \rfloor, \quad q = e^{\pi \mathrm{i}/N}, \quad Q = q^2, \quad \kappa = q^{d^2/2}, \quad F_r = (Q;Q)_r.$$

In this section x denotes a scaled Newton index, rather than the physical-frame scalar q^d. The sine factorial has the exact continuation

$$G_N(z) = \left(\frac{2\pi}{N}\right)^z \Gamma(z+1) \prod_{k=1}^{\infty} \frac{\Gamma(Nk+z+1)}{\Gamma(Nk-z)(Nk)^{2z+1}}, \qquad -1 < \Re z < N. \tag{B.2}$$

All positive-base powers use their real logarithms. The logarithm of the kth factor is $O_K(k^{-2})$ on compact subsets of the strip: in the Gamma-ratio expansion its k^{-1} term cancels. Thus the product is holomorphic and zero-free there. Gamma's functional equation and the convergent sine product give

$$\frac{G_N(z+1)}{G_N(z)} = 2\sin\frac{\pi(z+1)}{N}, \qquad G_N(0) = 1, \qquad G_N(r) = s_r \quad (0 \le r < N). \tag{B.3}$$

The marked quantum factorial is

$$\widehat{F}_N(z) = e^{\pi \mathrm{i} z(z+1)/(2N) - \pi \mathrm{i} z/2} G_N(z). \tag{B.4}$$

Consequently

$$\widehat{F}_N(z+1) = (1 - Q^{z+1})\widehat{F}_N(z), \qquad \widehat{F}_N(r) = F_r, \qquad Q^z = e^{2\pi \mathrm{i} z/N}.$$

These identities fix the continuation and its phase, not just its asymptotic expansion. On a compact subset of $0 < \Re z < 1$, direct Stirling expansion of (B.2) gives

$$\begin{aligned} \log G_N(Nz) &= NJ(z) + \frac{1}{2}\operatorname{Log}(2N\sin\pi z) + \frac{\pi\cot\pi z}{12N} + O(N^{-3}), \\ J(z) &= \int_0^z \operatorname{Log}(2\sin\pi u)\,\mathrm{d}u. \end{aligned} \tag{B.5}$$

Here the logarithm continues the real positive factorial branch. One way to verify the uniform remainder is to expand the kth Gamma ratio: its error is $O_K(N^{-3}k^{-3})$. These errors are summable; the leading and half-log terms sum by the sine product. The first correction sums by

$$z^{-1} + \sum_{k\ge 1}\bigl((k+z)^{-1} - (k-z)^{-1}\bigr) = \pi\cot\pi z.$$

Cauchy's estimate supplies the corresponding derivative bounds on smaller compact sets.

Let h be a physical color deficit and put $t = h/N$. At an integer index $r > h$ the pure core coefficient is the literal factorial expression

$$A_N(r,t) = \frac{N^2 s_{2r}}{s_r^2 G_N(r-Nt)G_N(r+Nt)}. \tag{B.6}$$

For a retained half-twist sign $\varepsilon \in \{1,-1\}$, the exact triangular Newton transport gives

$$d_N^{\varepsilon}(r,t) = \kappa^{\varepsilon}\sum_{v=r}^{M}(-1)^v q^{\varepsilon v(r+1)}\frac{s_v}{s_r s_{v-r}}A_N(v,t), \qquad d_N^0(r,t) = A_N(r,t). \tag{B.7}$$

For complex t in a sufficiently small fixed disk, only the two color-shifted factorials are continued. All their arguments remain in regular strips. Equation (B.7) is then an exact finite,

even holomorphic function. At every physical integer $h < r$ it is the original transported core, including all first normal directions.

The cancellation inside this finite sum is encoded by

$$A_{h,r} = \sum_{v=r}^{M} \frac{Q^{-vr} F_{2v}}{F_v \widehat{F}_N(v-h) \widehat{F}_N(v+h) F_{v-r}}, \tag{B.8}$$

$$D_{h,r} = \frac{Q^{-r^2} F_{2r}}{F_r \widehat{F}_N(r-h) \widehat{F}_N(r+h)}, \qquad U_{h,r} = A_{h,r}/D_{h,r}. \tag{B.9}$$

The sum starts at r in the present corridor. For physical $r < h$ its lower endpoint would be h; those low indices belong to the separate global envelope, not to the interpolation asserted here. The terminal source in (B.9) is exactly $U_{h,M} = 1$.

B.1.2 Deriving the recurrence, including its endpoint

The identity traditionally called a Ward equation here is simply a second-order recurrence obtained by telescoping the actual finite sum. A recurrence has many solutions, so we include its upper endpoint condition in the derivation. That condition is what selects the coefficient occurring in the invariant.

Put $H = Q^h = e^{2\pi i t}$, $T_h = H + H^{-1}$ and $Y = Q^r$. The literal finite sum satisfies

$$\begin{aligned} QY^2(1-Q^2Y)A_{h,r+2} + Y(T_h Q^3 Y^2 + Q + 1)A_{h,r+1} \\ - Q(QY+1)(QY^2-1)A_{h,r} = 0, \qquad A_{h,M+1} = 0. \end{aligned} \tag{B.10}$$

Here and below an absent term beyond the finite upper support is zero. We give the summation certificate, since the endpoint fixes which solution of this second-order equation is the physical one.

For the summand of (B.8), write $z = Q^v$. Its ratio under $v \mapsto v+1$ is

$$\frac{Y^{-1}(1-Qz^2)(1+Qz)}{(1-Qz/H)(1-QHz)(1-Qz/Y)}.$$

Use the rational certificate

$$g_h(z) = \frac{(1-z/H)(1-Hz)(1-z/Y)}{z^2}.$$

The telescoping difference, divided by the original summand, is

$$\left(\frac{1}{YQ^2} - 1\right) z^{-2} + \left(\frac{1}{YQ} + H + H^{-1} + \frac{1}{Y}\right) z^{-1} - \frac{1}{YQ} - 1 - \frac{H + H^{-1}}{Y}. \tag{B.11}$$

The lower endpoint has $z = Y$ and hence vanishes. The upper endpoint contains $F_{2M+2} = 0$. The ratio for the shift $r \mapsto r+1$ is $z^{-1} - Y^{-1}$. Expressing the first two z^{-1} moments in terms of $A_{h,r+1}$ and $A_{h,r+2}$ in (B.11) gives (B.10) by multiplication of its displayed denominators. This argument remains exact for complex $h = Nt$: the shifted factorial equations are exact and the lower zero is still $z = Y$. It does not use interpolation from finitely many colors.

Normalize by the actual $D_{h,r}$ and set

$$\mathrm{den}_r = (1 - QY/H)(1 - QYH).$$

Then

$$U_r = p_N(r,t)U_{r+1} + q_N(r,t)U_{r+2}, \tag{B.12}$$

where the exact rational coefficients are

$$p_N = -\frac{Q^{-2}Y^{-1}(T_hQ^3Y^2 + Q + 1)}{\mathrm{den}_r}, \tag{B.13}$$

$$q_N = -\frac{Q^{-4}Y^{-2}(1-Q^2Y)(1-Q^3Y^2)(1+Q^2Y)}{\mathrm{den}_r\,\mathrm{den}_{r+1}}. \tag{B.14}$$

All these denominators are bounded away from zero on the corridor used below. At the last recurrence row one has the exact wall

$$q_N(M-1,t) = 0, \qquad U_{M-1} = p_N(M-1,t), \qquad U_M = 1. \tag{B.15}$$

For odd $N = 2m+1$, $M = m$ and the factor $1-Q^3Y^2$ vanishes. For even $N = 2m$, $M = m-1$ and $1+Q^2Y$ vanishes. The zero is independent of H. In particular no nonexistent $D_{h,M+1}$ is divided out and no arbitrary terminal ratio is imposed.

B.1.3 The selected analytic root and a source-exact continued fraction

We solve the recurrence backwards from its finite upper endpoint. On the strict high-index interval, one characteristic root dominates the other. The resulting contraction permits a holomorphic continued fraction. Crucially, an exact zero in the last recurrence row erases the artificial tail of this continued fraction, so it agrees with the original finite solution on the integer grid.

Regard (B.13)–(B.14) as holomorphic functions $p(x,t,\varepsilon_N)$ and $q_c(x,t,\varepsilon_N)$, where $\varepsilon_N = N^{-1}$, $Y = e^{2\pi\mathrm{i}x}$ and $Q = e^{2\pi\mathrm{i}\varepsilon_N}$. At $\varepsilon_N = 0$, put

$$\begin{aligned} D_t &= (1-Y/H)(1-YH), \\ p_0 &= -Y^{-1}\frac{(H+H^{-1})Y^2+2}{D_t}, \qquad q_0 = -Y^{-2}\frac{(1-Y^2)^2}{D_t^2}. \end{aligned} \tag{B.16}$$

The two roots of $\mu^2 - p_0\mu - q_0 = 0$ at $t = 0$ are

$$\mu_s(x,0) = -Y^{-1}, \qquad \mu_f(x,0) = Y^{-1}\cot^2(\pi x). \tag{B.17}$$

On a compact real interval in $(1/4, 3/4)$ containing $[\tau_0, 1/2]$, the selected branch $\mu = \mu_s$ is nonzero, $\Re\mu > 0$, and $|\mu_f/\mu| < 1$. The same inequalities, with uniform strict constants, hold on a fixed complex neighborhood and a sufficiently small complex t disk. Indeed at $t = 0$ these assertions follow directly from (B.17); nonvanishing and a separated quadratic root persist on a small complex thickening. Write

$$\left|\frac{\mu_f}{\mu}\right| \le k_0 < 1, \qquad \Re\mu \ge m_0 > 0. \tag{B.18}$$

The endpoint $x = 1/2$ is included. Direct substitution gives $\mu(1/2,t) = 1$, while q_0 has a double zero there and its first ε_N derivative also vanishes. The selected root and its normalization are regular at the physical terminal point.

Choose $\sigma_0 > 0$ so small that $[\tau_0 - \sigma_0, 1 - \tau_0 + 4\sigma_0] \subset (1/4, 3/4)$. Use a fixed complex neighborhood of this larger interval on which (B.18) holds, and put

$$\mathcal{L}_N = \lceil (1/2 - \tau_0 + 2\sigma_0)N \rceil, \qquad \mathcal{F}_x(z) = p(x, t, \varepsilon_N) + \frac{q_c(x, t, \varepsilon_N)}{z}.$$

Define the finite continued fraction

$$v_N(x, t) = \mathcal{F}_x \circ \mathcal{F}_{x+\varepsilon_N} \circ \cdots \circ \mathcal{F}_{x+(\mathcal{L}_N-1)\varepsilon_N}(\mu(x + \mathcal{L}_N\varepsilon_N, t)). \tag{B.19}$$

This is defined on a fixed complex neighborhood of $[\tau_0, 1/2]$. For a constant C independent of N, the disks of radius $C\varepsilon_N$ around the respective selected roots are invariant under backward iteration. To prove this, note first that

$$\mathcal{F}_x(\mu(x + \varepsilon_N, t)) - \mu(x, t) = O(\varepsilon_N).$$

On those disks the derivative in the continued-fraction argument has modulus at most $k_0 + O(\varepsilon_N)$. Choose $k_1 \in (k_0, 1)$ and then C so that the $O(\varepsilon_N)$ error plus $k_1 C\varepsilon_N$ is at most $C\varepsilon_N$. This proves the invariance inductively. The disks avoid zero, so every iterate is holomorphic, nonzero, and has uniformly bounded inverse. In particular

$$v_N(x, t) = \mu(x, t) + O(N^{-1}) \tag{B.20}$$

uniformly holomorphically on a fixed smaller domain.

At a physical $x = r/N$, the continued fraction reaches the row $M - 1$. By (B.15), that row forgets its entire remaining tail and takes exactly the value $p_N(M - 1, t)$. Backward induction from the literal source $U_M = 1$ therefore proves

$$v_N(r/N, t) = \frac{U_{Nt,r}}{U_{Nt,r+1}}, \qquad \tau_0 N \le r < M. \tag{B.21}$$

This also proves that the displayed physical source values are nonzero. The equality holds uniformly for complex t in the fixed disk. There is no assertion about a nonexistent physical ratio at $r = M$.

Adding one more future step to (B.19) changes the future value by $O(\varepsilon_N)$, and the preceding $\mathcal{L}_N$ contractions reduce that change by $k_1^{\mathcal{L}_N}$. Thus its Riccati defect is exponentially small. This explains why the analytic tail used to define the interpolant cannot hide a macroscopic change of the original source.

B.1.4 First transport correction and absolute amplitude normalization

Let $p_1 = \partial_\varepsilon p|_{\varepsilon=0}$ and $q_1 = \partial_\varepsilon q_c|_{\varepsilon=0}$, differentiating the explicit rational coefficients. Define

$$\nu = \frac{p_1 + q_1/\mu - q_0\mu_x/\mu^2}{1 + q_0/\mu^2}. \tag{B.22}$$

The denominator is $1 - \mu_f/\mu$, bounded away from zero by (B.18). Substitution of $\mu + \varepsilon_N\nu$ into the one-step Riccati equation gives a residual $O(\varepsilon_N^2)$. The same contracting backward iteration then gives

$$v_N = \mu + N^{-1}\nu + O(N^{-2}) \tag{B.23}$$

on a fixed smaller complex neighborhood. For example, subtract the trial iterate from the exact one at each step. The resulting error is bounded by the geometric sum of its $O(N^{-2})$ local defects and the contracted terminal discrepancy. Cauchy estimates give the corresponding uniform derivative expansions.

Use the logarithm with positive real part in (B.18), continuing from $\mu(1/2,t)=1$, and set

$$\begin{aligned} S(x,t) &= \int_x^{1/2} \operatorname{Log}\mu(u,t)\,\mathrm{d}u, \\ C(x,t) &= \sqrt{\mu(x,t)}\exp\left(\int_x^{1/2}\frac{\nu(u,t)}{\mu(u,t)}\,\mathrm{d}u\right). \end{aligned} \tag{B.24}$$

The square root is one at $1/2$. The integral branches are fixed by the actual selected source; C is holomorphic and nonzero.

Summing logarithms of the exact ratios in (B.21), using (B.23) and the first Euler–Maclaurin endpoint term, gives

$$U_{Nt,r} = C(r/N,t)e^{NS(r/N,t)}[1+O(N^{-1})]. \tag{B.25}$$

The product stops at $M-1$, because $U_M=1$. Replacing the true endpoint M/N by $1/2$ costs only $O(N^{-1})$: the displacement is $O(N^{-1})$ at either parity, and $\operatorname{Log}\mu(1/2,t)=0$. The remaining endpoint terms in the correction are likewise $O(N^{-1})$. In particular there is no free terminal matching constant.

At $t=0$, direct differentiation of (B.13)–(B.14) yields

$$\operatorname{Log}\mu = \pi\mathrm{i} - 2\pi\mathrm{i}x, \qquad \frac{\nu}{\mu} = -\frac{\pi}{2}\cot(\pi x) - \frac{5\pi\mathrm{i}}{2}.$$

Therefore

$$S(x,0) = \pi\mathrm{i}(1/2-x)^2, \qquad C(x,0) = \sqrt{\sin\pi x}\,e^{-3\pi\mathrm{i}(1/2-x)/2}. \tag{B.26}$$

This calculation fixes the absolute amplitude from the finite source, rather than from the value of an unknown closed invariant.

B.1.5 Exponential interpolation before an oscillatory saddle

Equation (B.25) is not sufficient by itself for a complex wire saddle. Its relative error is only inverse polynomial and could exceed a fixed oscillatory cancellation. The exact ratio (B.21) supplies the needed stronger estimate.

Put $f_N(x,t)=\operatorname{Log}v_N(x,t)$ and $b_N=M/N$. On a fixed larger complex tube the f_N are uniformly bounded and holomorphic. On an inner tube, Cauchy's estimate gives constants $B,a>0$, independent of N,t, such that

$$|\partial_x^k f_N(x,t)| \le Bk!a^{-k}. \tag{B.27}$$

For $K_N=\lfloor c_aN\rfloor$, with a sufficiently small fixed $c_a>0$, form the holomorphic expression

$$\begin{aligned} E_N(x,t) = N\int_x^{b_N} f_N(u,t)\,\mathrm{d}u + \frac{f_N(x,t)-f_N(b_N,t)}{2} \\ + \sum_{k=1}^{K_N}\frac{B_{2k}}{(2k)!}N^{1-2k}\big[\partial_x^{2k-1}f_N(b_N,t) - \partial_x^{2k-1}f_N(x,t)\big]. \end{aligned} \tag{B.28}$$

The Bernoulli polynomial bound $|B_{2k}(u)| \le 4(2k)!/(2\pi)^{2k}$ together with (B.27) bounds the Euler–Maclaurin remainder by

$$CN\left(\frac{K_N}{\pi aN}\right)^{2K_N} \le Ce^{-cN}$$

after decreasing c_a. This is uniform in the complex color disk and on the physical high grid. Since the product of the exact ratios is precisely U_r, it follows that

$$U_{Nt,r} = e^{E_N(r/N,t)}[1 + R_{N,r}(t)], \qquad |R_{N,r}(t)| \le Ce^{-cN}. \tag{B.29}$$

The empty terminal product equals one. The residual is retained as a source term, rather than reinterpreted as a homogeneous solution.

The same explicit expression is holomorphic in the index x. Using (B.23) in (B.28), the higher Bernoulli terms sum to $O(N^{-1})$ on an inner fixed tube, and hence

$$e^{E_N(x,t)} = C(x,t)e^{NS(x,t)}[1 + O(N^{-1})] \tag{B.30}$$

uniformly holomorphically. Equations (B.29) and (B.30) have distinct roles: the former is an exponentially accurate comparison with the original finite source, and the latter is a leading expansion of a specified complex function.

B.1.6 Keeping the quadratic phase and determining the signed action

The half-twist coefficient contains a quadratic exponential in the color. Such a phase is sometimes called a chirp. Its continuation is part of the finite formula, not an arbitrary choice of interpolation. We now combine it with the recurrence solution and the continued factorials; this calculation determines the local action and the nonzero amplitude used in the saddle.

Remove only the known external phase by defining

$$\ell_N^{\varepsilon}(r,t) = e^{-\mathrm{i}\varepsilon\pi d/4}d_N^{\varepsilon}(r,t), \qquad \eta_N = \exp\left(\frac{\mathrm{i}\pi d}{4}\sum_{\ell}\varepsilon_\ell\right).$$

The positive sign is the coefficientwise conjugate of the negative sign on real parameters. Its holomorphic continuation uses Schwarz reflection, not conjugation of a free holomorphic variable.

The original factorial phase gives the exact identity

$$q^{h^2}\frac{D_{h,r}}{D_{0,r}} = \frac{G_N(r)^2}{G_N(r-h)G_N(r+h)}. \tag{B.31}$$

It follows immediately from (B.4); the linear and quadratic exponents in the two shifted arguments must both be kept. Combining (B.7)–(B.9) with (B.31) gives the negative-sign interpolant

$$\begin{aligned} A_N^{\mathrm{hol}}(x,t) &= \frac{N^2 G_N(2Nx)}{G_N(Nx)^2 G_N(N(x-t))G_N(N(x+t))}, \\ \nu_N(z) &= e^{-\pi\mathrm{i}z+\pi\mathrm{i}z(z+1)/N}, \qquad P_N^{\mathrm{ch}}(x) = \frac{\kappa^{-1}e^{\pi\mathrm{i}d/4}}{\nu_N(Nx)}, \\ I_N^{-}(x,t) &= A_N^{\mathrm{hol}}(x,t)P_N^{\mathrm{ch}}(x)e^{E_N(x,t)}. \end{aligned} \tag{B.32}$$

The superscript on P_N^{ch} distinguishes this chirp factor from the Casimir polynomial. At every integer r, $\nu_N(r) = (-1)^r q^{r(r+1)}$. Its displayed analytic continuation is fixed before any contour deformation. In particular,

$$P_N^{\mathrm{ch}}(x) e^{NS(x,0)} = e^{\pi \mathrm{i}(3/4 - x - 1/(2N))}. \tag{B.33}$$

Changing the continuation of $(-1)^r$ would change a lattice frequency; that change is not permitted. Equation (B.29) proves that $I_N^-(r/N, t)$ agrees with the actual $\ell_N^-(r, t)$ up to a relative exponential residual. For the positive sign set $I_N^+(x,t) = \overline{I_N^-(\bar{x}, \bar{t})}$; for the pure sign use $I_N^0 = A_N^{\mathrm{hol}}$, which agrees exactly on the grid.

Define

$$\begin{aligned} g(x,t) &= J(2x) - 2J(x) - J(x-t) - J(x+t), \\ f_0(x,t) &= g(x,t), \qquad f_-(x,t) = g(x,t) + S(x,t) - S(x,0), \\ f_+(x,t) &= \overline{f_-(\bar{x}, \bar{t})}. \end{aligned} \tag{B.34}$$

The factorial expansion and (B.33) yield

$$I_N^{\varepsilon}(x,t) = N^{1/2} b_{\varepsilon}(x,t) e^{N f_{\varepsilon}(x,t)} [1 + O(N^{-1})], \tag{B.35}$$

uniformly holomorphically on fixed smaller domains, with

$$\begin{aligned} b_0(x,t) &= \frac{\sqrt{\sin(2\pi x)}}{2^{3/2} \sin(\pi x) \sqrt{\sin(\pi(x-t)) \sin(\pi(x+t))}}, \\ b_-(x,t) &= b_0(x,t) e^{\pi \mathrm{i}(3/4 - x)} C(x,t), \qquad b_+(x,t) = \overline{b_-(\bar{x}, \bar{t})}. \end{aligned} \tag{B.36}$$

All square roots continue their positive complete factorial branches. At completeness,

$$\frac{b_{\varepsilon}(x,0)}{b_0(x,0)} = \sqrt{\sin(\pi x)} e^{-\mathrm{i}\varepsilon \pi x/2}, \qquad \varepsilon \in \{-1, 1\}. \tag{B.37}$$

These factors are nonzero. The remaining $e^{-\pi \mathrm{i}/(2N)}$ in (B.33) belongs to the stated inverse-power error; it is not an omitted constant or parity phase.

B.1.7 An exact source action identity and its Hessian

The action above can also be characterized without integrating the quadratic root. For each physical sign $\varepsilon \in \{0, 1, -1\}$ solve near $s = t = 0$ the equation

$$t = s - \frac{\varepsilon}{2\pi \mathrm{i}} g_s(x,s). \tag{B.38}$$

Its s derivative at zero is $1 - \varepsilon \mathrm{i} \cot(\pi x)$, which is nonzero. Then the source-selected action satisfies

$$f_{\varepsilon}(x,t) = g(x,s) - \frac{\varepsilon}{4\pi \mathrm{i}} g_s(x,s)^2, \qquad (f_{\varepsilon})_x = g_x(x,s), \qquad (f_{\varepsilon})_t = g_s(x,s). \tag{B.39}$$

The derivative identities follow by differentiating (B.38). To identify the value with the actual selected source, consider the negative sign. Put $E = e^{2\pi \mathrm{i} x}$, $H = e^{2\pi \mathrm{i} t}$ and $T = e^{2\pi \mathrm{i} s}$. Exponentiating (B.38) gives

$$H = \frac{T(E-T)}{ET - 1}, \qquad T^2 + E(H-1)T - H = 0, \qquad T = 1 \text{ at } H = 1.$$

For $K = e^{-g_x(x,s)} = -(1 - E/T)(1 - ET)/(1 + E)^2$, elimination of T gives

$$K^2(1 + E)^2 + K[(H + H^{-1})E^2 + 2] + (1 - E)^2 = 0. \tag{B.40}$$

On the other hand (B.34) and the selected quadratic give $e^{-(f_-)_x} = E\mu D_t/(1 + E)^2$, satisfying exactly (B.40). At $H = 1$, both constructions take the branch $K = \tan^2(\pi x)$; their logarithms agree by their complete marks. Hence their x derivatives agree. The possible remaining function of t is zero by the limit $x \to 1/2$: there $S = 0$, $J(1/2 + t) = -J(1/2 - t)$, and the implicit shear has $s = t$, $g_s = 0$. Schwarz reflection proves the positive sign. This fixes the source action, not merely a Hessian with the same entries.

In particular, all local actions are even in t, equal $g(x, 0)$ at zero, and

$$(f_\varepsilon)_{tt}(x, 0) = -\frac{2\pi \cot(\pi x)}{1 - \varepsilon \mathrm{i} \cot(\pi x)}, \qquad \varepsilon \in \{0, 1, -1\}. \tag{B.41}$$

Their real parts are strictly negative on the fixed high interval.

B.1.8 The exact finite wire window and its lattice saddle

We now specify the function to which the complex saddle argument is applied. Let the fixed completed-root expression have H leaves and $K = 2H - 3$ primitive free integer wires. The last forced product is removed using its exact relation $i + j = d$, whose coefficient is one. Every remaining wire has the affine form

$$r_w = NL_w(y) + \delta_w, \qquad y = k/N, \quad k \in \mathbb{Z}^K,$$

where the integer offsets δ_w are fixed by the original compiler. Take a small fixed rectangular window B about the strict complete optimizer. All leaf coordinates lie in intervals of the form (B.1), and every retained gate inequality is strict on a fixed neighborhood of the closure of B.

The exact finite window is

$$\mathcal{C}_N(\mathbf{t}) = \sum_{k/N \in B} \prod_{\ell=1}^{H} \ell_N^{\varepsilon_\ell}(r_\ell, t_\ell) \prod_{g=1}^{H-2} \mu_N(r_{g,1}, r_{g,2}, r_{g,3}). \tag{B.42}$$

Its support is independent of the complex colors. It is a finite, separately even holomorphic function. For all sufficiently large N, every state in this window is a true core state at every physical small color deficit. The factors μ_N are the original positive Newton coefficients, not their leading asymptotic replacements.

Replace each exact local factor by its canonical holomorphic interpolant only after keeping its residual. The gate interpolation is the same seven-factorial expression with G_N, and agrees exactly on its integer grid. For the signed leaves, use (B.32). The total error in the finite sum is bounded absolutely by

$$\mathrm{poly}(N) \exp\{N(F_0 + C\|\mathbf{t}\|^2 - c)\} \tag{B.43}$$

for some fixed $c > 0$. Indeed there are polynomially many states, all actions are even analytic perturbations of the common complete real action, and the relative leaf errors in (B.29) are exponentially small. This estimate is made before any oscillatory cancellation. Shrink the fixed color polydisk so that (B.43) is exponentially below the nearby selected critical action.

The interpolated summand has phase

$$\mathcal{F}(y,\mathbf{t}) = \sum_{\ell} f_{\varepsilon_\ell}(L_\ell(y), t_\ell) + \sum_{g} m(L_{g,1}(y), L_{g,2}(y), L_{g,3}(y)). \tag{B.44}$$

At zero color it is the original complete action. Its wire Hessian at the strict optimizer is negative definite. Thus it has one holomorphic critical branch $y_*(\mathbf{t})$, with

$$\Phi(\mathbf{t}) = \mathcal{F}(y_*(\mathbf{t}), \mathbf{t}).$$

The local gate amplitude is

$$b_\mu(x,y,z) = 2^{-3/2}\sqrt{\frac{\sin\pi(x+y)\sin\pi z}{\sin\pi x\sin\pi y\sin\pi(z-x)\sin\pi(z-y)\sin\pi(x+y-z)}}.$$

Multiply all leaf and gate amplitudes by the exact leading offset factor

$$\exp\left[\sum_{\ell}\delta_\ell\partial_x f_{\varepsilon_\ell} + \sum_{g}\nabla m(L_g(y))\cdot\delta_g\right] \tag{B.45}$$

to obtain the nonzero holomorphic density $b_E(y,\mathbf{t})$. The changed f_ε is used in (B.45), including at nonzero complex color; a pure offset factor would be incorrect. The interpolated summand is

$$N^{3-H} b_E(y,\mathbf{t}) e^{N\mathcal{F}(y,\mathbf{t})}[1+O(N^{-1})].$$

For clarity we give the finite-lattice contour step. Choose the vertical faces of the rectangle at half-grid values. In each coordinate the exact summation kernel is $\pi N\cot(\pi N z)$, with residue one at $z = k/N$. On a vertical half-grid line, $|\cot(\pi N z)| \le 1$. Those faces lie away from the strict real optimizer and retain a fixed phase gap. On a horizontal line of fixed height $a > 0$, the cotangent differs from its constant value $\mp\mathrm{i}$ by $O(e^{-2\pi N a})$. At complete color, the real part of the phase grows under this small imaginary displacement by at most Ca^2; the even color perturbation costs at most $C\|\mathbf{t}\|^2$. Choose a so $Ca^2 < \pi a$, and then choose the color polydisk small enough. Every nonconstant kernel term has an exponential gap, including all side connectors. Applying this argument successively in the K primitive coordinates retains the full product contour and its orientations. The constant horizontal terms give N^K times the ordinary integral. Replacing the moving half-grid rectangle by B costs only exponentially small boundary strips.

Translate that integral to the nearby complex critical point. The endpoint connectors remain below its action, and the real tangent Gaussian is uniformly nondegenerate. Extend its local Gaussian range with exponentially small error. The odd first correction integrates to zero, giving a relative $O(N^{-1})$ expansion. Thus

$$\mathcal{C}_N(\mathbf{t}) = N^{3/2} a_E(\mathbf{t}) e^{N\Phi(\mathbf{t})}[1+O(N^{-1})], \qquad a_E(\mathbf{t}) = \frac{(2\pi)^{K/2} b_E(y_*(\mathbf{t}),\mathbf{t})}{\sqrt{\det[-\mathcal{F}_{yy}(y_*(\mathbf{t}),\mathbf{t})]}}. \tag{B.46}$$

The determinant square root is selected from the positive complete Gaussian. The power is $3 - H + K/2 = 3/2$. The exponential grid error (B.43) is smaller than this answer by the prior choice of color disk, and is added back only now. Both level parities have the same primitive density and displayed normalization.

At completeness,

$$\Phi(0) = F_0, \qquad a_E(0) = a_E^{\mathrm{pure}}(0) \prod_{\varepsilon_\ell \neq 0} \sqrt{\sin(\pi x_{\ell,*})} e^{-\mathrm{i}\varepsilon_\ell \pi x_{\ell,*}/2} \neq 0. \tag{B.47}$$

Evenness gives $\mathcal{F}_{yt_\ell}(y,0) = 0$, so the wire Schur correction to the color Hessian vanishes. With $c_\ell = \cot(\pi x_{\ell,*})$, one obtains

$$\partial_{t_\ell}\partial_{t_j}\Phi(0) = -\frac{2\pi c_\ell}{1-\varepsilon_\ell \mathrm{i} c_\ell}\delta_{\ell j} = 2\pi \mathrm{i}\tau_\ell \delta_{\ell j}, \qquad \tau_\ell = \frac{\mathrm{i}c_\ell}{1-\varepsilon_\ell \mathrm{i}c_\ell}, \qquad \Im\tau_\ell > 0. \tag{B.48}$$

On a smaller fixed complex box,

$$|\mathcal{C}_N(u+\mathrm{i}v)| \le \mathrm{poly}(N) e^{N(F_0 - a\|u\|^2 + B\|v\|^2)}, \qquad a > 0. \tag{B.49}$$

This bound is for the exact finite function (B.42), by (B.46), not just its leading formal action.

B.1.9 Returning from the local sum to the original evaluation

We finish by stating precisely how the independent alias estimates are used. The compositional proof in the main text gives a uniform positive gap for every nonempty alias sector on the entire physical color cube. It also gives a concave all-core majorant with the unique strict complete maximizer used above. The latter uniqueness is global: it agrees with the strict complete action near that point, and a second maximum of a concave function would give a constant segment through that strictly concave neighborhood.

Hence all core states outside the fixed wire window have a fixed gap, uniformly in physical colors. Within that window the actual local far-color recurrence estimate loses a fixed positive amount whenever one physical deficit exceeds a fixed small corridor. Polynomial state counting and the finite number of sectors therefore give constants $\delta, \gamma > 0$ such that the literal mixed scalar $\mathcal{A}_N$ obeys

$$\max_{0 \le h_\ell \le \delta N} |\mathcal{A}_N(\mathbf{h}) - \eta_N \mathcal{C}_N(\mathbf{h}/N)| \le \mathrm{poly}(N) e^{N(F_0 - \gamma)}, \tag{B.50}$$

$$\max_{\exists \ell:\, h_\ell > \delta N} |\mathcal{A}_N(\mathbf{h})| \le \mathrm{poly}(N) e^{N(F_0 - \gamma)}. \tag{B.51}$$

The maximum in the second line is over the entire original physical color cube, not just the low-core band. No alias is folded into the even analytic packet. The external η_N is exactly the retained unit phase defined above.

The global coherent-core input here is its uniform exponential-rate envelope, of the form $\exp\{N(\widetilde{g} + o(1))\}$ with the $o(1)$ uniform over all physical indices and colors. It is not a polynomial-prefactor assertion through every spectral collision. To obtain (B.50)–(B.51), first fix the strictly positive variational and far-color margins, then absorb the uniform $o(N)$ errors into, for example, half their minimum. Polynomial counting and the remaining finite prefactors are harmless after that reduction of γ. The polynomial-prefactor local packet proved above is a separate estimate on its fixed regular high interval.

These estimates preserve the whole-source remainder through the finite filling sum. For each fixed signed filling tuple, the exact Hermite coefficient bound is $\sum_j |\beta_{j-1}^{-p_\ell}| \le 2N(1+|p_\ell|)$. Thus (B.50) and (B.51) incur only a polynomial multiplier in that whole sum. The finite analytic

packet, rather than its pointwise leading approximation, is then used for the Fourier endpoint and charge arguments. Its exact evenness, complex estimate (B.49), nonzero amplitude and Hessian (B.48) are the needed inputs. The retained filling analysis supplies the smaller filled action and a nonzero leading coefficient; the source gap is not itself a replacement for that analysis.

The argument in this section requires (B.1). No continuation through a quarter-index dominance boundary, no additional root-leaf closure, and no theorem for arbitrary tangle syntax is used in the manuscript's completed-root result.

B.2 A global coefficient bound, including repeated recurrence roots

The local interpolation of Section B.1 is used only on a fixed high-index interval. Here we prove the different estimate needed before restricting to that interval: a uniform exponential-rate bound for every physical color and every core index. The proof retains the other recurrence branch, including its repeated-root colors. Here C_j^{ε} always denotes the core polynomial of a single physical leaf of color j and half-twist sign ε, as in (2.17); it does not denote the whole analytic packet.

B.2.1 Normalization and the finite upper endpoint

Use q, Q, d, M, F_r as in Section B.1. For a physical integer $0 \le h \le M$, interpret the finite sum $A_{h,r}$ in (B.8) with lower endpoint $\max(r, h)$. Thus all its factorials have nonnegative arguments. Set

$$V_{h,r} = \frac{q^{h^2} A_{h,r}}{D_{0,r}}, \qquad \chi = \frac{\pi h}{N}, \qquad H = Q^h, \qquad Z = Q^{r+1}.$$

The denominator is the complete diagonal, which is nonzero at every $0 \le r \le M$. The Ward identity, or its polynomial version before cancellation of zero endpoint factors, gives

$$\begin{aligned} V_{h,r} &= p_{h,N}(r) V_{h,r+1} + q_N(r) V_{h,r+2}, \\ p_{h,N}(r) &= -\frac{Z^{-1}[(H + H^{-1})Z^2 + 1 + Q^{-1}]}{(1-Z)^2}, \\ q_N(r) &= -\frac{Q^{-2} Z^{-2}(1 - QZ^2)(1 + QZ)}{(1-Z)^2(1-QZ)}. \end{aligned} \tag{B.52}$$

It holds also when $r < h$: the omitted lower terms of the polynomial Ward telescope have their zero numerator factors, rather than divided negative factorials. More explicitly, the certificate $g_h(z)$ in Section B.1 vanishes at the lower endpoint $z = Q^{\max(r,h)}$: its factor $1 - z/Y$ applies when $r \ge h$, and its factor $1 - z/H$ applies when $r < h$. At $r = M - 1$ one has $q_N(r) = 0$ at both parities. We may therefore set the unused auxiliary value $V_{h,M+1} = 0$. The actual remaining terminal value obeys the exact identity

$$|V_{h,M}| = \frac{s_M^2}{s_{M-h} s_{M+h}} = \begin{cases} 1, & N \text{ odd}, \\ \cos\chi, & N \text{ even}. \end{cases} \tag{B.53}$$

Indeed $s_a s_{N-1-a} = N$. For odd $N = 2M+1$, both products in (B.53) equal N. For even $N = 2m$, $M = m-1$, $s_M^2 = N/2$ and $s_{M-h}s_{M+h} = N/(2\cos\chi)$. Thus the source at the wall has modulus at most one uniformly over the entire physical color range.

The finite triangular transport and the marked factorial phases give the following exact modulus in the normalized Newton basis:

$$\big|[\widehat{B}_r]C^{\varepsilon}_{N-h}\big| = \frac{N^2 s_{2r}}{s_r^4}|V_{h,r}|, \qquad 0 \le r \le M, \quad \varepsilon \in \{-1,1\}. \tag{B.54}$$

One can obtain this directly by interchanging the two finite sums in (B.7); the coefficient in the unnormalized basis B_r is $N^2 s_{2r} V_{h,r}/s_r^6$ times a unit phase. Multiplication by s_r^2 gives (B.54). The positive sign is the conjugate of the negative sign on the physical grid. Phases have been used only to compute the modulus, not deleted from the finite source.

B.2.2 The spectral ellipse

The characteristic roots of the limiting two-by-two recurrence matrix control the exponential size of a long matrix product. They may coincide for some colors. We first locate both roots using a quadratic equation and an ellipse in its coefficient plane. The next subsection bounds powers of the matrix directly, so no singular eigenvector basis is required at a collision.

Let $T_N(r)$ be the companion matrix of (B.52). Thus

$$(V_{h,r}, V_{h,r+1})^t = T_N(r)(V_{h,r+1}, V_{h,r+2})^t.$$

Freeze $u = (r+1)\pi/N$ and let $Q \to 1$ while $Z = e^{2\mathrm{i}u}$ and $H = e^{2\mathrm{i}\chi}$ stay fixed. The limiting matrix $T_0(u,\chi)$ has eigenvalues $Z^{-1}v$, where

$$v^2 - bv - c = 0, \qquad c = \cot^2 u, \qquad b = (c-1)\cos^2\chi - 2\mathrm{i}\sqrt{c}\,\sin^2\chi. \tag{B.55}$$

We claim

$$\rho(T_0(u,\chi)) \le \rho_0(u) := \max(1, \cot^2 u), \qquad 0 < u \le \pi/2, \quad 0 \le \chi \le \pi/2. \tag{B.56}$$

For $0 < c < 1$, a root on the unit circle would give $b = v - c/v$. The unit-root locus in the b plane is therefore the ellipse

$$\left(\frac{\Re b}{1-c}\right)^2 + \left(\frac{\Im b}{1+c}\right)^2 = 1.$$

For the value in (B.55), its left side is

$$\cos^4\chi + \frac{4c}{(1+c)^2}\sin^4\chi \le 1,$$

strictly less than one when $\chi > 0$. At $b = 0$ the roots have modulus $\sqrt{c} < 1$. The line segment to an interior point of this ellipse cannot cross the unit-root locus, so both roots remain in the open unit disk. Continuity treats the ellipse boundary. For $c > 1$, apply the same argument to v/c, with parameter $1/c$. At $c = 1$ the roots of $v^2 + 2\mathrm{i}\sin^2\chi\, v - 1$ have modulus one. At $c = 0$ they are 0 and $-\cos^2\chi$. This proves (B.56) without selecting an eigenbasis.

In particular, if $\chi_0 > 0$ and $I \subset (\pi/4, \pi/2)$ is a fixed closed interval, then

$$\sup_{\substack{u\in I\\ \chi_0\le\chi\le\pi/2}} \rho(T_0(u,\chi)) \le \theta < 1. \tag{B.57}$$

The endpoint color $\chi = \pi/2$ is included: there the two roots coalesce, but their common modulus is $\sqrt{c} < 1$.

B.2.3 Frozen blocks and the singular lower endpoint

Over a fixed number of consecutive indices, the exact recurrence matrix changes by $O(N^{-1})$. We compare that short product with a power of one constant matrix. The block length is chosen first and N is then taken large. A separate factorial bound treats the small indices where the coefficients are singular. This order of estimates gives a uniform exponential rate without asserting a polynomial prefactor at every color.

Define

$$E(u) = \begin{cases} \dfrac{2}{\pi}\displaystyle\int_u^{\pi/4} \log\cot v\,\mathrm{d}v, & 0 \le u < \pi/4,\\ 0, & \pi/4 \le u \le \pi/2. \end{cases} \tag{B.58}$$

We prove the two uniform assertions

$$\limsup_{N\to\infty} \max_{0\le h,r\le M} \left\{ N^{-1}\log|V_{h,r}| - E(\pi r/N)\right\} \le 0, \tag{B.59}$$

$$\limsup_{N\to\infty} \max_{\substack{\delta N\le h\le M\\ 0\le r\le\beta N}} \left\{ N^{-1}\log|V_{h,r}| - E(\pi r/N)\right\} \le -\gamma_{\delta,\beta}, \tag{B.60}$$

where $0 < \delta < 1/2$, $0 < \beta < 1/2$, and $\gamma_{\delta,\beta} > 0$. We use $\log 0 = -\infty$.

First fix a lower cutoff $a > 0$. On $a \le u \le \pi/2$ the exact coefficients and their first u derivatives are uniformly bounded, independently of the physical color. The exact normalized matrices $B_N(r) = T_N(r)/\rho_0(u_r)$, $u_r = (r+1)\pi/N$, differ by $O_a(N^{-1})$ from the bounded Lipschitz family $B(u_r,\chi) = T_0(u_r,\chi)/\rho_0(u_r)$.

For any bounded 2 by 2 matrix B whose eigenvalues have modulus at most one, Cayley–Hamilton gives

$$\|B^L\| \le L\|B\| + L - 1 \le C_a L. \tag{B.61}$$

The coefficient of B is $\sum_{j=0}^{L-1}\lambda_1^{L-1-j}\lambda_2^j$, of modulus at most L, and the constant coefficient has modulus at most $L-1$. These expressions remain valid at a double root. If both roots instead have modulus at most $\theta < 1$, the same calculation gives $\|B^L\| \le C_a L\theta^{L-1}$.

Choose a fixed block length L. Telescoping a product of L adjacent exact normalized matrices against its frozen value gives

$$B_N(r)\cdots B_N(r+L-1) = B(u_r,\chi)^L + O_{a,L}(N^{-1}),$$

uniformly in χ. For a prescribed $\epsilon > 0$, choose L so that $C_a L < e^{\epsilon L}$, and then take N sufficiently large. Partition any propagation product into these blocks. The incomplete end blocks cost a fixed constant, and all ordinary blocks together cost at most $C_{a,\epsilon}e^{2\epsilon N}$.

For the strict estimate, choose $I \subset (\max(\pi\beta, \pi/4), \pi/2)$. Every propagation from the physical wall to $r \le \beta N$ crosses this interval. By (B.57), increase the same fixed L until

each full block in I has norm at most $e^{-\gamma_0 L}$ for some $\gamma_0 > 0$. There are $N|I|/(\pi L) + O(1)$ such blocks. Choose $\epsilon < \gamma_0|I|/(8\pi)$. Their strict negative budget then exceeds the cost of all ordinary blocks.

Only $u = 0$ has a singular coefficient. Directly from (B.52), for $u_r \le a < \pi/8$,

$$\|T_N(r)\| \le C\frac{N^2}{(r+1)^2}. \tag{B.62}$$

Here $|H + H^{-1}| \le 2$, $|1 - Z| \ge 4(r+1)/N$, and the ratio $|1 - QZ^2|/|1 - QZ|$ is bounded by an absolute constant in this range. If $K \le aN/\pi$, the product of the bounds in (B.62) is at most $C^K N^{2K}/(K!)^2$. Hence the entire omitted initial segment costs

$$\exp\{O(Na\log(1/a)) + O(\log N)\},$$

uniformly in its starting index. This cost has rate tending to zero as $a \downarrow 0$.

Finally $\log \rho_0(u)$ is decreasing until it reaches zero, so the right-endpoint Riemann sum satisfies

$$\sum_{j=r}^{M-1} \log \rho_0((j+1)\pi/N) \le NE(\pi r/N). \tag{B.63}$$

Combine this scalar product with (B.53), the frozen blocks, and the endpoint bound. For (B.59), take $N \to \infty$ first and then let $a, \epsilon \downarrow 0$. For (B.60), first fix I and its negative budget, then make $a\log(1/a)$ and ϵ small compared with that budget, and finally take N large. This proves both estimates with their stated uniformity. There is no adiabatic eigenvector argument and no polynomial bound claimed at a spectral collision.

B.2.4 The actual coefficient envelopes and the macroscopic color loss

The elementary sine-factorial estimate is uniform up to both endpoints:

$$\log s_j = NJ(j/N) + O(\log N), \qquad 0 \le j < N. \tag{B.64}$$

For example, subtract $\log t + \log(1-t)$ from $\log(2\sin \pi t)$ on $[0,1]$. The remainder is continuously differentiable, so its Riemann-sum error is bounded; the two removed terms are ordinary factorials and have $O(\log N)$ errors by Stirling's inequalities. This also proves the uniform assertion when one of the factorial indices is zero.

The complete prefactor in (B.54) has rate $g(x) = J(2x) - 4J(x)$. Since $g'(x) = 2\log\cot \pi x$,

$$g(x) + E(\pi x) = g(\max(x, 1/4)) = \widetilde{g}(x). \tag{B.65}$$

Consequently (B.59) supplies a sequence $e_N \downarrow 0$ after taking its decreasing upper envelope, such that

$$|[\widehat{B}_r]C^{\varepsilon}_{N-h}| \le \exp\{N[\widetilde{g}(r/N) + e_N]\}, \qquad 0 \le h, r \le M, \quad \varepsilon \in \{-1, 1\}.$$

The same estimate holds for the pure sign. Its coefficient is zero for $r < h$ and otherwise is $N^2 s_{2r}/(s_r^2 s_{r-h} s_{r+h})$. Its rate is $g(x) - \Delta(x,t)$, where

$$\Delta(x,t) = J(x-t) + J(x+t) - 2J(x), \qquad 0 \le t \le x \le 1/2.$$

For $t > 0$ and $x < 1/2$,

$$\partial_t \Delta = \log \frac{\sin \pi(x+t)}{\sin \pi(x-t)} > 0,$$

because the difference of the two sines is $2 \cos \pi x \sin \pi t > 0$. Continuity includes the integrable boundary $t = x$. Thus pure cores have a uniform strict loss when x lies in a fixed compact high interval and $t \geq \delta > 0$. For signed cores the same conclusion follows from (B.60), choosing β above that high interval. Both parities, the largest physical color, and all zero coefficients are retained.

For completeness, the alias bound in (5.1) requires no recurrence. We first prove the finite bound on the untransformed alias. Write $\theta = \pi/N$. Its coefficient from (2.18), for $1 \leq j \leq \lfloor N/2 \rfloor$ and $0 \leq r < j$, has modulus

$$b_{j,r} = \frac{s_r^2}{s_{2r+1}} \prod_{u=-r}^{r} 2 \sin((j+u)\theta).$$

As a function of real j in $r < j < N - r$, this positive product is log concave and symmetric about $N/2$, so its maximum is there. Cancelling $(2 \sin u\theta)(2 \cos u\theta) = 2 \sin 2u\theta$ gives the upper bound

$$B_r = 2 \frac{\prod_{u=1}^{r} 2 \sin(2u\theta)}{\prod_{u=0}^{r} 2 \sin((2u+1)\theta)}, \qquad \frac{B_{r+1}}{B_r} = \frac{\sin((2r+2)\theta)}{\sin((2r+3)\theta)}.$$

The ratio is at most one and then at least one, with only one possible switch, at $(4r+5)\theta = \pi$. Therefore the maximum over $0 \leq r \leq \lfloor N/2 \rfloor - 1$ is at an endpoint. The first is $B_0 = \csc \theta$. At the other endpoint, the even and odd sine products give $B_{N/2-1} = N/2$ when N is even; when $N = 2m + 1$, reflection of the sine factors gives $B_{m-1} = \csc \theta$. The even product identities follow by evaluating the factorizations of $z^m - 1$ and $z^m + 1$ at $z = 1$. Finally $\sin(\pi/N) \geq 2/N$. We have proved $b_{j,r} \leq N/2$ at every color and both parities.

A triangular half-twist entry has modulus $s_v/(s_r s_{v-r})$, so (B.64) gives the rate

$$J(w) - J(u) - J(w-u), \qquad u = r/N, \quad w = v/N \geq \max(u, 1/2).$$

For $0 < u < 1$ its derivative in w is $\log[\sin(\pi w)/\sin(\pi(w-u))]$; it changes from positive to negative at $w = (1+u)/2$. Reflection $J(1-z) = -J(z)$ and duplication give the maximum

$$J((1+u)/2) - J(u) - J((1-u)/2) = -2J(u/2) = k(u).$$

The two endpoint cases follow by continuity. All factorial errors are $O(\log N)$, and there are at most N summands. This proves the alias bound with a polynomial prefactor, on its full original index range $0 \leq r \leq d$.

To pass from these leaf estimates to the complete source, first fix the positive compositional alias margins and the compact core-state gap in Chapter 5. Every fixed tree has finitely many leaves and gates and only polynomially many lattice states. Absorb their uniform e_N and the $O(\log N)/N$ gate errors into half the fixed margin. Inside the remaining strict state window, the macroscopic-color loss just proved supplies the other half of the argument. This gives the whole-source bounds (B.50)–(B.51). The result is a uniform exponential-rate envelope at all colors; the polynomial leading packet remains the separate regular local theorem of Section B.1.

B.3 Nonlocal filling charges and local contour connectors

We give the additional contour details used after the infinite tails have been removed in Chapter 8. All constants below are chosen from one fixed source neighborhood. They do not depend on the relative sizes of different filling slopes.

The exact finite Fourier polynomial satisfies a strict real gap whenever one coordinate lies outside a fixed small interval about zero. Its fixed derivatives have the same property. This follows from the cotangent rectangle estimate (8.2); summing the other physical colors in absolute value costs powers of N under the source Gaussian bound. The estimate persists in a small complex strip in that coordinate, since an imaginary displacement of height v costs at most $e^{2\pi\delta N|v|}$ on the physical support.

Fix a sufficiently small local ratio radius t_0. For a finite charge with $|m_i/p_i| \geq t_0$, split the real interval in coordinate i into $|z_i| \leq t_0/4$ and its complement. In the first interval,

$$|m_i + 2p_i z_i| \geq |p_i| t_0/2.$$

Choose the sign of the imaginary shift to decrease the real part of $-2\pi\mathrm{i}(p_i z_i^2 + m_i z_i)$. On this interval its derivative in the shift direction is bounded above by $-\pi|p_i|t_0$. The Fourier source grows at most $e^{2\pi\delta N|v|}$, and its analytic local action has bounded derivatives on the fixed tube. For sufficiently long $|p_i|$, the chirp decrease dominates uniformly. A fixed small shift gives a strict gap on the displaced segment.

The two joining segments start at $z_i = \pm t_0/4$, where the real Fourier source already has a fixed loss. Choose their common height small enough that this source loss is preserved. On the remainder of the real interval the source Fourier gap applies directly, so it need not be displaced. This proves a strict nonlocal-charge loss. The other variables remain real at this stage; their chirps have modulus one. The finite number of charges depends on the fixed slopes but is independent of N.

For a local charge, its critical point satisfies the separately even factorization in Chapter 8. For real $\mathbf{t}, \mathbf{e}$, analyticity and the real complete coefficients give

$$|\Im z_i(\mathbf{t}, \mathbf{e})| \leq C|e_i||t_i|, \qquad |\Re z_i + t_i/2| \leq C|e_i||t_i|.$$

Translate a small real box to this critical point. On its real boundary the Fourier action has a quadratic loss relative to its critical value. The chirp's real change on a connector is bounded by a constant times $|p_i||\Im z_i|$ times the boundary displacement; the critical equation cancels the linear term at the saddle. Shrinking the fixed local ratio box first and then increasing the slope threshold leaves the quadratic boundary loss strict. The real part of the Hessian on the translated real tangent space is close to that of $\Psi''(0)$ and remains negative definite. Taylor expansion gives the Gaussian with its branch fixed by this translated contour. Odd cubic terms integrate to zero; the first relative correction is $O(N^{-1})$.

These connector estimates are applied only after the tails have been removed by the real-spectator telescoping argument. They therefore introduce no uncontrolled exponential factor from a larger slope at another cusp. Finally, the selected action tends to F_0 as all reciprocal slopes tend to zero. A single threshold absorbs the fixed source and contour losses. The proof never exchanges the limit $N \to \infty$ with a slope tending to infinity.

Appendix C

The exact certificate arithmetic

The inequalities in Chapter 5 concern continuous domains. Their certificates are finite because concavity bounds each alias action by a tangent plane, while a feasible pure point bounds the pure maximum from below. The remaining difference is affine in the domain variables and slope parameters. It is therefore enough to check finitely many vertices and corners, provided coverage and all transcendental errors are controlled. This appendix supplies those controls.

C.1 A finite proof object

The two main certificates consist of rational points and exact coverage data. The two-leaf block table has sixteen closed slope intervals, one rational feasible pure point and two strict alias tangent points per interval. The attachment table is a recursively subdivided dyadic cover of $[-2, 0]^2$ with 55 leaves, one pure point and one alias tangent point per leaf. The verifier checks feasibility and coverage before evaluating an inequality. It then checks every slope corner and every vertex of the appropriate product polytope.

The mathematical reason this finite object is sufficient is independent of the way candidate points were found. A feasible pure value is a lower bound for the pure maximum. A tangent to a concave alias action is a global upper bound, including at boundary points and on the core plateau. The difference is affine in the slope parameters and the product-domain point. Corner and vertex checks therefore prove the continuum inequality. An approximate optimizer could produce a poor candidate, but could not create a false verification result.

The exact files are

- `experiments/conway_two_leaf_block_alias_certificate.json`;
- `experiments/conway_single_leaf_attachment_certificate.json`;
- the corresponding `verify_conway_..._certificate.py` programs and recorded result files.

The reproduction runner supplies the complete, unambiguous path list and verifies fresh results against the committed records. The optional endpoint certificate strengthens the attachment margin but is not required by Theorem 1.2.

C.2 Outward rational intervals

Every interval endpoint is an integer multiple of 10^{-40}. Addition and subtraction are exact on this grid. Products and quotients use floor and ceiling integer division in the outward directions, with sign cases handled explicitly. A denominator interval must avoid zero. Intersections and monotone range reductions are checked, rather than inferred from floating-point values.

The constant π is enclosed by Machin's formula

$$\pi = 16\arctan(1/5) - 4\arctan(1/239).$$

Forty terms of each alternating arctangent series, with the next-term remainder, suffice for the recorded precision. Sine arguments are reflected into $(0, \pi/2]$. The Taylor polynomial through degree 41 has error at most the next degree-43 term there. Cosine is evaluated as the sine of the complementary angle.

For a positive logarithm argument, scale by an explicitly recorded power of two. On the resulting range let $z = (x-1)/(x+1)$, with the verifier checking $0 \le z < 1/2$. Then

$$\log x = 2\sum_{j=0}^{79} \frac{z^{2j+1}}{2j+1} + R, \qquad 0 \le R \le \frac{2z^{161}}{161(1-z^2)}.$$

The same series at $z = 1/3$ encloses $\log 2$. Negative logarithms are handled by reciprocal and sign, with outward intervals preserved.

C.3 The integrated sine product

For $0 < t < 1/2$, Euler's sine product gives

$$J(t) = t\log(2\pi t) - t - \sum_{n=1}^{\infty} \frac{\zeta(2n)t^{2n+1}}{n(2n+1)}. \tag{C.1}$$

To justify termwise integration, expand $\log(1 - t^2/k^2)$ on compact subintervals below one, where the absolute double series converges; the initial logarithm is integrable at zero. The remaining terms all have the same sign. Use $\zeta(2) = \pi^2/6$, and for $2 \le n \le 20$ the rational bounds

$$\sum_{k=1}^{64} k^{-2n} \le \zeta(2n) \le \sum_{k=1}^{64} k^{-2n} + \frac{64^{1-2n}}{2n-1}.$$

Since $\zeta(2n) < 2$, the tail after $n = 20$ in (C.1) is bounded by

$$\frac{2t^{43}}{21 \cdot 43(1-t^2)}.$$

The exact identities $J(0) = J(1/2) = J(1) = 0$ and $J(1-t) = -J(t)$ reduce every action argument to this range. Thus every action and derivative used by the certificates has a proved rational enclosure.

C.4 Gradients and coverage

The derivatives evaluated by the verifier are

$$g'(y) = 2\log\cot\pi y, \qquad k'(x) = -\log(2\sin(\pi x/2)),$$
$$m_x = \log\frac{\sin\pi(x+y)\sin\pi(z-x)}{\sin\pi x\sin\pi(x+y-z)}, \qquad m_y = \log\frac{\sin\pi(x+y)\sin\pi(z-y)}{\sin\pi y\sin\pi(x+y-z)},$$
$$m_z = \log\frac{\sin\pi z\sin\pi(x+y-z)}{\sin\pi(z-x)\sin\pi(z-y)}.$$

Every tangent point is strict, and the core tangent point in the block certificate lies above $1/4$, so these derivatives are finite and select the intended branch. The entire action remains concave even where a maximizer could lie on the boundary; differentiability is needed only at the chosen tangent point.

For the square certificate, each nonterminal rectangle is split into its four dyadic children. The verifier checks equality with these children, their endpoints and their depth. Thus the 55 leaves cover the square exactly. It does not infer coverage from a list of sampled centers or from an approximate total area. The 880 corner/vertex inequalities are then enclosed from below. Their minimum is the rational number printed in Chapter 5; outward error can only weaken the lower bound.

C.5 What execution establishes

A successful replay establishes the finite supporting inequalities, their exact domains, the recorded rational margins and consistency of the immutable data. It does not numerically establish strict geometry, the complex recurrence asymptotic, or the filling theorem. Those are analytic arguments in the preceding chapters and Appendix B. The optional finite matrix audits check independent instances of the exact identities; their logical role is supplementary, not an extrapolation from tested levels.

Appendix D

Face maps, peripheral markings, and the absolute volume

This appendix supplies the complete rational dictionary behind Chapter 7. Its scope is the completed grammar in Definition 1.1, with two completed cylindrical root children. The quadrilateral gluing faces are auxiliary cut data; no physical four-punctured-boundary representation is inferred from their moduli. The original closed cut fixes the link group and every peripheral path. At each crossing circle its two shaded triangles meet at their common ideal apex; their union is the configuration called a bowtie. We retain both triangles and their labels. All matrices below are projective: multiplying a matrix by a nonzero scalar does not change its Möbius transformation. All logarithms continue from the strict complete chamber. An unfilled main cusp remains unfilled after all auxiliary crossing-circle fillings.

D.1 Two global incidence checks for the complete anchor

The complete cells and ordered stack maps are constructed in Chapter 4. Two points in that construction deserve an explicit check before one uses the group presentation.

First, strict output dihedral angles keep the matching quadrilateral a maximal supported face. Cells on opposite supporting sides intersect only in that face inside hyperbolic space. Their ideal closures can meet only on its supporting circle. An additional ideal vertex on that circle would enlarge the supporting face beyond a quadrilateral edge, contradicting its strictly nonflat angle. Thus there are no unrecorded ideal coincidences.

Second, at every open stage the four output corners are four unpaired original terminal singletons. Product consumes F_2/G_1 and F_3/G_0 and retains F_0, F_1, G_2, G_3; turns only permute these singletons. Consumed pairs never reappear among the exposed vertices of a gluing face. At the root, all four pairs under $C_j^F \mapsto C_{3-j}^G$ join the disjoint root subtrees. Therefore every main vertex is exactly a two-terminal pair, not a chain of identifications, and a leaf's four corners stay distinct. Its two shaded triangles share only their original apex. An edge and its partner in the other shaded triangle are distinct, so the complete four-step cycle contains four distinct edge occurrences. The horospherical links are positive rectangles, and the canonical shaded-side decoration gives the actual complete cusp tori described in Chapter 4.

D.2 The selected color action and the complete face-pairing maps

Write the pure leaf action and its color derivative as

$$g(x,t) = J(2x) - 2J(x) - J(x-t) - J(x+t), \qquad v = g_t = \operatorname{Log}\frac{\sin\pi(x-t)}{\sin\pi(x+t)}.$$

For physical sign $\varepsilon \in \{0, 1, -1\}$, the source-selected shear is

$$\chi = t - \frac{\varepsilon v}{2\pi\mathrm{i}}, \qquad f_\varepsilon(x,\chi) = g(x,t) - \frac{\varepsilon v^2}{4\pi\mathrm{i}}, \qquad \mathrm{d}f_\varepsilon = g_x\,\mathrm{d}x + v\,\mathrm{d}\chi. \tag{D.1}$$

At completeness its implicit derivative is $1 - \varepsilon\mathrm{i}\cot\pi x \neq 0$. Here $J(u) = \int_0^u \operatorname{Log}(2\sin\pi s)\,\mathrm{d}s$, on the continued complete branch. The finite packet identifies the shear and its constant. An explicit algebraic check is useful: put $E = e^{2\pi\mathrm{i}x}$, $H = e^{2\pi\mathrm{i}\chi}$, and $K = e^{-\partial_x f_-}$. The selected source quadratic gives

$$K^2(1+E)^2 + K[(H+H^{-1})E^2 + 2] + (1-E)^2 = 0. \tag{D.2}$$

At $H = 1$, it factors as $(K+1)[K(1+E)^2 + (1-E)^2]$, and the source branch is $K = \tan^2(\pi x)$. For the negative shear in (D.1), setting $T = e^{2\pi\mathrm{i}t}$ gives

$$H = \frac{T(E-T)}{ET-1}, \qquad T^2 + E(H-1)T - H = 0, \qquad K = \frac{(E-T)(1-ET)}{T(1+E)^2}.$$

Eliminating T gives (D.2) with the same branch. This identifies the x derivative. The remaining χ-dependent constant is zero by the actual endpoint normalization. For fixed small χ, at $x = 1/2$ one has $v = 0$, so the hidden pure color tends to $t = \chi$, not to zero. On the continued branch $J(1) = J(1/2) = 0$ and $J(1/2+t) = -J(1/2-t)$, hence $g(1/2,t) = 0$ and the shear quadratic vanishes. The finite-source action is $g(x,\chi) + S(x,\chi) - S(x,0)$, where $S(x,\chi) = \int_x^{1/2} \operatorname{Log}\mu(u,\chi)\,\mathrm{d}u$. The selected root is regular with $\mu(1/2,\chi) = 1$; both integrals therefore tend to zero. This verifies the fixed-color constant on the same holomorphic branch. Schwarz reflection gives the positive sign. Thus the geometry below does not select a new source solution or discard a quadratic color phase.

Normalize the ordered gluing face of an elementary pyramid to $(0, \infty, 1, R)$, put $E = e^{2\pi\mathrm{i}x}$, $T = e^{2\pi\mathrm{i}t}$, and select

$$Z = \frac{1+E}{1+T}, \qquad R = \frac{T(1+E)^2}{E(1+T)^2}, \qquad Z^2 - (1+E)Z + ER = 0.$$

The two projection charts have apices Z and Z/E. Their four maps are listed below. A white map pairs pieces coming from the projection sphere. A completion map is the same algebraic expression at an attachment edge where the adjacent prism supplies the actual face. We distinguish these roles before using the maps in a group relation:

$$\begin{aligned} W_0 &= \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}, & W_1 &= \begin{pmatrix} 1 & E-1 \\ 0 & E \end{pmatrix}, \\ W_2 &= \begin{pmatrix} E-R & R(1-E) \\ E-1 & 1-ER \end{pmatrix}, & W_3 &= \begin{pmatrix} R & 0 \\ 1-E & ER \end{pmatrix}. \end{aligned} \tag{D.3}$$

At an elementary pyramid, W_0, W_2 are its actual white-face maps. The expressions W_1, W_3 become actual maps when the adjacent prism is attached at the edges called cap edges in Chapter 4. All matrix identities here are projective. The pure shaded map and the two odd cross-copy maps are

$$S = \begin{pmatrix} 0 & -ER \\ 1 & -1-E \end{pmatrix}, \qquad A = \begin{pmatrix} R & -R \\ 1 & ER-1-E \end{pmatrix}, \qquad B = \begin{pmatrix} R/E-1-1/E & R \\ -1 & R \end{pmatrix}.$$

They send the original labeled triangle vertices to their prescribed partners. The apex equation verifies the fourth flag without any unit-modulus assumption.

For a prism, transport all W_j to the coordinates in (4.6). Direct rational substitution gives

$$\begin{aligned} W_{F,1} = W_{G,1} = W_{p,1} = W_{\mathrm{top}}, \qquad & W_{F,3} = W_{G,3} = W_{p,3} = W_{\mathrm{bottom}}, \\ W_{F,2} = W_{G,0}, \qquad & W_{p,0} = W_{F,0}, \quad W_{p,2} = W_{G,2}. \end{aligned} \tag{D.4}$$

Here p denotes the output gluing face, not a filling slope. The nine identifications are identities of entire maps, not just cross ratios. They can be verified without geometric continuation: use independent phase variables $\mathsf{a} = e^{2\pi i p}$, $\mathsf{b} = e^{2\pi i r}$, $\mathsf{c} = e^{2\pi i q}$, and the rational coordinates

$$\begin{aligned} &v_0 = 1, \quad v_1 = \mathsf{a}, \quad v_2 = \mathsf{ab}, \quad v_3 = \mathsf{abc}, \\ &P = 1 + \mathsf{a} - \frac{\mathsf{a}[\mathsf{ab}(1+\mathsf{c}) - 1 - \mathsf{a}]}{\mathsf{a}^2\mathsf{b}^2\mathsf{c} - \mathsf{a}}. \end{aligned}$$

The three wire phases are $\mathsf{bc}, \mathsf{ab}, \mathsf{abc}$. Normalize each ordered quadrilateral gluing face by

$$M_{a,b,c}(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}, \qquad R = M_{a,b,c}(d).$$

In world coordinates its white map is $\overline{M}^{-1} W_j M$, where the bar replaces $\mathsf{a}, \mathsf{b}, \mathsf{c}$ by their reciprocals, as prescribed by the second chart. Inserting these formulas into (D.4) gives equality entry by entry up to a common scalar. Thus no unit-circle restriction or missing translation is hidden in the identities.

The turn map $T_Q(z) = (R-1)/(z-1)$ and closing map $C_Q(z) = (z-R)/(z-1)$ satisfy

$$T_Q W_{j+1}(E,R) = W_j(E^{-1}, 1-R) T_Q, \qquad C_Q W_j(E^{-1}, R) = W_{2-j}(E,R) C_Q.$$

The identities show that the map on a white face is independent of which constituent cell is used to compute it. Thus every triangular piece of one original white face has the same map, even when its deformed image is not planar. For pure leaves, writing $\overline{S}$ for S with E replaced by E^{-1}, the edge words are

$$W_0 S = \overline{S} W_0, \quad W_2 S = \overline{S} W_2, \quad W_3 S = \overline{S} W_1.$$

For odd leaves they are

$$W_0^{-1} B W_2^{-1} A = I, \quad W_2^{-1} B W_0^{-1} A = I, \quad W_1^{-1} B W_3^{-1} A = I.$$

All hold for independent nearby E, R. They include translation and dilation, and the reverse words are inverses. The original closed cut therefore supplies a representation of the actual

half-twist partner group: use its dual two-skeleton presentation, with exactly these white-face seams and shaded four-step relations. No representation is inferred from the shape parameters of the gluing faces alone. The second charts depend holomorphically on the source parameters by the reflected-minus convention, rather than by conjugating those parameters.

A main meridian is $W_{j-1}^{-1}W_j$ at the corresponding corner. The two maps fix that corner and have equal derivatives there: E^{-1} at C_0, C_2, and E at C_1, C_3, using the finite coordinate at infinity. Thus its multiplier is one identically. It is nonidentity at completeness and stays parabolic nearby. Its commuting peripheral partner is consequently a translation as well. Every main cusp remains parabolic.

D.3 Primitive logs, the absolute regulator, and filling

We next compute the holonomy of the two actual peripheral curves. Their being primitive is a topological statement about the cusp lattice, so we check their complete translations before using their logarithms. The logarithms below are those of similarity multipliers; they are twice eigenvalue logarithms. This convention explains the factors of two in the volume differential and in the filling equation. Put

$$\rho = \frac{Z}{1+E-Z}, \qquad Q = \frac{E-Z}{Z-1}.$$

The original disk longitude is $W_2^{-1}W_0$, with multiplier Q^2 in the selected flag-at-infinity frame. The pure meridian is S^{-1}, with multiplier ρ. The two actual odd meridians are $A^{-1}W_0$ and $W_0^{-1}B$, with multipliers ρQ and ρ/Q. Returning to the original complete leaf coordinate, their translations are -2 and $2E$, while the disk longitude is $-2(E+1)$. The two-rectangle lattice fixes their primitive integral marking and orientation. The sign is fixed by the original threading convention: for the negative coupon, the typed loop $[2,1,1,2]$ followed by R^{-1} has auxiliary matrix

$$M_{\mathrm{aux}} = I - \frac{P+I}{1+E}, \qquad \operatorname{Tr} P = -2,$$

where P is the actual disk-product matrix and the preferred projective longitude is $-P$. Its translation ratio is $1+E$, identifying the first path above. Reversing the retained half twist changes the meridian by one disk longitude and gives the second. Thus the physical R^{-1} coupon uses the first path and R the second, with no sign inferred solely from a Hessian. Consequently

$$\rho = e^{-2\pi \mathrm{i} t}, \qquad Q = e^{-v}, \qquad U = -2\pi \mathrm{i} t - \frac{\varepsilon V}{2} = -2\pi \mathrm{i}\chi, \qquad V = -2v.$$

The half-longitude shear is a relation between the two actual primitive cusp markings. It does not assert that half a longitude is an integral loop or replace the actual link by a pure rational filling. The centered similarity multipliers used for U, V are squares of the corresponding eigenvalues. The original lift and pivot character are retained in the finite source; no additional square-root choice changes a peripheral word here.

Let $\Phi_\varepsilon(\chi)$ be the selected stationary action in all free wires. The envelope identity in (D.1) gives

$$U_\ell = -2\pi \mathrm{i}\chi_\ell, \qquad V_\ell = -2\partial_\ell \Phi_\varepsilon, \qquad G = 2\pi \mathrm{i}\Phi_\varepsilon, \qquad \mathrm{d}G = \frac{1}{2}\sum_\ell V_\ell \,\mathrm{d}U_\ell. \tag{D.5}$$

The source primitive includes the local term $-\varepsilon_\ell V_\ell^2/8$. It is retained before taking volume. At the complete point, $\operatorname{Im} G(0) = 2\pi F_0$ is the actual half-twist partner volume from the explicit complete cells and original pairings.

For the nearby framed representation, triangulate the same original cut balls and retain their peripheral decoration. The ideal-tetrahedron variation follows directly from the Bloch–Wigner formula

$$\mathrm{d}D(z) = \log|z|\,\mathrm{d}\arg(1-z) - \log|1-z|\,\mathrm{d}\arg z.$$

In the sum over the original oriented tetrahedra, face terms have equal flags and opposite orientations and cancel. On a peripheral torus, cut along the recorded primitive μ, λ to a parallelogram. The opposite sides cancel except for their two eigenline scale factors; the remaining alternating product is $-2\,\mathcal{M} \wedge \mathcal{L}$, where $\mathcal{M} = e^u, \mathcal{L} = e^v$ and $u = U/2, v = V/2$. Here the sign is the boundary orientation of the truncated original fundamental chain, and the factor two retains both eigenline flags. Applying the displayed differential to that alternating product gives $2(\Re v\,\mathrm{d}\Im u - \Re u\,\mathrm{d}\Im v)$. Different boundary tori have zero mutual intersection, and the parabolic main tori have zero log contribution. Thus

$$\mathrm{d}\operatorname{Vol}(\rho) = \frac{1}{2}\sum_\ell \bigl(\operatorname{Re} V_\ell\,\mathrm{d}\operatorname{Im} U_\ell - \operatorname{Re} U_\ell\,\mathrm{d}\operatorname{Im} V_\ell\bigr).$$

Using (D.5), this is the differential of $\operatorname{Im} G - \frac{1}{2}\sum \operatorname{Re} U \operatorname{Im} V$. The complete anchor fixes the constant on the same connected marked branch. Therefore the absolute identity is

$$\operatorname{Vol}(\rho_\chi) = \operatorname{Im} G - \frac{1}{2}\sum_\ell \operatorname{Re} U_\ell \operatorname{Im} V_\ell. \tag{D.6}$$

This uses the original relative fundamental chain and its actual nearby representation, not equal differentials on unrelated components.

D.4 Smooth completion on the actual primitive filled topology

We give the metric argument used here, including its uniformity in the independently chosen slopes. Let $M = Y_E^\varepsilon$ carry the oriented complete metric constructed above, with holonomy ρ_0. A nearby *framed* representation consists of the actual face-paired representation ρ and, at each cusp, its selected common fixed flag. In a frame sending that flag to infinity, the two peripheral maps are complex similarities. Their uniquely small multiplier logs are the U, V just computed. We construct one neighborhood of this framed complete point before choosing the filling slopes.

D.4.1 An actual positive metric on the core and its ends

The first task is to turn nearby holonomy data into local hyperbolic coordinate charts on the same manifold. On a compact truncated part we average small changes of the complete developing map. On each end we use an equivariant affine construction. Both maps are close to the original one on their common compact collar, allowing them to be joined without losing local invertibility. Choose a compact cusp-truncated core with a slightly larger collar, finitely many evenly covered balls, and a smooth partition of unity. Write $\eta_{j,\gamma}$ for its lifts, with one

chosen lift of each ball and $\gamma \in \pi_1(M)$. In the future unit hyperboloid model of hyperbolic space put

$$Y_\rho(x) = \sum_{j,\gamma} \eta_{j,\gamma}(x)\rho(\gamma)\rho_0(\gamma)^{-1}x, \qquad E_\rho(x) = \frac{Y_\rho(x)}{\sqrt{-\langle Y_\rho(x), Y_\rho(x)\rangle}}.$$

Every summand is future timelike; their convex average is future timelike too, so the normalization is defined. Reindexing the sum proves $E_\rho(\rho_0(\delta)x) = \rho(\delta)E_\rho(x)$. Only finitely many terms occur on a compact fundamental part, and at ρ_0 the map is the identity. Thus it is C^1-close to the identity and is an orientation-preserving local diffeomorphism in one fixed representation neighborhood. This does not assume that the deformed action is already discrete or free.

For one complete cusp write its universal coordinates as $(w, s) \in \mathbb{C}\times[s_0, \infty)$, with translation lattice Λ, and complete development (w, e^s) in upper half-space. Denote the deformed affine peripheral action by $A_\lambda(z) = a_\lambda z + b_\lambda$. Choose a nonnegative compactly supported smooth η with $\sum_{\lambda\in\Lambda} \eta(w - \lambda) = 1$, and define

$$d_\rho(w) = \sum_{\lambda\in\Lambda} \eta(w - \lambda)A_\lambda(w - \lambda).$$

Then $d_\rho(w + \lambda) = A_\lambda d_\rho(w)$. On a fixed closed fundamental parallelogram it is C^1-close to w, so its real Jacobian is positive; equivariance propagates that fact everywhere. Let ξ_ρ be the real-linear extension of $\lambda \mapsto \log|a_\lambda|$. The map

$$F_\rho(w, s) = \big(d_\rho(w), e^{s+\xi_\rho(w)}\big)$$

is exactly peripheral-equivariant. Its differential has positive determinant $e^{s+\xi_\rho(w)} \det_{\mathbb{R}} Dd_\rho(w)$ for every $s \geq s_0$. Restore the moving flag frame and extend to all cusp lifts by equivariance. Geodesically interpolate E_ρ and F_ρ on the compact collar with an invariant smooth cutoff. There both maps are C^1-close to the complete development, so the interpolation remains a positive local diffeomorphism after one further uniform shrink. Pullback of the hyperbolic metric now gives an actual oriented nonsingular metric on the original manifold with its ideal ends deleted.

D.4.2 The logarithmic link and its primitive angular period

At a filled cusp, the nearby peripheral similarities have two common fixed points. Sending one to infinity and translating the other to zero makes them multiplications. Their logarithms then act by translations on a Euclidean plane. We must show that the translations are exactly the selected small logs U, V, with no unnoticed multiple of 2πi. Only after this check may the large integer p be used in the relation $U + pV = -2\pi$i. At a cusp to be filled, the equation $U + pV = -2\pi$i ensures that at least one multiplier differs from one. Commuting affine maps then have a common finite fixed point q, in addition to infinity. For either generator with $a \neq 1$, $q = b/(1 - a)$. Its numerator stays away from zero near completeness, whereas both generator multipliers tend to one; hence $|q| \to \infty$ uniformly, even if one multiplier equals one. The developed fundamental parallelogram and fixed generator paths have uniformly bounded images and lengths. After shrinking the neighborhood they miss q, and equivariance implies that the entire development misses q.

On the simply connected link cover choose $f = \operatorname{Log}(d_\rho - q)$. Its differential is invertible as a real map and is invariant under deck transformations. Consequently $|\mathrm{d}f|^2$ descends to a nonsingular Euclidean metric on the compact torus, hence is complete. Lifting Euclidean segments proves that its developing local isometry f covers $\mathbb{C}$; because the plane is simply connected, this is a global isometry. Its periods form an injective real rank-two lattice. A fixed generator path of developed length at most L, contained in $|d_\rho| \le R$, satisfies

$$|\Delta f| \le \frac{L}{|q| - R} \longrightarrow 0.$$

Exponentiating its increment gives the corresponding similarity multiplier. The increment and its prescribed small log therefore differ by an integer multiple of $2\pi\mathrm{i}$ which is exactly zero near completeness. Thus the actual two log-link periods are U, V, before they are multiplied by any large slope. This is why the same representation neighborhood works for all slopes.

Translate q to zero. On the compact link quotient the function $e^{s_0+\xi_\rho(w)}/|d_\rho(w) - q|$ is bounded above. The explicit end therefore contains an entire equivariant epigraph $y > A|d_\rho - q|$. With

$$r = \frac{y}{|d_\rho - q|}, \qquad R = \operatorname{arsinh}(1/r), \qquad \theta = \Im f, \qquad X = \Re f + \tfrac{1}{2}\log(1 + r^2),$$

direct substitution into the upper-half-space metric gives

$$\mathrm{d}s^2 = \mathrm{d}R^2 + \cosh^2 R\,\mathrm{d}X^2 + \sinh^2 R\,\mathrm{d}\theta^2. \tag{D.7}$$

The epigraph contains a whole punctured tube $0 < R < R_0$. A log period L acts by $(X, \theta) \mapsto (X + \Re L, \theta + \Im L)$. The actual primitive curve $\mu + p\lambda$ has period $-2\pi\mathrm{i}$, exactly one angular turn. Its complementary primitive period can be taken to be V, which has nonzero real part since U, V are real-linearly independent. Adding the axis $R = 0$ in (D.7) is therefore a smooth hyperbolic solid torus, whose meridian is exactly $\mu + p\lambda$. Its affine holonomy is the identity, since it fixes both q and infinity with multiplier one. The representation descends to the actual filled group. Reversing the angular sign changes the parametrization of the meridian, not the orientation of the metric.

At an unfilled main cusp both multipliers are one. The same construction gives a translation-equivariant Euclidean link development, globally the plane by completeness of the compact link; $\xi_\rho = 0$. Its end contains a full horoball quotient, so it is complete and has finite volume. The remaining region is compact. Attaching the finitely many bounded-radius solid tori therefore gives a complete finite-volume hyperbolic metric on precisely the filled topology. Its complete universal developing map covers simply connected hyperbolic space, so the holonomy is discrete and faithful. This proof used one fixed framed neighborhood and no comparison between different slope sizes.

D.4.3 The source saddle and its absolute knot-complement volume

The full source filling transform selects a critical point of

$$S_{\varepsilon,\mathbf{p}}(\chi) = \Phi_\varepsilon(\chi) + \frac{\pi\mathrm{i}}{2}\sum_\ell \frac{(\chi_\ell - 1)^2}{p_\ell}.$$

Its equations are exactly $U_\ell + p_\ell V_\ell = -2\pi\mathrm{i}$. For existence set $r_\ell = 1/p_\ell$ and write them as $V_\ell + r_\ell U_\ell = -2\pi\mathrm{i} r_\ell$. At $r = 0$ their χ-Jacobian is $-4\pi\mathrm{i}\,\mathrm{diag}(\tau_\ell)$, with $\Im\tau_\ell > 0$, by the source packet. It is invertible. The holomorphic implicit-function theorem thus gives one nearby solution for all r in a fixed polydisk, independently of their relative sizes and signs. It lies in the framed neighborhood just constructed after that polydisk is shrunk once more. For independently sufficiently long fixed signed integral p_ℓ, the marked geometric completion gives smooth angle 2π on these primitive filled curves, with the unfilled main cusp parabolic. The physical leaf power is $R^{\varepsilon_\ell - 2p_\ell}$, once, in its original disk framing. At this saddle, (D.6) is

$$\mathrm{Vol}\big(S^3 \setminus K_{F,G,\varepsilon}(\mathbf{p})\big) = 2\pi\,\mathrm{Re}\,S_{\varepsilon,\mathbf{p}}(\chi_*).$$

All auxiliary circles have been filled and the chosen main component is the knot. The knot cusp is unfilled. The separate full-source alias, complex-packet and entire Fourier-period arguments are what identify this geometric action with the original invariant's limiting rate.

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